E-flow: Intelligent Document Tracking and workflow management system for universities

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DECLARATION

We are students of Adama Science and Technology University in the School of Electrical Engineering and Computing in the Department of Computer Science and Engineering. The information found in this project is our original work. And all sources of materials that will be used for the project work will be fully acknowledged.

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This project has been submitted for examination with our approval as a university advisor.

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Acronomy

- **E-flow** – Electronic Flow
- **OCR** – Optical Character Recognition
- **DMS** – Document Management System
- **EDMS** – Electronic Document Management System
- **BPM** – Business Process Management
- **WfMS** – Workflow Management System
- **RBAC** – Role-Based Access Control
- **2FA** – Two-Factor Authentication
- **OTP** – One-Time Password
- **PKI** – Public Key Infrastructure
- **API** – Application Programming Interface
- **REST** – Representational State Transfer
- **SSL** – Secure Sockets Layer
- **TLS** – Transport Layer Security
- **AES** – Advanced Encryption Standard
- **ISO** – International Organization for Standardization
- **UI** – User Interface
- **UX** – User Experience
- **ERD** – Entity Relationship Diagram
- **UML** – Unified Modeling Language
- **JSON** – JavaScript Object Notation
- **JWT** – JSON Web Token

- **SLA** – Service Level Agreement
- **DPI** – Dots Per Inch
- **LCNC** – Low-Code / No-Code
- **EUD** – End-User Development
- **HEI** – Higher Education Institution
- **PII** – Personally Identifiable Information
- **CIA** – Confidentiality, Integrity, and Availability
- **TAM** – Technology Acceptance Model
- **ROI** – Return on Investment
- **ICT** – Information and Communication Technology

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Abstract

This project presents **E-flow**, an intelligent document tracking and workflow management system designed to digitize and automate administrative processes within universities. Addressing the inefficiencies of manual paper-based handling, E-flow introduces a visual drag-and-drop workflow builder, allowing administrators to customize approval chains dynamically. The system integrates Optical Character Recognition (OCR) to bridge the gap between physical and digital records, alongside a secure two-factor authentication (2FA) module for digital signatures. By providing real-time tracking, role-based access control, and comprehensive audit trails, E-flow enhances transparency, reduces processing delays, and supports the digital transformation goals of modern higher education institutions.

Chapter 1

1. Introduction

Universities and higher education institutions have made significant progress in digitalizing their academic and administrative operations. Yet, despite these advancements there are many opportunities remain to enhance efficiency, transparency, and interdepartmental coordination through smarter automation tools. This Project introduces “E-flow: Intelligent Document Tracking and workflow management system for universities” a web- based platform designed to complement and enhance existing digital ecosystems. This system will serve as an intelligent automation layer, providing a unified platform for document routing, secure approval, and transparent monitoring. A key feature is the inclusion of Optical Character Recognition (OCR), allowing the system to scan and digitalize workflow. This capability helps institutions transition away from paper-based systems without disrupting established procedures.

Unlike conventional digital tools they rely on rigid, preconfigured workflows, E-Flow’s primary innovation is the Visual workflow builder . This drag and drop interface enables administrators to design, adjust , and manage complex approval chains with ease. Departments can create workflows tailored to their unique needs without writing a single line of code making the system accessible to both technical and non-technical users .

By introducing a flexible and secure approach to workflow automation, E-Flow aims to enhance accountability,minimize processing delays, and strengthen coordination across the university’s digital ecosystem. Ultimately, the system supports institutions in becoming more responsive, organized, and transparent in their day to day operations.

1.2 Background of the project

Organizations in recent years increasingly rely on Document management systems (DMS) to keep information structured , secure and easily accessible . These systems strengthen collaboration across departments by automating repetitive processes and minimizing manual workloads. As a result, institutions can focus more on decision-making and less on administrative overhead.

Within higher education institutions, digital transformation initiatives have already created a strong foundation for technology-driven administration. The proposed project builds upon this progress by introducing an intelligent visual workflow automation layer. Through this addition, universities will be able to design, visualize, and manage approval flows in real time. The combination of a secure DMS, OCR capabilities for digitizing paper documents, and an interactive workflow builder enables administrators and staff to define sequences such as Dean → Registrar → President for document approvals, while maintaining version control, verified digital signatures, and comprehensive audit trails.

Rather than disrupting existing platforms, this solution extends their functionality by introducing higher levels of automation, consistency , and transparency in document lifecycle management. From the initial submission of a document to its final approval, every step becomes traceable, efficient, and less prone to human error.The system will be developed using React.js with React Flow for the front-end, Node.js/Express for the back-end, PostgreSQL for the database, and an OCR library (like Tesseract.js) , all of which are open-source and scalable technologies suited for institutional use.

1.3 Statement of the problem

Despite the growing need for streamlined administrative processes, many higher education institutions still rely on manual or partial methods for handling documents. These approaches are often inefficient . Without a centralized intelligent workflow system it will be difficult to monitor documents and maintain consistent service quality across departments.

The primary problems addressed by the project include:

1. **Lack of Document Traceability:**

- Once a document is submitted (whether paper or digital), there is often no clear or standardized system to monitor its movement between offices.
- Students and staff are required to make frequent follow-ups in person or via email to determine their request's status. This leads to uncertainty, wasted time, and operational inefficiency.

➢ **Data Inaccessibility and Processing Delays:**

- Physical documents are vulnerable to being misplaced, filed incorrectly, or left unattended.
- Furthermore, existing digital documents (such as scanned PDFs) are often stored as static, non-searchable images. This "dead data" makes it impossible to find documents by searching for their content, creating digital "black holes" and forcing manual data re-entry.
- Without digital timestamps or action history, it is difficult to identify where or why a document is delayed, resulting in missed deadlines and dissatisfaction.

➢ **Limited Accountability and Security Risks:**

- When actions are not logged in a secure, centralized system, it is difficult to hold individuals responsible for workflow disruptions.
- More importantly, this creates a significant security risk. Without a verifiable audit trail or a system for verifying a signer's identity (beyond just clicking "approve"), there is no true non-repudiation. Sensitive documents are routed without proof of who approved them and when, posing a high risk to institutional integrity.

➢ **Static and Inflexible Digital Systems:**

- Even when digital systems are in place, they are often "hard-coded" with rigid, static workflows.

- When an administrative process changes (e.g., a new approval step is added), these systems cannot adapt without complex and costly reprogramming.
- This lack of flexibility forces administrators and staff to create "workarounds" (like reverting to email or paper), breaking the digital workflow and re-introducing all the problems of manual processing.

To overcome these persistent issues, universities must adopt a secure and flexible workflow management system. Such a solution enhances transparency, minimizes risk, and ensures more efficient experience for both staff and students.

1.4. Objectives

1.4.1 General Objectives

The general objective of this project is to develop E-flow, an intelligent document tracking and workflow management system designed to streamline document handling and enhance institutional transparency through automated approval processes and real-time status monitoring.

1.4.2 Specific Objectives

- **To design a centralized, role-based architecture** that facilitates secure document submission and management while ensuring data integrity through robust access control mechanisms.
- **To develop a dynamic workflow engine** utilizing a visual drag-and-drop interface to allow administrators to define and modify departmental approval sequences.
- **To implement an automated routing and notification system** that ensures seamless document progression and real-time stakeholder engagement.
- **To integrate Optical Character Recognition (OCR) technology** for the transformation of physical and image-based documents into searchable digital formats.
- **To establish a comprehensive tracking and audit framework** that provides real-time status updates, version control, and time-stamped processing histories.
- **To incorporate a secure approval verification system** employing multi-factor authentication to ensure the authenticity and accountability of digital signatures.
- **To evaluate the system's performance and usability** through rigorous testing to ensure it meets the administrative needs of the university environment.

1.5 Scope and Limitation

1.5.1 Scope of the project

The scope of this project is to design and implement a web- based systems that digitizes document management processes and introduces a visual workflow builder for defining, automating, and tracking approval sequences across university departments. The system's primary goal is to complement a university's existing digital digital infrastructure by introducing a layer of intelligent process automation rather than replacing any of its current platforms.

Target Users:

- **Administrators and Department Heads:** Can design and customize workflows for various document types (e.g., student requests, memos, purchase approvals).
- **Administrative Staff:** Handle assigned tasks, review documents, and track approval progress.
- **Students and Faculty:** Submit forms and monitor status updates through the portal.

Functional Scope:

- **OCR-based Document Ingestion:** Ability to scan, digitize, and process paper documents or image-based PDFs, extracting text to make them searchable within the system.
- Document upload, retrieval, and version control with metadata tagging.
- Workflow creation and automation using a drag-and-drop interface (powered by React Flow).
- **Verified Digital Signature Module:** A secure signature function where approvals are confirmed using a Two-Factor Authentication (2FA/OTP) process, ensuring signer identity and non-repudiation.
- Role-based access control (RBAC) and notification mechanisms for each workflow stage.
- Real-time progress tracking with comprehensive audit trails.
- Reporting dashboard for performance and activity summaries.

Non-Functional Scope:

- **Performance:** The system will be optimized for reliability, supporting concurrent users without significant latency.
- **Security:** The system will be designed to be "very secure." All data in transit (SSL/TLS) and at rest (e.g., AES-256) will be encrypted. It will enforce granular RBAC and secure authentication protocols to maintain confidentiality and integrity.
- **Scalability:** Built with modular architecture to allow future integration with other institutional systems such as the eStudent or staff portal.
- **Usability:** The system will feature an intuitive interface, designed to require minimal training.

System Architecture: The project employs a three-tier architecture comprising a React.js front-end, a back-end (Node.js) which includes an OCR processing engine (e.g., Tesseract.js), and a PostgreSQL database. RESTful APIs will facilitate data exchange, and the visual workflow engine will operate on top of this infrastructure to manage dynamic process flows.

Development Methodology: The system will be developed following an Agile approach, with iterative cycles of design, development, and user feedback. Regular reviews with key stakeholders will ensure continuous alignment with institutional needs. The system's design will be guided by principles from ISO 15489 (Records Management) for document lifecycles and ISO 27001 (Information Security) for the security framework.

Evaluation Criteria: The system’s success will be measured based on usability, system responsiveness, workflow accuracy, OCR accuracy, error rates, and user satisfaction obtained through testing and feedback sessions.

1.5.2 Limitations

- **Integration with Institutional Systems:** The "E-flow" system will operate as a standalone platform. Future integration with existing institutional systems (e.g., student finance, grading databases) is possible but outside this project's initial scope.
- **No Native Mobile Application:** (Only a responsive web interface will be provided).

- **OCR Accuracy:** The Optical Character Recognition (OCR) module's accuracy is dependent on the quality of the scanned document (e.g., clarity of text, handwriting, document layout). 100% accuracy on all document types is not guaranteed.
- **Signature Verification Scope:** The system implements strong identity verification using Two-Factor Authentication (email OTP). It does not implement a full PublicKey Infrastructure (PKI) for cryptographic certificates, which is considered beyond the scope of this project.

1.6 Deliverables

Upon completion, the project will deliver the following key component

- A fully functional web application titled "E-flow: Intelligent Document Tracking and workflow management system for universities."
- A Visual Workflow Builder module that allows administrators to create and modify document approval routes.
- An Optical Character Recognition (OCR) module integrated into the back-end for digitizing, indexing, and extracting text from scanned paper documents and PDFs.
- A Secure Signature Verification module that, before applying any approval, confirms the user's identity via a Two-Factor Authentication (2FA/OTP) challenge sent to their official email.
- An administrative dashboard for real-time monitoring and reporting.
- Comprehensive project documentation, including user manuals, system architecture diagrams, and testing results.
- The complete source code and deployment scripts for the application.

1.7 Feasibility Study

1.7.1 Technical Feasibility

The project is technically feasible as it relies on open-source, widely-used, and well-documented technologies. The development team has foundational knowledge of the core stack (React.js, Node.js , and PostgreSQL) from their academic coursework.

The project's advanced components are also feasible due to the availability of established libraries:

1. **Visual Workflow Builder:** React Flow, the component for the workflow builder, is a well-documented open-source library, making this complex feature achievable.

2. **Optical Character Recognition (OCR):** The OCR requirement will be met using an established open-source engine like Tesseract.js. This library integrates with Node.js, allowing the system to perform text extraction from scanned documents without requiring the team to build a complex AI model from scratch.

3. **Secure Signature Verification:** This "very secure" requirement will be implemented via a Two-Factor Authentication (2FA/OTP) mechanism. This involves generating a secure token and sending it to a user's registered email. This is a standard and achievable feature using common back-end libraries (e.g., Nodemailer, crypto) and is far more feasible than implementing a full Public Key Infrastructure (PKI).

The university’s existing IT infrastructure (servers, network) is sufficient for deployment, although the OCR module will require adequate server-side processing resources. No specialized or proprietary hardware is required, making the technical barriers manageable.

1.7.2 Operational Feasibility

Operationally, the system assesses whether the system will effectively function within the university's environment and whether stakeholders will adopt and benefit from it Based on current workflows and stakeholder needs, the "E-flow" system is highly practical and aligns with an institution’s operational goals.

1. **Usability and Accessibility:** The system will feature an intuitive and responsive web interface accessible through any modern browser. The Visual Workflow Builder allows administrators to define processes without technical expertise, promoting ease of use. End users, including staff, students, and management, can interact with the platform using simple dashboard controls and receive real-time notifications about document progress.

2. **User Acceptance and Training:** The system's introduction will be accompanied by a brief training and demonstration phase for administrators and staff. This training will specifically cover how to use the new OCR scanning feature for paper documents and how to use the 2FA/OTP process for signature verification. Because the interface mimics familiar digital patterns, users are expected to adapt quickly. The solution complements existing digital systems rather than replacing them, ensuring minimal disruption.

3. **Maintenance and Support:** The system will be designed for long-term maintainability. Clear documentation will accompany the project, enabling future developers or IT staff to manage updates and enhancements easily. Using open-source tools eliminates dependency on external vendors, and modular design simplifies troubleshooting.

4. **Security and Privacy:** Given the sensitivity of academic and administrative documents, a "very secure" design is a core operational requirement.

    - **Encryption:** The system will implement encryption for data at-rest (e.g., AES-256) and in-transit (SSL/TLS).

    - **Signature Verification:** A secure signature verification module (2FA/OTP) will be implemented to ensure the identity of all approvers.

    - **Access Control:** Access will be strictly role-based, ensuring each user can only view or act on documents relevant to their position.

    - **Standards:** The entire security framework will be guided by principles from the **ISO 27001** standard for information security management.

5. **Organizational Alignment:** The "E-flow" system aligns directly with a university's vision of becoming a technologically advanced and digitally efficient institution. It integrates

seamlessly into existing workflows, including new digital (OCR) processes for paper-based tasks, enhancing collaboration and transparency. The system's flexibility allows departments to tailor workflows to their unique processes, and its design will be guided by ISO 15489 (Records Management) principles to ensure proper document handling, fostering ownership and long-term engagement.

1.7.3 Economic Feasibility

The economic feasibility of the proposed "E-flow" system is highly favorable. The project utilizes open-source tools, existing institutional infrastructure (like labs and internet), and the technical expertise of the student development teams. This approach significantly reduces financial barriers while still demonstrating a project with high institutional benefit.

1. **Cost Analysis:** The estimated project budget is approximately 3,000 ETB, covering internet access for research, printing of documentation, and miscellaneous contingency costs. The primary cost-saving is the use of open-source technologies: there are no licensing fees for React.js, Node.js, PostgreSQL, the Tesseract.js OCR engine, or the React Flow library. Development and demonstration will be handled on the developers' personal laptops, eliminating the need for any cloud hosting or new hardware costs for this prototype phase.
2. **Benefit Analysis:** The system's potential benefits extend well beyond its modest development costs.
    - **Direct Cost Savings:** The OCR module directly addresses the high cost of paper-based processes. A deployed version of this system would lead to measurable reductions in institutional spending on paper, toner, printers, and physical file storage.
    - **Operational Efficiency:** By automating workflows, the institution would save thousands of administrative hours. The OCR feature eliminates time-consuming, error-prone manual data-entry from paper forms, freeing staff for higher-value tasks.
3. **Return on Investment (ROI):** Although this is an academic project focused on a prototype, its potential institutional ROI is significant. If implemented, an institution could see a direct, measurable ROI from drastic reductions in printing, paper, and physical

storage costs. This is in addition to the "soft ROI" from improved process turnaround times, enhanced data security, and full accountability. These outcomes easily justify the modest investment.

4. **Resource Allocation and Utilization:** The development team efficiently utilizes human and technological resources, dividing tasks according to specialization (frontend, backend, database, security/2FA implementation, OCR integration, and UI/UX). Furthermore, relying on personal laptops and existing university facilities ensures that the project timeline estimated at six months is achievable without additional financial burden.

1.8 Significance of the project

The proposed E-flow system carries important academic, institutional, and social value. It strengthens a university's digital ecosystem and demonstrates how technology can enhance efficiency, security, transparency, and collaboration in higher education administration.

Institutional Significance: E-flow supports digital transformation by integrating intelligent workflow automation and OCR, closing the gap between digital systems and traditional paper processes. It ensures all document stages are traceable and auditable, while aligning with standards such as ISO 15489 and ISO 27001 to reinforce trust, accountability, and service quality. Its paperless approach also contributes to sustainability efforts.

Academic and Research Value: The system serves as a practical case study in modern software engineering, combining database management, web development, UX design, and information security with applied AI through OCR. It provides a foundation for future research in workflow optimization, secure digital governance, and data-driven analysis of digitized records.

Social and Environmental Impact: By digitizing documents and reducing printing, the system lowers paper waste and promotes environmentally responsible practices. Secure verification mechanisms, such as 2FA/OTP signatures, foster accountability and strengthen institutional communication.

Broader Contribution: With its modular, open-source design, E-flow can serve as a model for other Ethiopian universities and public organizations seeking to modernize their document workflows in an efficient, secure, and eco-friendly manner.

1.9 Beneficiaries of the project

The implementation of the E-flow system will benefit a wide range of stakeholders within a university and can also be adopted by other institutions seeking improved transparency, security, and administrative efficiency.

University Administration: Leaders and decision-makers gain real-time visibility into document movement, approval delays, and workflow performance. This supports data-driven decision-making and strengthens institutional oversight. The inclusion of 2FA/OTP verification and ISO 27001–aligned security ensures every approval is authenticated, auditable, and compliant, reducing organizational risk.

Departmental Staff: Administrative staff—who handle most document processing—will see a significant reduction in manual workload.

Automated routing and notifications: remove the need for constant follow-ups. OCR integration eliminates manual data entry from paper documents, enabling staff to search, manage, and process legacy paper records as easily as digital ones.

Students and Faculty: End users benefit from faster request processing and greater transparency. They can submit forms online, check approval status in real time, and receive automated updates, reducing the need for physical visits or repeated inquiries.

It and System Administrators: Technical teams benefit from E-flow's modular architecture, maintainable codebase, and thorough documentation. Its adherence to ISO 15489 and ISO 27001 principles makes oversight, security management, and future system expansion more efficient and standardized.

The Institution as a Whole: The university gains improved operational efficiency, cost reductions, and environmental benefits. Digitizing paper archives and limiting new paper use

contributes to sustainability goals, frees storage space, and advances the institution's broader digital transformation strategy.

1.10 Methodology

The methodology of this project follows a systematic and iterative approach based on the principles of Agile software development. This approach allows for continuous collaboration with stakeholders, adaptability to evolving requirements, and consistent delivery of functional modules. The project progresses through several well-defined phases, ensuring clarity, quality, and alignment with institutional goals.

1. **Planning and Requirement Analysis:** The project begins with gathering and analyzing requirements from administrative staff, faculty, and university management. Interviews and workflow observations are conducted to understand existing digital and paper-based document processes and identify automation opportunities. The team then defines functional and non-functional requirements, such as usability, performance, and "very secure" design. This phase includes a risk assessment and defining compliance requirements, guided by principles from ISO 15489 (Records Management) and ISO 27001 (Information Security).
2. **System Design:** In this phase, the system's architecture, database schema, and user interface are designed. UML diagrams, Entity-Relationship Diagrams (ERD), and wireframes are developed to visualize the data flow and user interaction. The architecture adopts a three-tier model (frontend, backend, database) and includes the design for:
    - The secure 2FA/OTP signature verification module.
    - The processing pipeline for the OCR engine.
    - The Role-Based Access Control (RBAC) model.
    - The design will adhere to the ISO 27001/15489 principles defined in the planning phase.
3. **Implementation and Development:** The development process follows an incremental model. Each module (Visual Workflow Builder, OCR engine integration, 2FA/Signature module, document tracker, user management, etc.) is developed and integrated

progressively. The frontend is built using React.js and React Flow for a dynamic interface. The backend is powered by Node.js with Express for robust API and server-side logic, integrating an OCR library (e.g., Tesseract.js). PostgreSQL manages all document metadata, workflow configurations, and user data.

4. **Testing and Quality Assurance:** Testing is performed at multiple stages, including unit testing, integration testing, and system testing.
    - Functional tests verify that features meet specifications.
    - Usability tests ensure intuitive navigation.
    - OCR accuracy testing will be conducted using scanned documents of varying quality.
    - Security testing is critical and will focus on data encryption, access control vulnerabilities, and verifying the integrity of the 2FA signature module.
    - Feedback from stakeholders will guide iterative improvements.
5. **Deployment and Evaluation:** Upon successful testing, the system is deployed on a secure cloud or on-premise server. Training sessions are provided for users, emphasizing how to use the OCR scanning feature, how to create workflows, and how to use the 2FA signature process. Post-deployment evaluations are conducted to measure performance, user satisfaction, and overall system reliability.

This methodological framework ensures that the project remains goal-oriented, user-centered, and adaptable while maintaining a strong focus on technical precision and academic rigor.

1.11 Development Tools

The development tools and technologies selected for this project are chosen for their reliability, open-source nature, and compatibility with modern web development standards. Each tool contributes to a specific layer of the system, ensuring efficient design, implementation, and maintenance.

1. **Frontend Development:**

- **React.js:** Provides a component-based framework for building dynamic, responsive, and interactive interfaces.
- **React Flow:** Enables the drag-and-drop workflow builder functionality for designing visual approval flows.
- **Tailwind CSS / Material UI:** Used for styling the application to ensure a professional and user-friendly appearance.

2. **Backend Development:**
    - **Node.js with Express.js :** Handles server-side logic, API development, user authentication, and workflow automation logic.
    - **Tesseract.js** : The core Optical Character Recognition (OCR) engine for extracting text from scanned documents and PDFs.
    - **Nodemailer** : For sending the One-Time Password (OTP**)** emails required for 2FA signature verification.
    - **Crypto (Node.js)** : For generating secure cryptographic tokens for the 2FA/OTP process.
    - **JWT (JSON Web Token):** Ensures secure authentication and role-based access control.
3. **Database Management:**
    - **PostgreSQL:** Used for storing user accounts, document metadata, workflow structures, audit logs, and approval history. It supports JSON fields, making it ideal for dynamic workflow storage.
4. **Development and Collaboration Tools:**
    - **Git & GitHub:** Version control and team collaboration.

- **VS Code / PyCharm:** Integrated Development Environments for coding and debugging.
- **ClickUp / Trello:** Project management tools for tracking progress and assigning tasks.

5. **Testing and Deployment Tools:**
   - **Jest / PyTest:** For automated and manual testing of modules, including security and OCR accuracy tests.
   - **Yegara Host or Cloud Deployment Platform:** For hosting the system securely with SSL encryption, a critical component of the "very secure" requirement.

These tools collectively support an efficient, scalable, and maintainable system, ensuring the project's success from development to deployment.

1.12 Required resources with cost

The "E-flow: Intelligent Document Tracking and workflow management system for universities" will be developed using open-source technologies on the developer's existing laptop, with all required software already installed. The project relies on university facilities for internet access and presentation purposes.

Since no new hardware or paid software will be required, the project cost is limited to minimal operational expenses. The estimated total cost is 3,000 ETB.

Resource	**Estimated Cost (ETB)**	**Description**
Internet and Research Access	500	Occasional use of university lab or library internet for research, updates, and collaboration.
Printing and Documentation	1,000	Printing hard copies of reports, mockups, and project documentation for defense and submission.

Miscellaneous Expenses	1,500	Contingency for small, unforeseen needs such as USB drives, cables, or data backup devices.
Total Estimated Cost	3,000 ETB	

Table 1: Cost Break Down Table

Resource Utilization:

- **Human Resources:** The student team will complete all project tasks, including analysis, design, coding, and testing, as part of academic coursework. No external personnel or financial compensation is required.
- **Material Resources**: The project will be developed and hosted on the developer's personal laptop, eliminating the need for new hardware or cloud hosting services for this prototype phase. University laboratories and facilities may be used for collaborative testing and evaluation.
- **Software Resources:** All technologies are open-source and freely available. No licensing fees or subscriptions are necessary. This includes:
    - Core Stack: React.js, Node.js, PostgreSQL, and Git.
    - Workflow Engine: React Flow.
    - OCR Engine: Tesseract.js.
    - Security Libraries: JWT, Node.js Crypto, and Nodemailer (for 2FA/OTP).

This lean cost reflects the project's academic purpose and reliance on existing personal and institutional resources, ensuring cost-effectiveness while maintaining full technical feasibility for local demonstration and evaluation.

1.13 Task and Schedule

The development of **E-flow** follows an **Agile Scrum methodology** spanning six months. The project is divided into monthly sprints, prioritizing the delivery of a Minimum Viable Product (MVP) by the end of Month 2, followed by iterative feature additions and refinements based on continuous feedback.

Sprint 1: Inception & Foundation (Month 1)

- **Focus:** Core Requirements & Architecture Setup.
- **Tasks:**
  - Conduct initial requirement gathering with administrative staff to map core workflows.
  - Set up the development environment (Node.js/Express, PostgreSQL).
  - Design the initial database schema (ERD) and basic UI wireframes.
  - Implement User Authentication and basic Role-Based Access Control (RBAC).
- **Deliverable:** Validated Requirements & System Skeleton.

Sprint 2: MVP Development & Release (Month 2)

- **Focus:** Core Document Management (The MVP).
- **Tasks:**
  - Develop the basic Document Management System (DMS) allowing file upload and retrieval.
  - Implement basic workflow tracking (status updates).
  - Conduct unit testing on core functions.
- **Deliverable: Minimum Viable Product (MVP)** – A functional system allowing users to log in, upload documents, and track basic status.

Sprint 3: Intelligent Features Integration (Month 3)

- **Focus:** OCR & Workflow Builder.
- **Tasks:**

    - Integrate Tesseract.js for the OCR pipeline to automate data extraction.
    - Develop the visual Workflow Builder using React Flow.
    - Refine the MVP UI based on initial user feedback.
- **Deliverable:** Beta Version 1.0 (with OCR and Visual Workflow capabilities).

Sprint 4: Security & Compliance Hardening (Month 4)

- **Focus:** Security (ISO 27001) & Advanced Logic.
- **Tasks:**
    - Implement the 2FA/OTP verification module for sensitive actions.
    - Audit system against ISO 15489 (Records Management) and ISO 27001 (Security) standards.
    - Optimize database queries for performance.
- **Deliverable:** Beta Version 2.0 (Secure & Compliant).

Sprint 5: System Integration & Testing (Month 5)

- **Focus:** Quality Assurance & User Acceptance.
- **Tasks:**
    - Conduct full Integration Testing and Security Penetration Testing.
    - Perform User Acceptance Testing (UAT) with administrative staff.
    - Fix bugs and finalize UI/UX polish.
- **Deliverable:** Production-Ready Candidate.

Sprint 6: Deployment & Closure (Month 6)

- **Focus:** Final Presentation & Handoff.
- **Tasks:**
    - Deploy the system locally for final demonstration (within 3,000 ETB budget).
    - Finalize user manuals, technical documentation, and the final project report.
    - Final presentation and project evaluation.
- **Deliverable:** Final Submission & Live Demonstration.

1.14 Team Composition

No	Student Name	ID. No.	Role
1	Maraki Teferi	UGR/25729/14	Backend Developer
2	Leul Negesse	UGR/25673/14	Frontend Developer
3	Beamlak Tesema	UGR/25545/14	System Analyst & Document Management Lead
4	Kidus Asnake	UGR/25665/14	Database Administrator

5	Selamu Feleke:	UGR/25478/14	UI/UX Designer and Tester

Table 2: Team composition

Chapter 2: Description of Existing System and Literature Review

2.1. Major Function of the Existing System

Currently, manual or semi-digital document processing technologies are used by the majority of universities. These systems incorporate informal communication channels, simple computing tools, and tangible documentation. The following are the main purposes of the current system:

Submission and Gathering of Documents: Staff members and students can email or physically submit scanned PDFs or paper-based forms. These records are manually entered and sent to the appropriate offices.

Approval and Routing: Documents are physically delivered or forwarded via email between offices (e.g., Department → Registrar → Dean → President). Usually, handwritten signatures or casual email confirmations are used for approval.

Document Filing and Storage: Documents are stored in physical file cabinets or scattered digital folders. Some documents are scanned, but many remain as static image-based PDFs with no searchable text.

Monitoring and Tracking: Because there is no centralized monitoring mechanism, students and staff must regularly follow upin person, over the phone, or via email to track document status.

2.2 Users of the Current System

Students: Among the main users are students. They often submit academic and administrative requests such as clearance forms, grade reports, and approval letters. They rely on the system to

provide timely responses, transparency, and a reduced need for physical follow-ups, which are currently common due to the manual nature of the system.

Academic staff: Academic staff also depend heavily on the system. They review, approve, or provide comments on documents related to students. Their work needs fast access to documents, and delays or misplacements can reduce their effectiveness, so smooth processes are important.

Administrative staff: Administrative staff handle the bulk of document intake, filing, forwarding, and communication between departments. They rely on organized workflows to handle large numbers of documents and require systems that reduce their manual workload while maintaining accuracy and efficiency.

Department Heads, Deans, and other higher officials: Higher officials serve as the key decision-makers in the approval hierarchy. They are responsible for authorizing various documents and therefore require secure, verifiable, and reliable approval mechanisms. They also expect clear oversight of workflow bottlenecks to maintain accountability across administrative processes.

University management and ICT offices: University management and ICT offices are responsible for the broader performance of the institutional system. They oversee security, data integrity, and operational functionality. They also rely on accurate reporting and analytics to support decision-making and long-term institutional improvement.

2.3 Drawbacks of the Current System

The current document processing system used by universities has many limitations. It lacks efficiency, transparency, and the ability to meet the speeding demands of modern institutions. These issues cause delays, increase workloads, and hinder students and staff from completing routine processes smoothly. Our analysis highlights the main drawbacks of the existing system:

- The lack of real-time tracking and centralized monitoring makes it impossible to know where a document is or who currently has it.
- Moving documents physically slows down processing times, especially when the location of the offices are far apart and when offices are busy or staff members are not available.

- Documents can be easily misplaced, damaged, or lost because of a reliance on paper filing systems without a proper digital backup.
- Scanned documents are often stored as non-searchable images, which slows down retrieval and forces staff to re-enter information manually.
- The absence of standardized workflows across departments leads to inconsistencies, delays, and communication gaps during document processing.
- Approval processes depend on handwritten signatures or informal emails, providing no secure verification or complete record of actions taken.
- Sensitive information is shared through physical delivery or unprotected emails, putting documents at risk of unauthorized access and security threats.

2.4 Literature Review

2.4.1 Introduction

The global landscape of higher education is undergoing a seismic shift, characterized by a transition from traditional, analog administrative models to dynamic, digital-first ecosystems. This literature review provides an exhaustive analysis of the research, theoretical frameworks, and technological paradigms underpinning the proposed "E-flow: Intelligent Document Tracking and Workflow Management System for Universities." As Higher Education Institutions (HEIs) grapple with increasing enrollment numbers, regulatory complexities, and the demand for operational transparency, the reliance on manual, paper-based document processing has emerged as a critical vulnerability. The persistence of physical archives and disjointed workflows not only impedes efficiency but also obscures accountability and exposes institutions to significant security risks.

This review synthesizes knowledge across four primary domains relevant to the E-flow project: (1) Electronic Document Management Systems (EDMS) and the challenge of hybrid record-keeping.

(2) Business Process Management (BPM) with a focus on visual, low-code workflow automation and End-User Development (EUD).

(3) Optical Character Recognition (OCR) technologies for bridging the physical-digital divide.

(4) Information Security frameworks, specifically centering on Role-Based Access Control (RBAC) and Two-Factor Authentication (2FA) within the context of Ethiopian legal standards. By critically examining these areas, this review establishes the academic and practical necessity

for a unified, intelligent system that empowers administrative staff while ensuring data integrity and legal compliance.

Furthermore, this analysis situates the technological discussion within the specific socio-technical context of developing nations, with a particular focus on Ethiopia. It moves beyond a generic discussion of technology to address the unique constraints of infrastructure, digital literacy, and the specific legal requirements of the Ethiopian Electronic Signature Proclamation No. 1072/2018. The review demonstrates that while disparate tools for storage, OCR, and workflow management exist, there is a distinct gap in the availability of integrated, cost-effective, and user-centric platforms designed specifically for the resource-constrained and legally specific environments of African universities.

2.4.2 Electronic Document Management Systems (EDMS) in Higher Education

2.4.2.1 The Evolution of Academic Record Keeping

Historically, universities have acted as stewards of vast quantities of physical information. From the moment a student applies for admission to decades after their graduation, the institution generates a continuous stream of records: application forms, transcripts, disciplinary records, financial aid applications, and faculty tenure reviews. For much of the 20th century, the management of these records was synonymous with physical custody—warehousing paper in filing cabinets and basements. This model, while tangible, suffers from inherent inefficiencies: physical records can only be in one place at a time, they are vulnerable to environmental degradation (fire, flood, pests), and their retrieval is labor-intensive and slow.

The advent of Electronic Document Management Systems (EDMS) in the late 1990s promised a revolution. Early EDMS implementations focused primarily on digitization for preservation—scanning documents to create static digital archives. However, the literature indicates that modern EDMS has evolved from simple "scan-and-store" repositories into sophisticated platforms that manage the entire lifecycle of information. Contemporary EDMS solutions integrate metadata indexing, version control, and access auditing, transforming documents from static artifacts into active data points that drive institutional decision-making.

Despite this technological maturation, the transition in the higher education sector has been uneven. While universities in the Global North have largely migrated to fully digital Student Information Systems (SIS), institutions in the Global South often operate in a "hybrid" reality. Research highlights that in many African universities, digital systems are often superimposed onto existing paper workflows rather than replacing them, leading to a duplication of effort where staff must manage both the physical file and its digital shadow. This phenomenon creates "information silos" where critical data is trapped in unlinked systems or, worse, exists only on paper in a locked

office, inaccessible to the broader administrative ecosystem.

2.4.2.2 The Persistence and Perils of Hybrid Records Management

A recurring theme in the library and information science literature is the resilience of paper in administrative cultures. Even with the availability of digital tools, many institutions maintain "hybrid records management systems" where paper and electronic records coexist. This hybridity is often driven by legal uncertainties regarding digital signatures, a lack of trust in digital storage reliability, or simply organizational inertia.

However, the literature warns that hybrid systems are fraught with operational risks. The "duality of existence" for a single record—living as both a physical paper and a digital file—creates synchronization challenges. If a grade change is noted on the physical file but not updated in the digital system (or vice versa), the integrity of the student's academic history is compromised. Furthermore, hybrid systems are notoriously inefficient regarding retrieval times. Staff must search across two distinct domains (physical and digital) to locate information, leading to what researchers describe as "search friction," which significantly hampers administrative responsiveness.

The E-flow project addresses this specific vulnerability by proposing an active digitization strategy. By integrating OCR at the point of ingestion, the system aims to minimize the lifespan of the "paper" version of a document. Rather than letting paper accumulate in a hybrid limbo, the proposed workflow forces the conversion of physical artifacts into searchable digital assets immediately upon entry into the system. This aligns with best practices identified in archival research, which suggests that the only way to escape the inefficiencies of hybridity is to treat the digital record as the "master" and the paper as merely a backup or temporary input source.

2.4.2.3 Operational Efficiency and Cost-Benefit Analysis

The economic and operational arguments for EDMS adoption in universities are robustly supported by empirical studies. A primary driver is cost reduction. The direct costs associated with paper—purchasing paper, toner, printers, and filing cabinets—are substantial. However, the *indirect* costs are often higher. Research indicates that knowledge workers can spend up to 40-50% of their time looking for information rather than acting on it. By indexing documents and making them searchable via keywords (a core feature of E-flow enabled by OCR), universities can reclaim thousands of man-hours annually.

Metric	Traditional Paper System	Hybrid System (Paper + Digital)	Integrated EDMS (E-flow Model)
Searchability	Physical search required; time-consuming (hours/days).	Partial searchability; disjointed across silos.	Instant keyword search via OCR metadata (seconds).
Storage Cost	High (physical space, cabinets).	Medium (physical + server costs).	Low (cloud/server only); physical space reclaimed.
Access	Single-user access (one file, one person).	Limited multi-user access.	Concurrent multi-user access with RBAC.
Security	Low (physical theft, loss, damage).	Medium (digital security + physical vulnerability).	High (Encryption, 2FA, Audit Trails).
Traceability	Low (manual logbooks).	Medium (some digital logs).	High (Automated, immutable audit trails).

Table 3: Operational Efficiency and cost benefit Analysis

Beyond efficiency, the literature emphasizes "Disaster Recovery" as a critical benefit. Physical archives are uniquely vulnerable to catastrophic loss from fire, flood, or civil unrest—risks that are particularly pertinent in some developing regions. Digital systems, when properly backed up and encrypted, offer a resilience that paper cannot match. The E-flow system's reliance on standard database technologies (PostgreSQL) facilitates automated backup strategies, ensuring institutional memory is preserved against physical calamities.

2.4.3 Workflow Management Systems (WfMS) and Process Automation

2.4.3.1 Business Process Management (BPM) in Academia

While EDMS addresses the *storage* of information, Workflow Management Systems (WfMS) address the *process*. The theoretical underpinning of WfMS is Business Process Management (BPM), a discipline that views an organization not as a hierarchy of people, but as a collection of processes that can be modeled, analyzed, and optimized. In the context of a university, a "workflow" is any sequence of tasks required to achieve an administrative goal—approving a curriculum change, processing a leave request, or finalizing a procurement order.

Traditional university administration is often characterized by ad-hoc, informal workflows. A document might be walked from one office to another, or emailed to a personal inbox, with no central visibility into its status. This lack of structure leads to "process opacity," where stakeholders cannot determine where a request is stuck or who is responsible for the delay. WfMS introduces formal logic to these interactions, enforcing rules that dictate exactly who must act on a document and in what order. Research confirms that implementing WfMS in educational settings leads to measurable reductions in "lead time" (the total time from initiation to completion) and "wait time" (the time a task sits idle).

2.4.3.2 The Rigidity Trap: Hard Coded vs. Flexible Workflows

A critical insight from the literature is the failure of rigid, "hard-coded" software in dynamic university environments. Many commercial ERP systems come with pre-defined workflows that are embedded in the application code. To change the process for example, adding a new vice-dean to an approval chain requires software developers to rewrite code, recompile the application, and redeploy it. This high technical barrier leads to "process stagnation," where the software dictates how the university operates, rather than the university adapting its tools to its needs.

In contrast, the "Flexible Workflow" paradigm, which E-flow adopts, decouples the process logic from the application code. By storing workflow definitions as data (e.g., JSON objects in a database) rather than code, the system allows for runtime modifications. This means that if a university policy changes, an administrator can update the workflow model immediately without needing IT intervention. The literature cites this flexibility as a key determinant of long-term system success and user adoption.

2.4.3.3 Democratizing Development: Low-Code and Visual Modeling

The E-flow project’s inclusion of a **Visual Workflow Builder** aligns with the burgeoning

movement of **End-User Development (EUD)** and **Low-Code/No-Code (LCNC)** platforms. EUD aims to empower non-programmers (domain experts like Registrars or Deans) to act as "citizen developers," creating or modifying software behaviors to suit their specific needs.

Research into cognitive dimensions of notations suggests that visual programming languages (VPLs)—which use nodes and edges to represent logic—are significantly more accessible to non-technical users than text-based scripts. The use of **React Flow**, a library specifically designed for node-based diagrams, allows E-flow to present complex routing logic (parallel branches, conditional approvals) in a format that mirrors the flowcharts administrators are already familiar with.

Benefits of the Visual/Low-Code Approach:

- **Agility:** Processes can be adapted in minutes rather than months. If a department head goes on leave, the workflow can be visually rerouted to a delegate instantly.
- **Contextual Fidelity:** Administrators, who understand the nuances of university policy, build the workflows themselves, eliminating the "translation error" that often occurs when explaining requirements to IT developers.
- **Reduced IT Backlog:** By offloading the task of workflow creation to the end-users, IT departments are freed to focus on infrastructure and security rather than routine process maintenance.

The literature validates that visual workflow modeling is not just a UI feature but a strategic enabler of institutional agility, particularly in resource-constrained environments where developer time is scarce.

2.4.4 Optical Character Recognition (OCR): The Bridge Technology

2.4.4.1 Digitizing the Legacy Archive

For universities in developing nations, the "paperless office" remains an aspiration rather than a reality. A vast amount of historical data student records from the pre-digital era, government correspondence, and legal contracts exists solely on paper. **Optical Character Recognition (OCR)** is the technological bridge that allows this "dark data" to be illuminated and integrated into the digital ecosystem.[18]

OCR is not merely about creating a picture of a document (which a simple scanner does); it is about extracting meaning. By converting pixel data into machine-encoded text, OCR allows documents to be indexed, searched, and analyzed. This transformation is critical for "accessibility" and "preservation," ensuring that legacy records are not just stored but are retrievable and usable within the same workflow as digital-born documents.

2.4.4.2 Tesseract.js: Capabilities and Implementation Strategies

The E-flow project leverages **Tesseract.js**, a WebAssembly port of the renowned Tesseract OCR engine. Tesseract is widely cited in computer vision literature as the gold standard for open-source OCR, offering support for over 100 languages and respectable accuracy rates.The choice of Tesseract.js is particularly strategic for a web-based system, as it allows for client-side processing (in the browser), reducing the computational load on the server and enhancing privacy by keeping data on the user's machine during the initial processing phase.

However, the literature also highlights significant challenges in applying OCR to real-world administrative documents, often referred to as "dirty data."

- **Accuracy vs. Image Quality:** Tesseract's accuracy is highly sensitive to image resolution (DPI) and noise. Scans of old, yellowed, or stained paper can result in high error rates.
- **Handwriting Recognition:** While Tesseract excels at printed text, it struggles with handwritten notes or signatures, which are common on older administrative forms. Research indicates that preprocessing techniques—such as binarization (converting to black and white), deskewing (straightening), and noise reduction—are essential to achieving acceptable recognition rates.

Operational Implication: The literature suggests that OCR cannot be a fully automated "black box" for archival data. It requires a "human-in-the-loop" workflow where the user verifies and corrects the extracted text before it is committed to the database. E-flow's design must accommodate this verification step to ensure data integrity.

2.4.5 Information Security Architecture: Trust and Compliance

2.4.5.1 The Security Imperative in Academic Systems

HEIs are repositories of high-value data, including Personally Identifiable Information (PII), financial records, and intellectual property. Consequently, they are prime targets for cybercrime. The literature emphasizes that security in an academic EDMS cannot be an afterthought; it must be architected into the core of the system following the principles of **Confidentiality, Integrity, and Availability (CIA)**.

Role-Based Access Control (RBAC):
RBAC is the industry standard for managing authorization in complex organizations. It decouples access rights from individual users, assigning them instead to "roles" (e.g., Dean, Registrar, Student). This approach is highly scalable; when a staff member changes positions, their access

rights are automatically updated based on their new role, reducing the risk of "permission creep" where users accumulate unnecessary access over time. For E-flow, RBAC is critical for enforcing privacy—ensuring, for instance, that a student worker in the admissions office cannot access faculty disciplinary records.

2.4.5.2. Two-Factor Authentication (2FA) and Non-repudiation

A central feature of the E-flow proposal is the use of **Two-Factor Authentication (2FA)** via One-Time Passwords (OTP) for document approval. This mechanism addresses the critical security concept of **non-repudiation** the ability to prove that a specific user performed a specific action and cannot later deny it.

Standard password authentication is increasingly viewed as insufficient due to the prevalence of phishing attacks. By requiring a second factor—something the user *has* (a phone or email account receiving the OTP) in addition to something they *know* (password)—the system significantly raises the bar for attackers.

- **The Trust Layer:** In a digital workflow where face-to-face verification is absent, the 2FA signature acts as the digital equivalent of a notary seal. Research indicates that while 2FA adds a layer of friction to the user experience, in high-stakes contexts (like grade approval or degree signing), users perceive this friction as a necessary indicator of security and institutional integrity.

2.4.5.3 Legal Validity in the Ethiopian Context

The viability of E-flow in its target environment (Ethiopia) depends on its alignment with local legal frameworks. The **Electronic Signature Proclamation No. 1072/2018** is the governing legislation. This proclamation explicitly grants legal recognition to electronic signatures, stating that they cannot be denied admissibility solely on the grounds of their digital nature.

Crucially, the law defines a "reliable" electronic signature as one where the means of signature creation is "linked to the signatory and not any other person" and is "under the control of the signatory" at the time of signing. The E-flow system’s use of OTPs sent to a personal device directly satisfies this "control" requirement. By tying the digital signature to a temporal code accessible only via the user's personal device, the system creates a legally defensible audit trail that aligns with the specific requirements of Ethiopian law. This legal alignment is a critical differentiator for E-flow compared to generic open-source tools that may not support such specific compliance workflows.

2.4.6 Contextual Analysis: Digital Transformation in Ethiopian Higher Education

2.4.6.1 Infrastructure Constraints and Resilience

Implementing advanced digital systems in the developing world requires a sober assessment of infrastructure realities. The literature on digital transformation in Ethiopia highlights significant barriers, including intermittent power supply, unstable internet connectivity, and a scarcity of high-end server hardware.

- **Architectural Implications:** These constraints dictate that E-flow must be lightweight and efficient. A web-based architecture (React/Node.js) is advantageous as it centralizes maintenance on the server while allowing access via low-powered client devices (standard laptops or smartphones(By using responsive website)).
- **The Connectivity Challenge:** While the current scope of E-flow relies on connectivity, the literature suggests that future iterations of such systems in Africa often need "offline-first" capabilities—allowing users to queue actions during outages that synchronize when connectivity is restored.

2.4.6.2 The Human Factor: Digital Literacy and Adoption

Technology adoption is as much a social process as a technical one. The **Technology Acceptance Model (TAM)** offers a theoretical lens for predicting user behavior. TAM posits that **Perceived Usefulness (PU)** and **Perceived Ease of Use (PEOU)** are the primary drivers of adoption.

- **Perceived Usefulness:** For E-flow to succeed, administrative staff must perceive it as a solution to their pain points (lost files, slow approvals) rather than an additional burden. The "real-time tracking" feature is a strong driver of PU, offering visibility that paper lacks.
- **Perceived Ease of Use:** The "Visual Workflow Builder" is critical here. By presenting workflows as intuitive diagrams rather than code, the system lowers the cognitive load for non-technical staff, directly enhancing PEOU.
- **Cultural Resistance:** Literature on Ethiopian public sector digitization often cites "resistance to change" as a major hurdle. Staff accustomed to the authority and tangibility of paper may view digital systems with suspicion. Successful implementation requires not just software, but "change management"—training, support, and demonstrating value to the end-users

2.4.7 Gap Analysis: The Need for E-flow

The review of existing solutions reveals a distinct gap in the market, particularly for African universities:

1. **High Cost of Commercial Solutions:** Enterprise tools like SAP or Oracle are prohibitively expensive for many institutions in developing economies
2. **Complexity of Open Source:** While open-source tools exist, they often require significant developer resources to customize and maintain resources that are scarce in many universities.
3. **Fragmentation:** Most tools are either storage systems (DMS) or process engines (WfMS). Few integrate both with OCR in a single, cohesive platform
4. **Lack of Contextual Fit:** Generic tools rarely account for the specific "hybrid" reality or legal compliance needs (like Proclamation 1072/2018) of the Ethiopian context.

E-flow is strategically positioned to fill this void. By offering a unified, low-cost, open-source solution that prioritizes **usability (Visual Builder)**, **legacy integration (OCR)**, and **security (2FA)**, it addresses the specific triad of challenges—efficiency, trust, and context—that define the current landscape of higher education administration in developing nations.

2.4.8 Theoretical Framework: The Integrated Process Automation Model

Based on the synthesized literature, this project is grounded in an **Integrated Process Automation Model**. This framework suggests that successful digital transformation in administration requires the convergence of three pillars:

1. **Digitization (OCR):** The conversion of physical atoms to digital bits to ensure accessibility.
2. **Orchestration (Workflow):** The intelligent routing of those bits through business logic to ensure efficiency.
3. **Verification (Security):** The cryptographic assurance of identity and integrity to ensure trust.

The literature indicates that removing any one of these pillars compromises the system: digitization without workflow leads to "data dumping"; workflow without digitization leaves paper behind; and both without verification create a system that no one trusts. E-flow's comprehensive approach aims to unify these elements into a resilient, legally compliant, and operationally efficient platform.

2.4.9 Conclusion

The review of the literature confirms that the "E-flow" project is not merely a software development exercise but a strategic intervention aligned with global trends in BPM, EDMS, and Information Security. By addressing the specific friction points of hybrid record-keeping and the

need for user-centric agility via visual modeling, the project offers a viable path toward the modernization of higher education administration. The integration of robust security measures like 2FA ensures that this efficiency does not come at the cost of integrity, making E-flow a potentially transformative tool for the Ethiopian higher education sector.

CHAPTER 3: Proposed System

3.1 Overview

The proposed system, E-flow: Intelligent Document Tracking and Workflow Management System for Universities, is a web-based platform designed to digitalize, automate, and streamline document-related processes within higher education institutions. The system acts as an intelligent layer that integrates with existing administrative workflows and transforms traditional paper-based or semi-digital processes into a unified digital ecosystem.

E-flow provides a centralized platform where students, academic staff, administrative personnel, and institutional leaders can submit, process, track, verify, and approve documents securely and efficiently. A key innovation of the system is the Visual Workflow Builder, a drag-and-drop interface that allows administrators to design and modify approval chains dynamically without writing code. This flexibility enables departments to adapt workflows to their unique processes while maintaining consistency and transparency across the institution.

Additionally, the system integrates Optical Character Recognition (OCR) to convert scanned or image-based documents into searchable digital text. This feature bridges the gap between paper and digital records, reducing manual data entry, enabling full-text search, and ensuring that legacy paper documents can participate in automated workflows.

To ensure institutional integrity and accountability, E-flow implements a secure Two-Factor Authentication (2FA/OTP) signature verification mechanism, ensuring that every digital approval is authentic, traceable.

Overall, the system provides a scalable, secure, and user-friendly solution that enhances efficiency, reduces delays, strengthens auditability, and modernizes administrative operations across universities.

3.2 Functional Requirements

The functional requirements describe the essential actions, behaviors, and capabilities that the E-flow system must support to achieve its intended purpose.

1. User Authentication and Role Management

- The system must allow users to log in securely using unique credentials.
- The system shall implement Role-Based Access Control (RBAC) to restrict access to features and documents based on user roles (e.g., Student, Staff).
- The system must allow administrators to manage user accounts and assign roles dynamically.

2. Visual Workflow Management

- The system must provide a Visual Workflow Builder utilizing a drag-and-drop interface (React Flow).
- Administrators must be able to create, edit, and save approval chains by connecting "nodes" (approvers) and "paths" (sequences) without writing code.
- The system shall allow workflows to be customized for different document types (e.g., Clearance Forms, Grade Reports).

3. Document Ingestion and OCR Processing

- The system must allow users to upload digital files (PDF, DOCX,JPG) directly.
- The system must include an OCR Module (Tesseract.js) to scan and process images of paper documents.
- The system shall extract text from scanned images to make the content searchable and editable within the database.
- The system must allow users to verify and correct OCR-extracted text before final submission (Human-in-the-loop).

4. Document Routing and Approval

- The system must automatically route submitted documents to the next approver defined in the workflow sequence.
- Approvers must be able to perform actions such as Approve, Reject (with comments), or Request Revision.
- The system must support Version Control, maintaining a history of document revisions.

5. Secure Signature Verification (2FA)

- The system must enforce a Two-Factor Authentication (2FA) challenge when a user attempts to approve a sensitive document.
- The system shall generate and send a One-Time Password (OTP) to the approver’s registered email.
- The system must only register the approval as valid once the correct OTP is entered, ensuring ver.

6. Tracking, Notifications, and Auditing

- The system must provide a Real-Time Tracking Dashboard where students and staff can view the current status (e.g., "Pending at Dean's Office") of their requests.
- The system shall send automated Notifications (email or in-app) to users when a document requires their attention.
- The system must maintain a comprehensive Audit Trail, logging every action (submission, approval, rejection) with a timestamp and user ID.

3.3. Non-Functional Requirements

This section specifies the quality attributes, performance standards, and security constraints the system must adhere to.

1. Security and Privacy

- Data Encryption: All sensitive data must be encrypted at rest (using AES-256 standards) and in transit (using SSL/TLS protocols).
- Compliance: The system's security framework shall be guided by ISO 27001 principles for information security management.
- Identity Verification: The system must ensure "very secure" approval processes through mandatory 2FA/OTP for signatures, satisfying the requirement for non-repudiation.

2. Performance and Responsiveness

- The system must be optimized to support concurrent users (students and staff) without significant latency.
- The OCR processing engine must process scanned documents efficiently, balancing recognition accuracy with server processing time.

3. Usability and Accessibility

- The user interface must be intuitive and responsive, accessible via standard web browsers on both desktops and laptops.
- The Visual Workflow Builder must be designed to be usable by non-technical administrators with minimal training.
- The system must provide clear feedback to users (e.g., success messages, error alerts) during interactions.

4. Reliability and Availability

- The system must ensure high availability during university working hours.
- The system must maintain data integrity, ensuring that no document metadata is lost during routing or approval processes.
- The system relies on the email server reliability for the delivery of OTPs.

5. Maintainability and Scalability

- The system shall use a modular architecture, allowing future developers to update individual components (like the OCR engine or 2FA module) without affecting the entire system.

- The system must be scalable to accommodate future integrations with other institutional systems (e.g., Student Finance or Grading databases).
- The codebase must follow standard documentation practices to facilitate easy handover and maintenance.

6. Records Management Standards

- The system's document lifecycle management (creation, maintenance, and use) shall be guided by ISO 15489 (Records Management) principles.

3.4. System model

3.4.1. Scenario

Scenario 1: Authenticate User

Actor: Department Head

Precondition:

– User is registered.

– User has valid login credentials.

Normal Flow:

1. The Department Head opens the E-Flow web portal. He enters his university email and password.
2. He clicks **"Login."**
3. The system validates the credentials using secure authentication.
4. The system verifies the user's role (Department Head).
5. The system creates a secure session and redirects the user to the Department Head Dashboard displaying pending approvals.

User Experience: Fast, secure login with direct access to relevant tasks.

Extensions (What Can Go Wrong):

1. **Incorrect password (three attempts):** System temporarily suspends login attempts.

2. **Inactive/disabled account:** System notifies user to contact the Administrator.

Postcondition:

– Successful: User is authenticated and active session is established.

– Failure: User remains on login page with an appropriate error message.

Scenario 2: Create Workflow

Actor: System Administrator

Precondition: Admin is authenticated.

Normal Flow:

1. Admin opens the **Workflow Builder**.
2. Admin drags workflow nodes (Library, Proctor, Department, Registrar) onto the canvas.
3. Admin connects nodes to define the approval sequence.
4. Admin assigns the name **"Standard Clearance."**
5. Admin saves the workflow.
6. System validates workflow logic (no loops, no disconnected nodes).
7. Workflow becomes available for document routing.

User Experience: Simple drag-and-drop builder that requires no technical skills.

Extensions:

1. **Circular or invalid logic detected:** System prevents saving and highlights the error.
2. **Workflow name missing:** System prompts user to enter a name.

Postcondition:

– A valid approval workflow is stored and ready for use.

Scenario 3: Submit Document Request (with OCR)

Actor: Student

Precondition:

– Student is authenticated.

Normal Flow:

1. Student selects **"Submit Request."**
2. Student chooses **"Medical Justification."**
3. Student uploads a scanned copy of the medical certificate.
4. Student clicks **"Extract Text."**
5. System uses an OCR engine to extract text from the image.
6. Student reviews extracted text, edits mistakes (e.g., date), and submits the request.
7. System stores the document and assigns it to the configured workflow.

User Experience: Smooth conversion of a physical document into a digital, searchable record.

Extensions:

1. **Blurry or unreadable image:** System requests re-upload with clearer quality.
2. **File size too large:** System displays size-limit warning.

Postcondition:

– Document is submitted and marked *Pending Review.*

– (Direct continuation to Scenario 4.)

Scenario 4: Approve Document (with 2FA)

Actor: Department Head

Precondition:

– Document submitted in Scenario 3.

– Department Head is authenticated.

Normal Flow:

1. Department Head receives notification of a new document to review.

2. He opens the request and views the scanned image and extracted text side-by-side.
3. He clicks **"Approve."**
4. System prompts for a One-Time Password (2FA).
5. Department Head enters the OTP received via email.
6. System validates the OTP, logs the approval in the audit trail, and forwards the document to the Registrar.

User Experience: Secure approval with clear visibility of document details.

Extensions:

1. **OTP expired:** System requests a new OTP.
2. **Incorrect OTP:** System prompts user to retry.

Postcondition:
– Document status updates to *Approved by Department Head.*
– Workflow progresses to the Registrar.

Scenario 5: Reject or Request Revision

Actor: Registrar
Precondition:
– Document approved in Scenario 4.
– Registrar is authenticated.

Normal Flow:

1. Registrar opens the forwarded document.
2. Registrar notices incorrect or invalid information (e.g., wrong date).
3. Registrar selects **"Reject / Request Revision."**
4. Registrar enters a mandatory comment explaining the issue.
5. System halts the workflow and sends the student a revision notification.

User Experience: Clear feedback with precise instructions for correction.

Extensions:

1. **Comment field empty:** System prevents submission until feedback is entered.

Postcondition:
– Document moves back to the student's *Draft* stage for correction.

Scenario 6: Track Document Status

Actor: Student
Precondition: Student has submitted a request.

Normal Flow:

1. Student navigates to **"My Requests."**
2. System displays a visual progress tracker (e.g., Dept Head ✓ → Registrar Pending).
3. Student views timestamps and workflow steps.

User Experience: High transparency; no need to visit offices physically.

Extensions:

1. **No submitted requests:** System displays: *"You have no active requests."*

Postcondition:
– Student understands exact workflow status.

Scenario 7: Notifications & System **Automated SLA Monitoring & Escalation**

Actor: System (Automated Background Process)

Preconditions:

1. Workflow steps have defined time limits (SLA).

2. Hierarchy is defined in the system (e.g., The system knows the "President" is the superior of the "Dean").

Normal Flow (The Logic):

1. **System** identifies a document in the "Pending" state that has exceeded its SLA time limit.
2. **System** calculates the duration of the delay.
3. **System executes Decision Logic:**
    - **IF delay is within Grace Period (e.g., < 24 hours):**
        - **Action:** Sends a **Warning Notification** to the **Current Assignee** (e.g., The Dean) reminding them to take action.
    - **IF delay exceeds Grace Period (e.g., > 24 hours):**
        - **Action:** Triggers an **Escalation Event**.
        - **Recipient:** Sends an **Escalation Alert** to the **Assignee's Supervisor** (e.g., The President) or System Admin, notifying them of the bottleneck.
4. **System** logs the specific type of notification (Warning vs. Escalation) in the audit history.

User Experience:

- **Current Actor:** Receives a helpful nudge before getting in trouble.
- **Management:** Gains visibility into bottlenecks without manually checking every queue.

Postcondition:

- Relevant parties are notified.
- Workflow status is updated to "At Risk" or "Escalated."

Scenario 8: View Audit Trail

Actor: Auditor (Read-Only Role)

Precondition: Auditor is authenticated and has authorization.

Normal Flow:

1. Auditor searches for a document using its ID.
2. Auditor selects **"View Audit Trail."**
3. System displays an immutable, time-stamped list of all actions (submission, approvals, rejections) with roles, timestamps, and action descriptions.
4. Auditor reviews the full chain of custody.

User Experience: Complete transparency and accountability.

Extensions:

1. **Unauthorized user attempts access:** System denies permission.

   Postcondition:

   – Audit trail is securely displayed.

   – non-repudiation is maintained.

3.4.2. Use case model

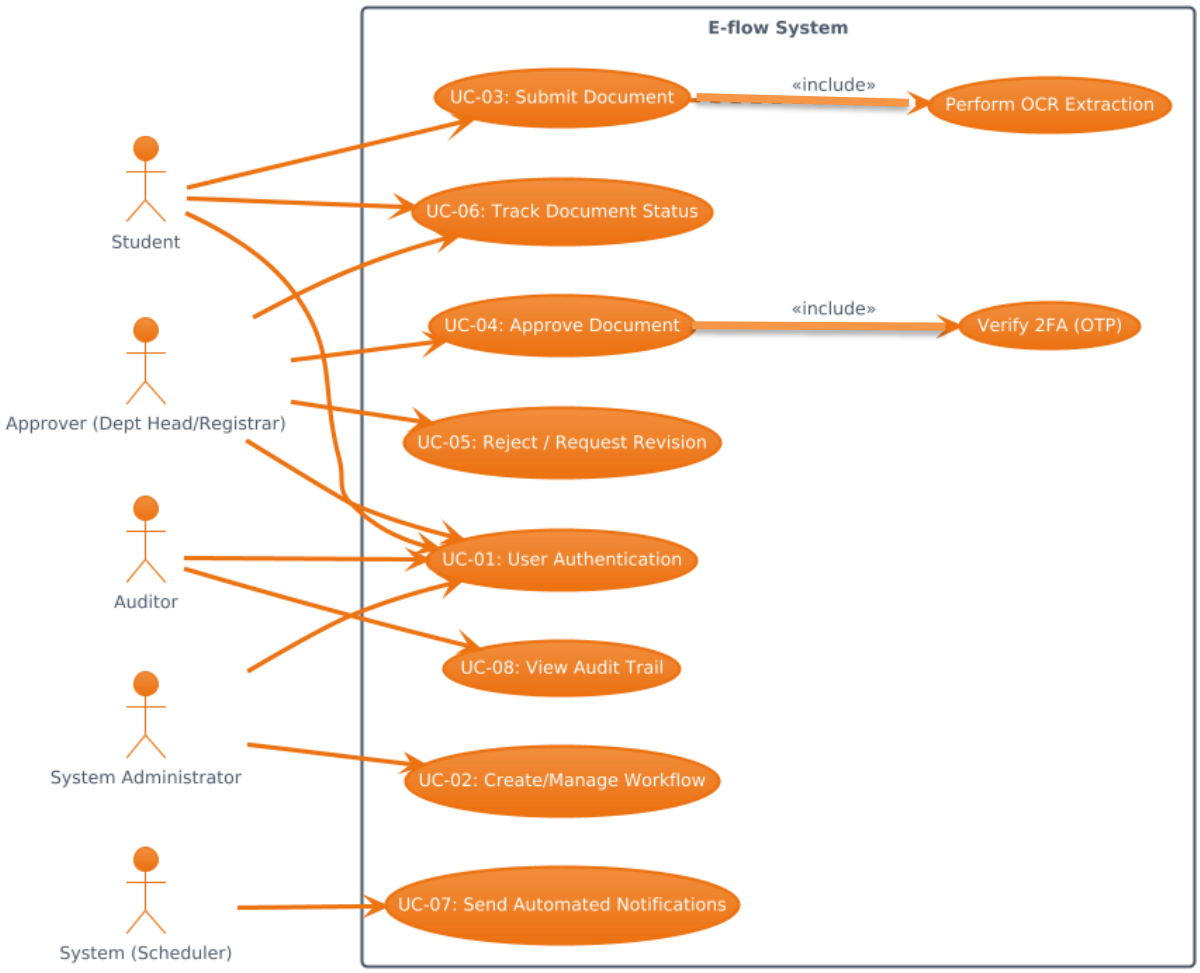

Figure 3.1: use case Model

3.4.2.1. Use case description

Field	Description
Use Case ID	UC-01
Use Case Name	User Authentication
Actors	Student, Approver, Administrator, Auditor
Description	Allows registered users to log in securely to access their role-specific dashboard.
Preconditions	User is registered in the system database.
Postconditions	User is authenticated, and a secure session is established.
Main Flow	1. User enters email and password. 2. System validates credentials. 3. System identifies user role. 4. System redirects user to the appropriate dashboard.

Alternative Flows	*Invalid Credentials:* System displays error and prompts retry. *Account Locked:* After 3 failed attempts, account is locked for 15 minutes.

Table 4: **UC-01 User Authentication**

Field	Description
Use Case ID	UC-02
Use Case Name	Create Workflow
Actors	System Administrator
Description	Allows the admin to visually design and save a new approval routing path.
Preconditions	Admin is logged in with management privileges.
Postconditions	A new workflow template is saved and active.

Main Flow	1. Admin opens Visual Workflow Builder. 2. Admin adds approver nodes and connects them. 3. Admin sets conditional logic for routing. 4. Admin saves the workflow configuration. 5. System validates logic and stores the template.
Alternative Flows	*Circular Logic Error:* System detects an infinite loop and prevents saving.

Table 5: **UC-01 User Authentication**

Field	**Description**
Use Case ID	UC-03
Use Case Name	Submit Document
Actors	Student
Description	Allows a student to upload a physical document scan, digitize it via OCR, and submit it for approval.
Preconditions	Student is logged in; Document scan is ready.

Postconditions	Document is stored, indexed, and routed to the first approver.
Main Flow	1. Student uploads document image. 2. System performs OCR extraction. 3. Student verifies and corrects extracted text. 4. Student submits the document. 5. System creates a new workflow instance.
Alternative Flows	*Upload Failed:* File format not supported or file too large. System alerts user.

Table 6: UC-03 Submit Document (with OCR)

Field	Description
Use Case ID	UC-04
Use Case Name	Approve Document
Actors	Approver (Dept Head, Registrar, etc.)

Description	Allows an approver to review a document and authorize it using Two-Factor Authentication.
Preconditions	Document is pending in the Approver's queue.
Postconditions	Document status updates to "Approved" and moves to the next node.
Main Flow	1. Approver reviews document content. 2. Approver clicks "Approve." 3. System sends OTP to approver's email. 4. Approver enters OTP. 5. System verifies OTP and signs the approval.
Alternative Flows	*Invalid OTP:* System denies approval and requests re-entry. *Expired OTP:* System generates a new token.

Table 7: **UC-04 Approve Document (with 2FA)**

Field	Description
Use Case ID	UC-05

Use Case Name	Reject or Request Revision
Actors	Approver
Description	Allows an approver to return a document to the student for corrections.
Preconditions	Document is pending in the Approver's queue.
Postconditions	Document status updates to "Returned"; Workflow resets to draft.
Main Flow	1. Approver identifies an issue with the document. 2. Approver selects "Reject/Revision." 3. Approver enters mandatory feedback comments. 4. System notifies the student and halts the workflow.
Alternative Flows	*Missing Comment:* System prevents rejection without feedback.

Table 8: **UC-05 Reject / Request Revision**

Field	**Description**

Use Case ID	**UC-06**
Use Case Name	**Track Document Status**
Actors	**Student, Approver**
Description	**Allows users to view the real-time progress and location of a document.**
Preconditions	**User has at least one active document.**
Postconditions	**User is informed of the current status.**
Main Flow	**1. User navigates to "My Documents."** **2. System retrieves workflow status from database.** **3. System displays a visual progress bar indicating current location.**
Alternative Flows	***No Data:*** **System displays "No active requests."**

Table 9: **UC-06 Track Document Status**

Field	Description
Use Case ID	UC-07
Use Case Name	Send Automated Notifications
Actors	System (Scheduler)
Description	Automatically reminds users of pending actions to prevent workflow bottlenecks.
Preconditions	A document has exceeded the defined wait time (SLA).
Postconditions	Reminder email is sent to the responsible party.
Main Flow	1. System checks for idle documents. 2. System identifies the current owner of the document. 3. System triggers an email reminder. 4. System logs the notification.
Alternative Flows	*Email Service Down:* System logs error and retries later.

Table 10: **UC-07 Send Automated Notifications**

Field	Description
Use Case ID	UC-08
Use Case Name	View Audit Trail
Actors	Auditor
Description	Allows inspection of the complete, immutable history of a document for compliance.
Preconditions	Auditor is authenticated with read-only access.
Postconditions	Complete history log is displayed.
Main Flow	1. Auditor searches for a document ID. 2. Auditor selects "View Logs." 3. System displays chronological list of all actions, timestamps, and actors.
Alternative Flows	*Unauthorized Access:* System denies access if user is not an Auditor.

Table 11: **UC-08 View Audit Trail**

3.5. Object Model

The Object Model identifies the key entities within the E-flow system and describes their responsibilities, attributes, and relationships. This model serves as the blueprint for the database schema and class structure during implementation.

Key Objects:

- User: Represents all system actors (Students, Staff, Administrators) who interact with the platform. This object handles authentication, including the specific Two-Factor Authentication (2FA) requirement for secure signatures.
- Document: The central entity of the system. It represents both digital-native files and digitized paper documents processed via the OCR module. It tracks the file's lifecycle from submission to final approval.
- Workflow: Represents the dynamic approval paths created by the Visual Workflow Builder. Unlike static systems, this object stores the logic for how a document moves between nodes (e.g., Dean $\rightarrow$ Registrar).
- ApprovalTransaction: Represents the security layer of the system. It records the specific action of a user signing a document, linking the action to a timestamp and an OTP verification status to ensure non-repudiation.
- Notification: Manages communication, ensuring users are alerted via email or in-app updates when a document requires their attention.

3.5.1. Data Dictionary

The following table defines the classes, attributes, and operations for the proposed E-flow system.

Class	Attributes	Operations	Descriptions
User	userId (String) fullName (String) email (String) passwordHash (String) role (Enum: Student, Staff, Admin) twoFASecret (String)	login() register() generateOTP() verifyOTP() updateProfile()	Represents an entity authorized to access the system. It includes methods for the 2FA/OTP verification required for secure document signing.
Document	documentId (String) title (String) filePath (String) ocrContent (Text) status (Enum: Pending, Approved, Rejected)	uploadDocument() scanOCR() extractText() updateStatus() download()	Represents the file being processed. The ocrContent attribute stores text as TSVector extracted by Tesseract.js to make the document searchable.

	uploaderId (String) currentStageId (String)		
Workflow	workflowId (String) name (String) structureDefinition (JSON) creatorId (String) department (String)	createWorkflow() modifyStructure() validatePath() assignToDocument()	Stores the visual layout of approval chains. The structureDefinition stores the node and edge configuration generated by the React Flow builder.

ApprovalLog	logId (String) documentId (String) approverId (String) timestamp (DateTime) action (String) otpVerified (Boolean) comments (String)	recordAction() verifySignature() getHistory()	Serves as the audit trail. It ensures traceability by logging exactly who approved a document and confirming that 2FA was used[11].
OCRProcessor	engineConfig (String) supportedLanguages (List)	initializeEngine() processImage() cleanupText()	A utility class responsible for interfacing with the OCR library to digitize paper documents and extract text for indexing[12].
Notification	notificationId (String) recipientId (String)	sendEmail() sendInApp()	Handles alerting users when a document arrives in their queue or when a request status changes[13].

	message (String) isRead (Boolean) triggerEvent (String)	markAsRead()	

Table 12: **Data Dictionary**

3.5.2. Class diagram

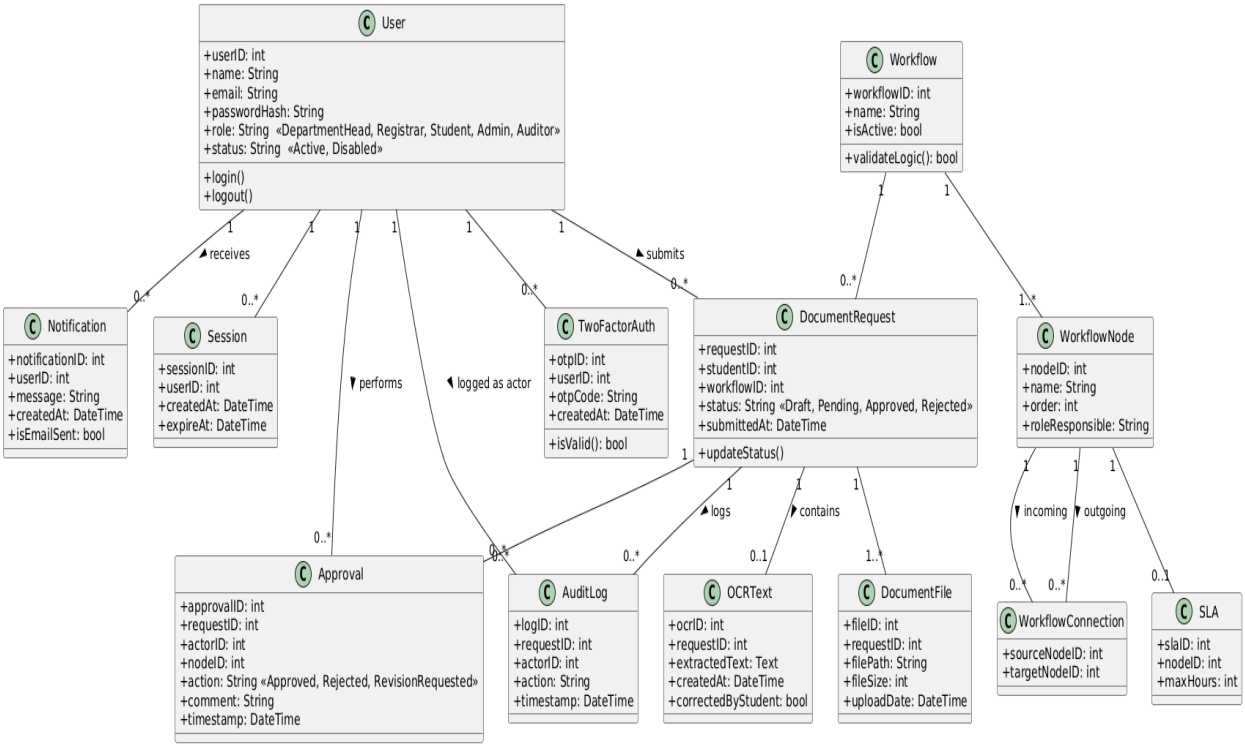

Figure 3.2: class diagram

3.5.3 Dynamic model

3.5.4 Sequence diagram

User authentication

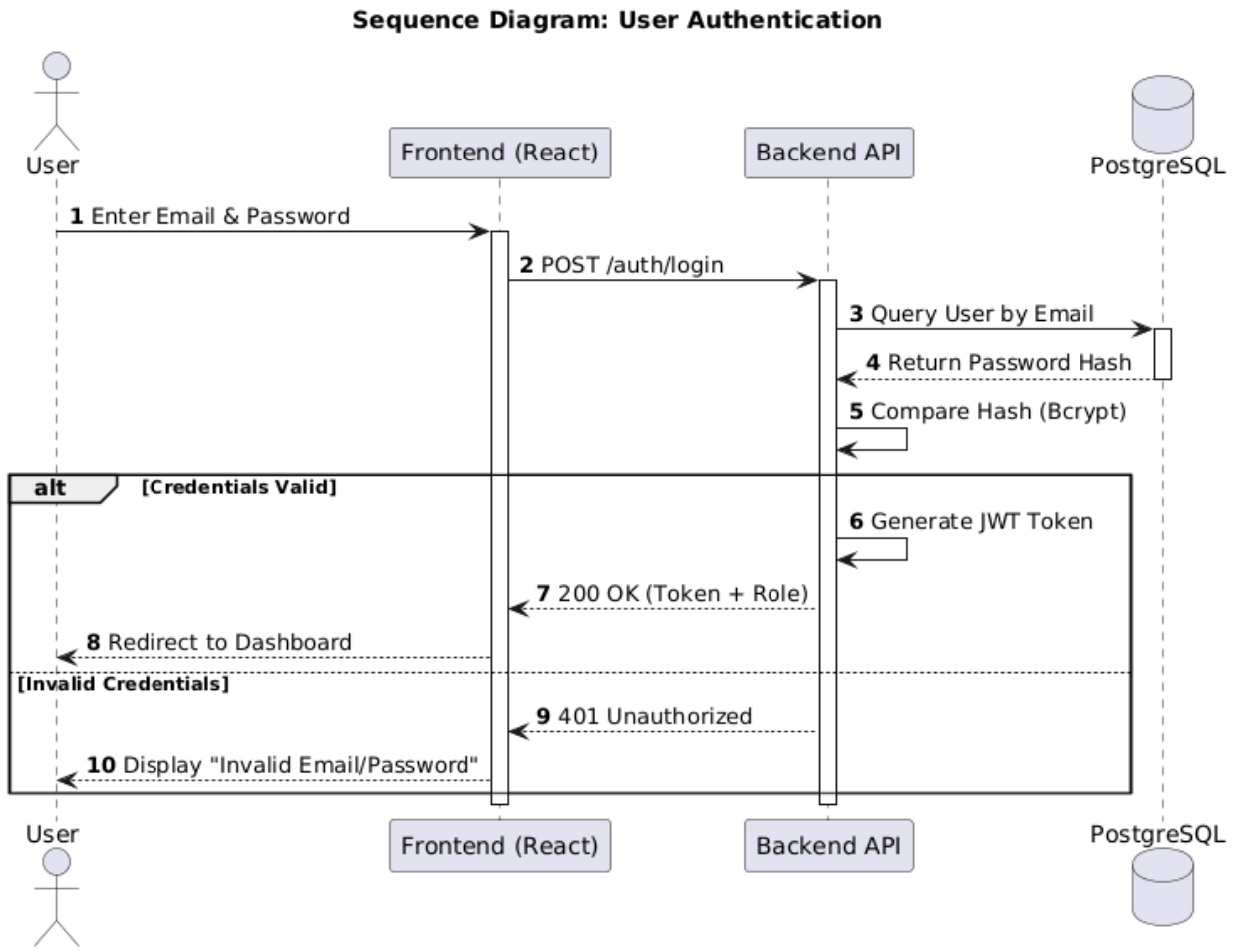

Figure 3.3: User Authentication Sequence diagram

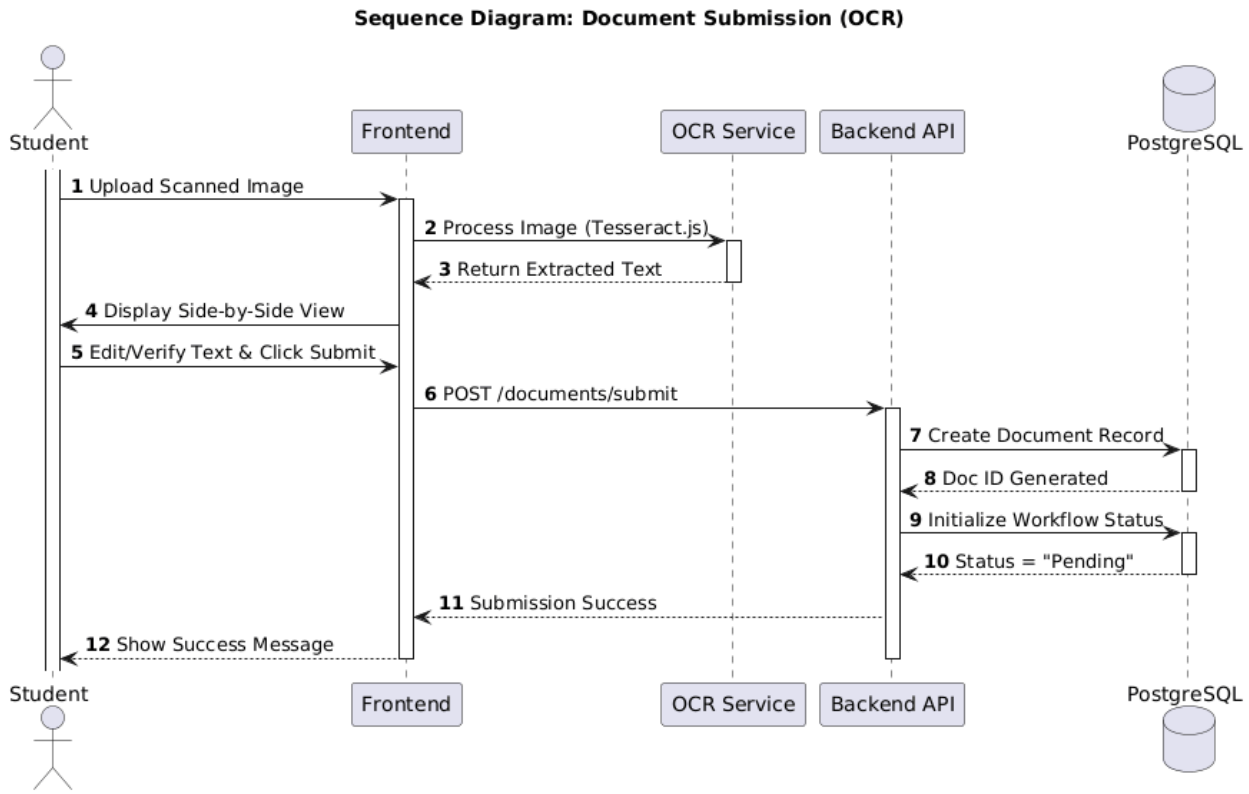

Figure 3.4: Document submission (OCR) Sequence diagram

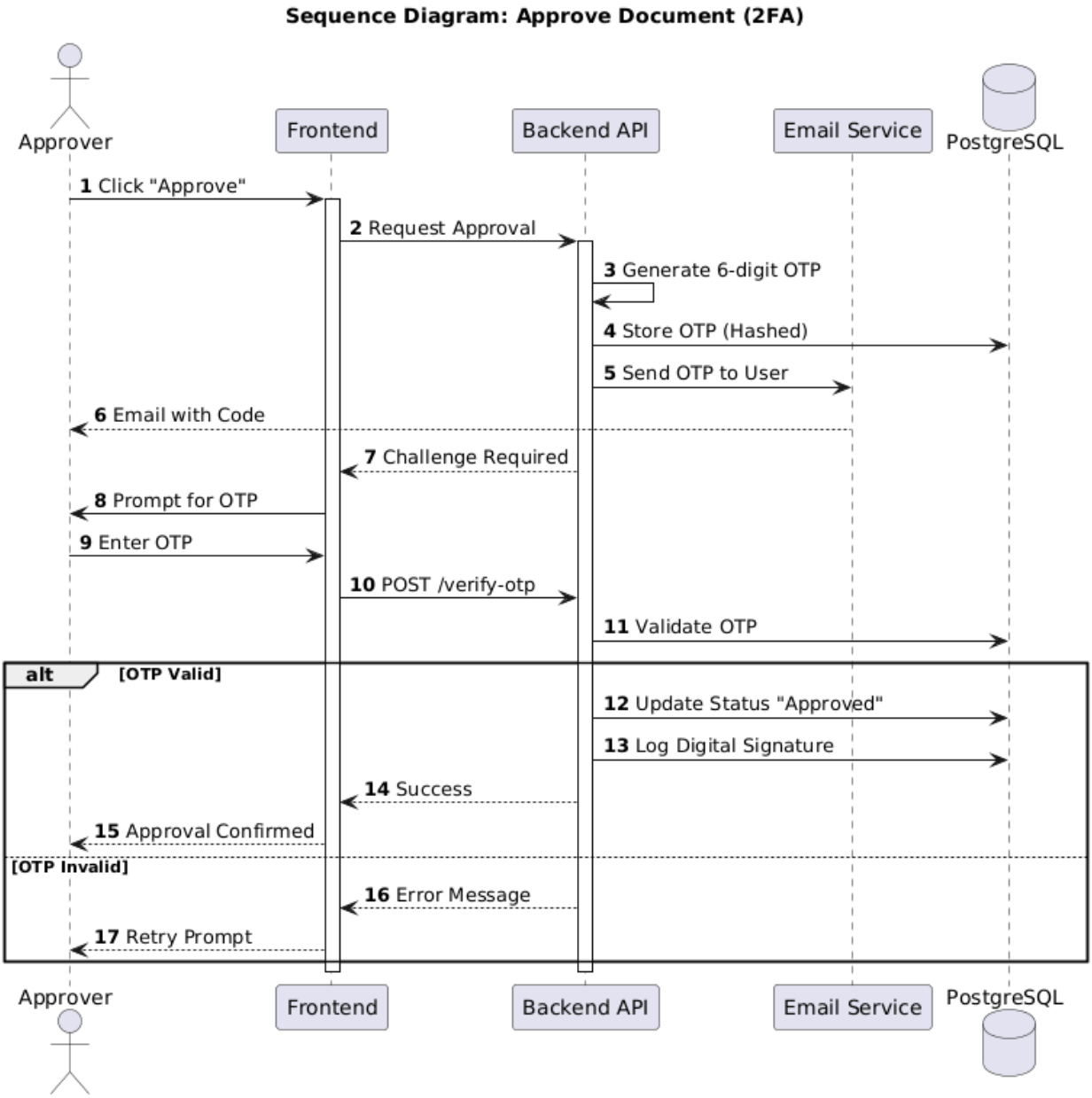

Figure 3.5: Approve Document Sequence diagram

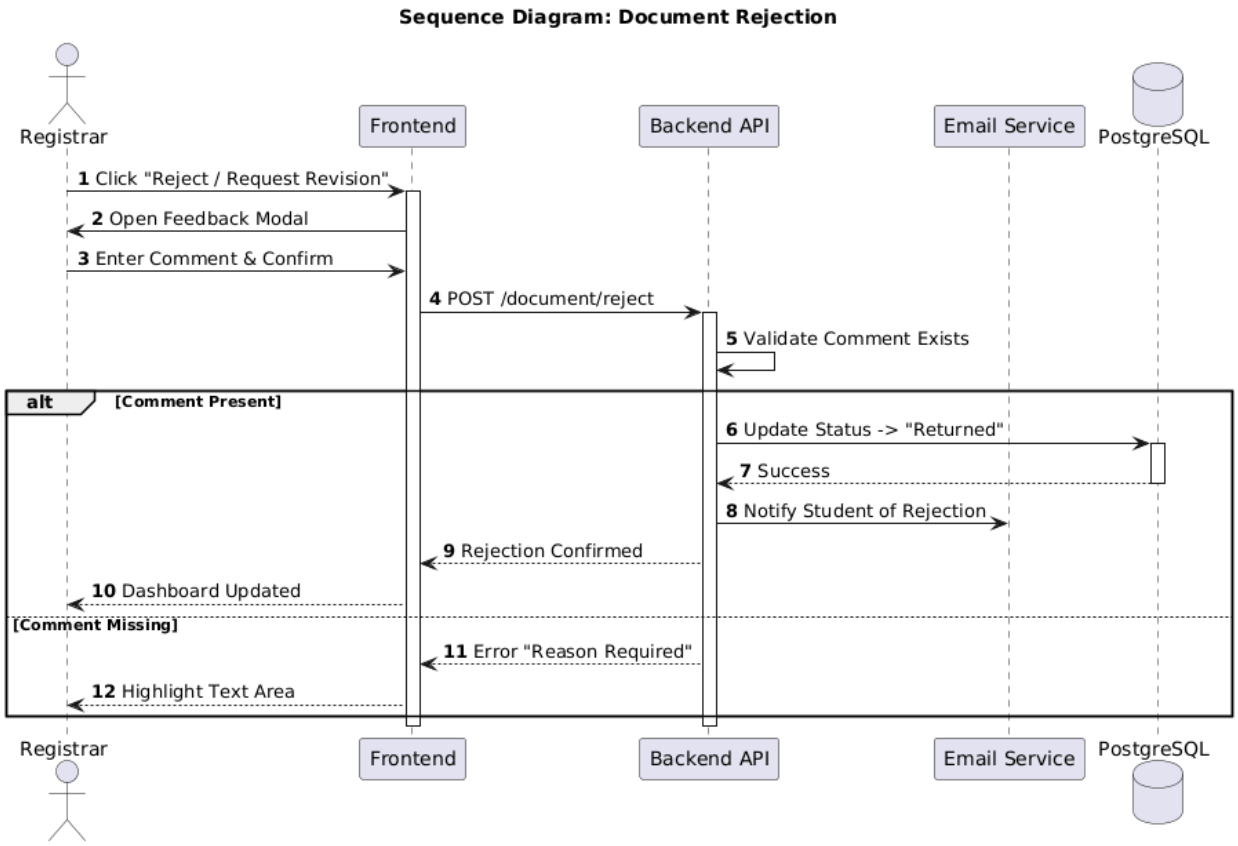

Figure 3.6: Document rejection sequence diagram

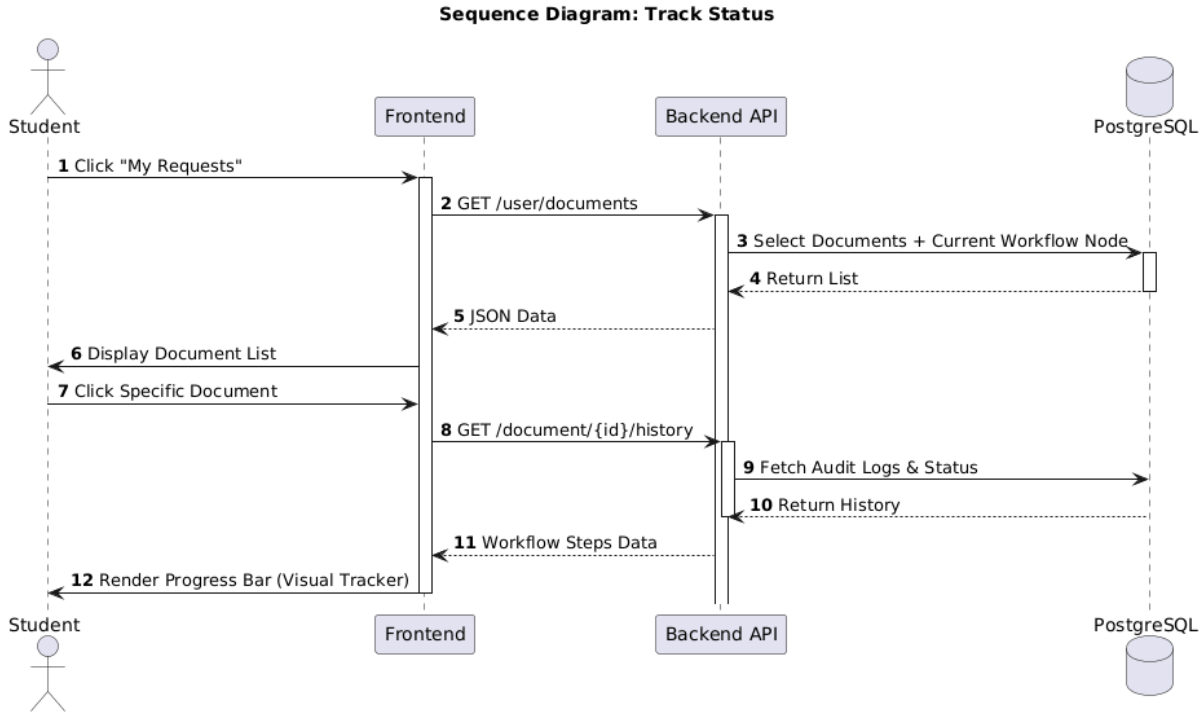

Figure 3.7: Track Status Sequence Diagram

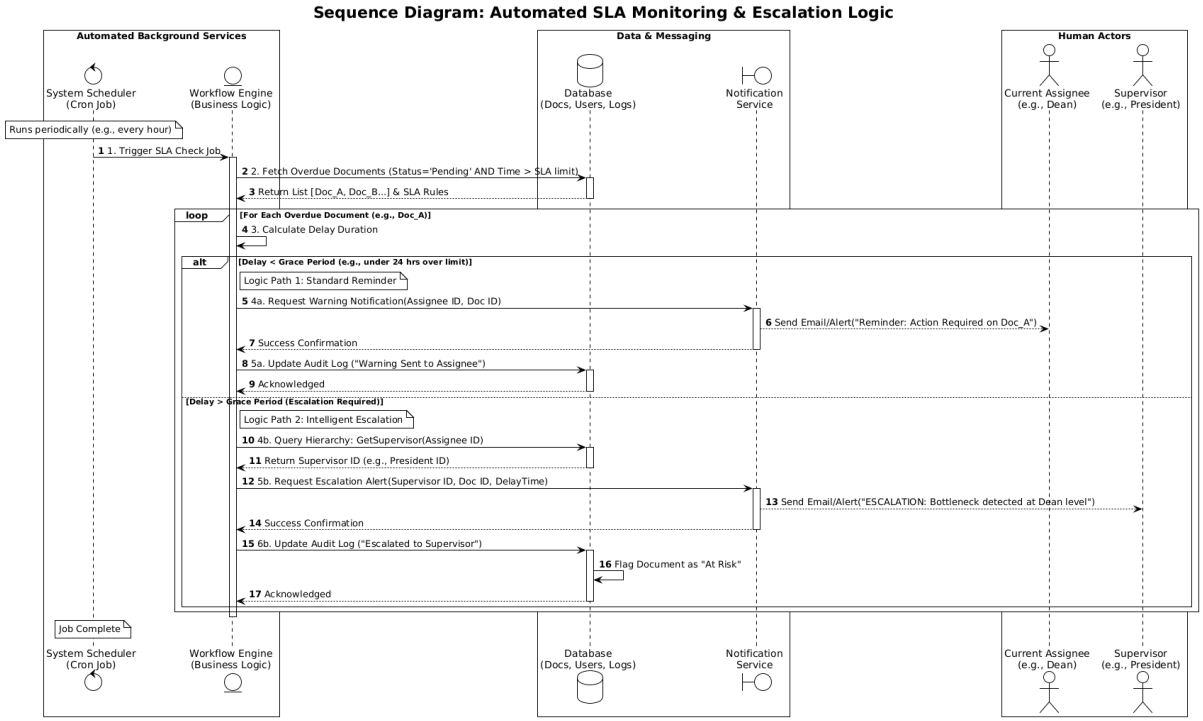

Figure 3.8: Automated SLA Monitoring & Escalation Logic

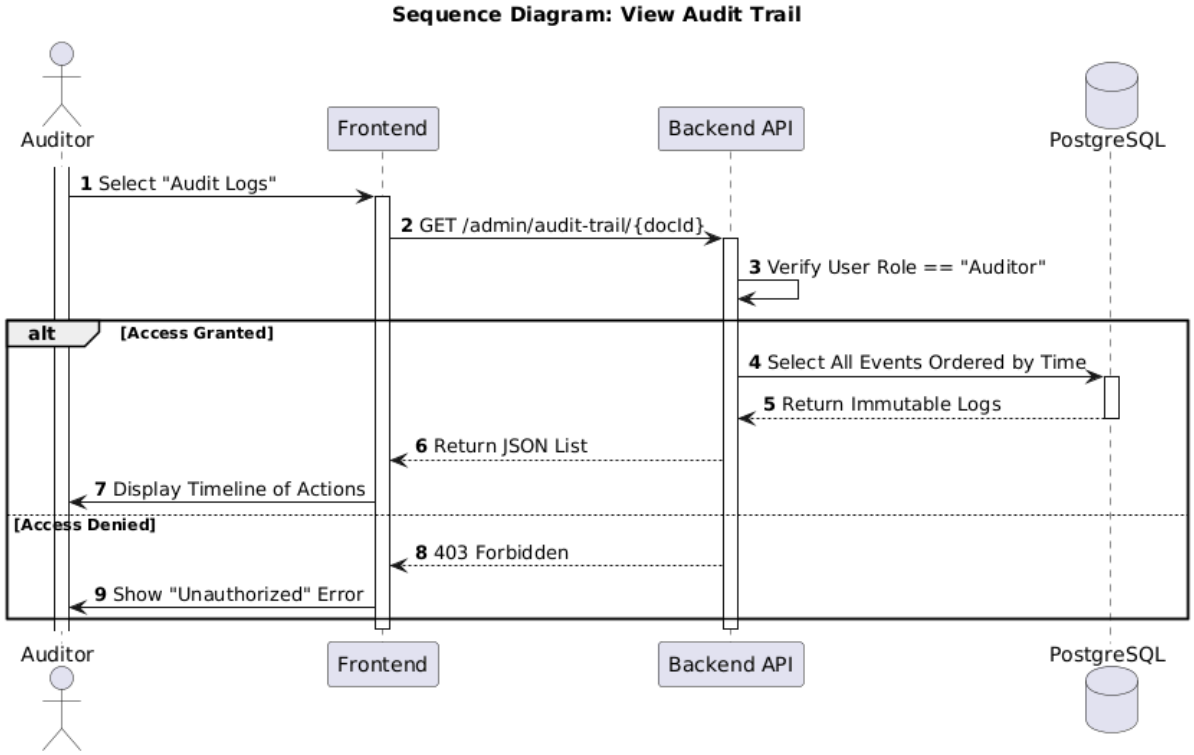

Figure 3.9: View Audit Trail sequence diagram

3.5.5. Activity diagram

Activity Diagram for User Login and Two-Factor authentication(2FA)

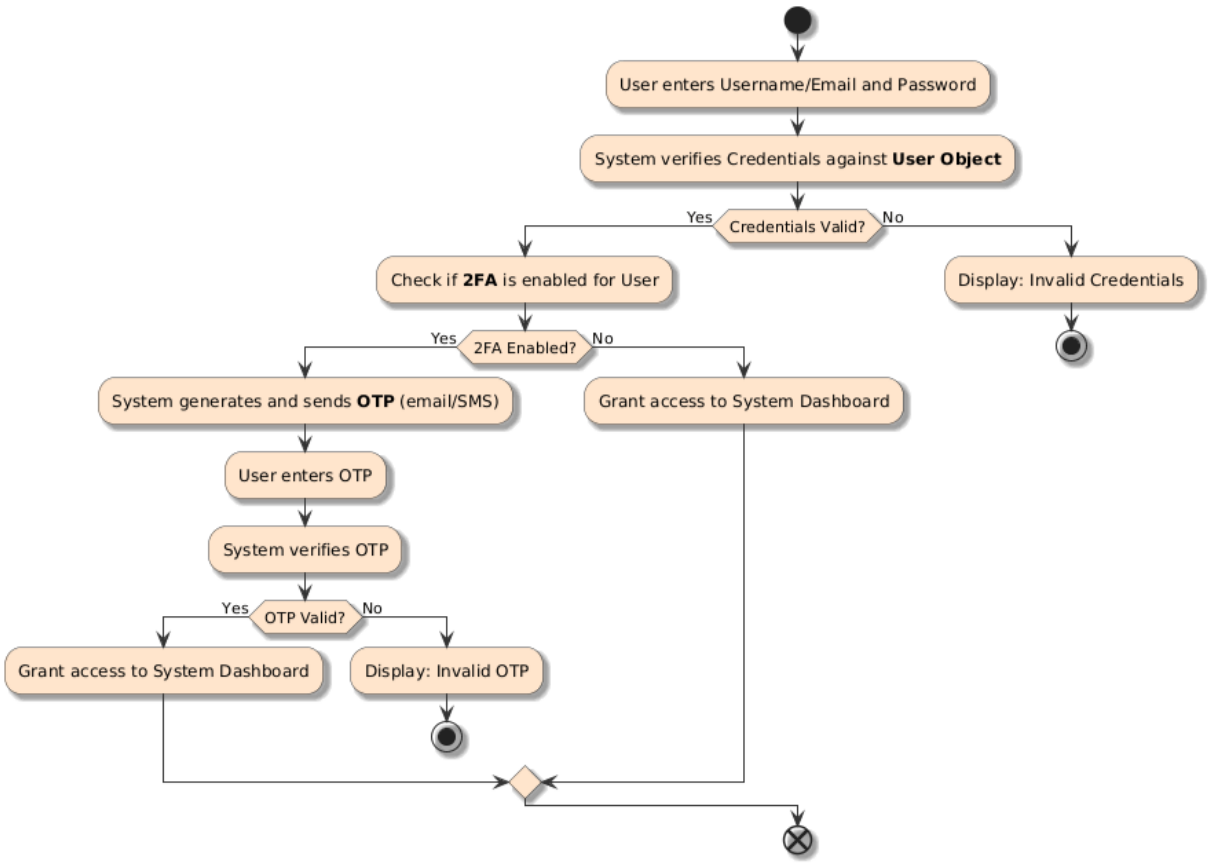

Figure 3.10: User Login and Two-Factor authentication(2FA) Activity Diagram

Activity Diagram for Document Submission and Approval Process

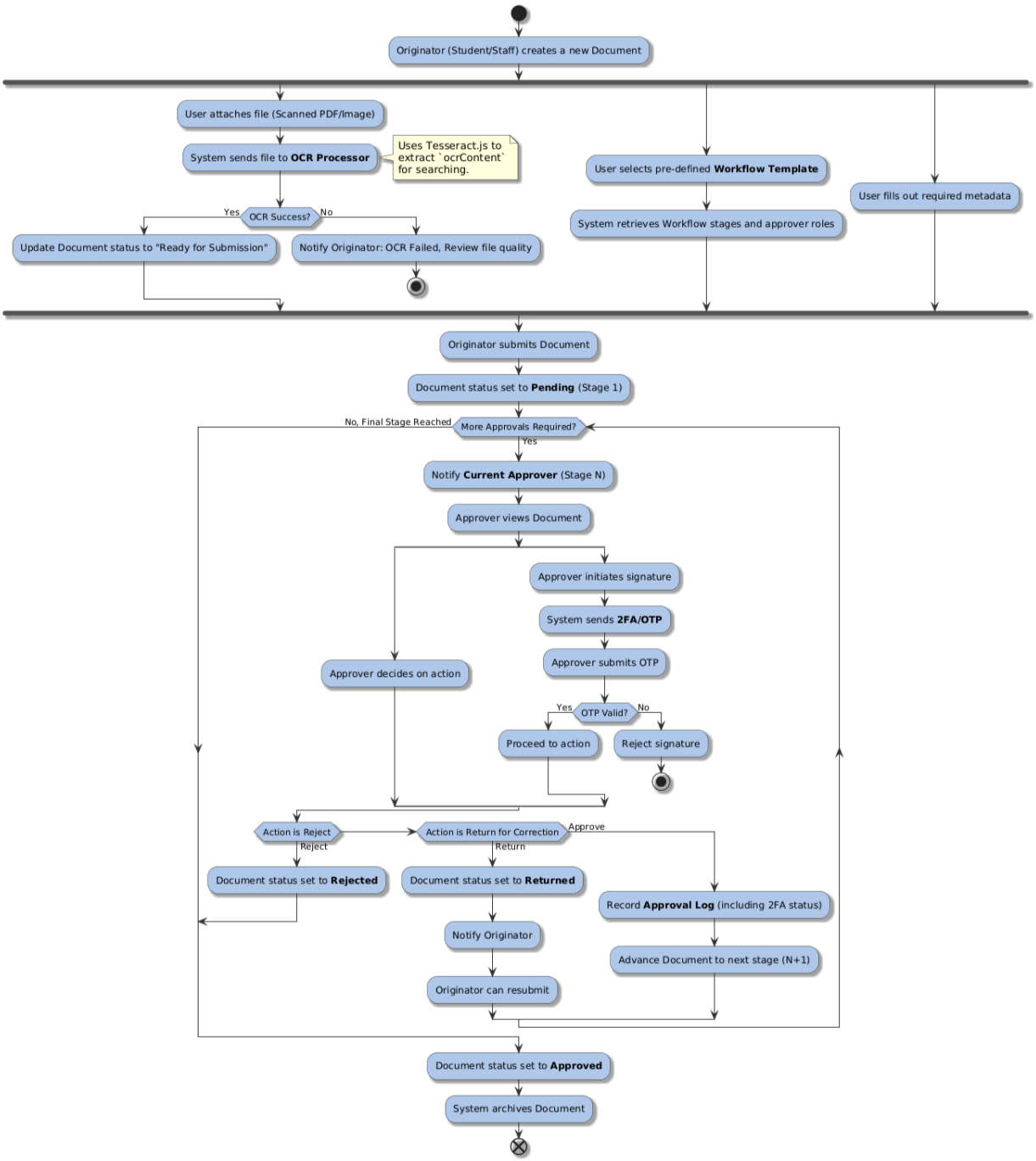

Figure 3.11: Document Submission and Approval Process Activity diagram

Activity Diagram for Workflow Configuration

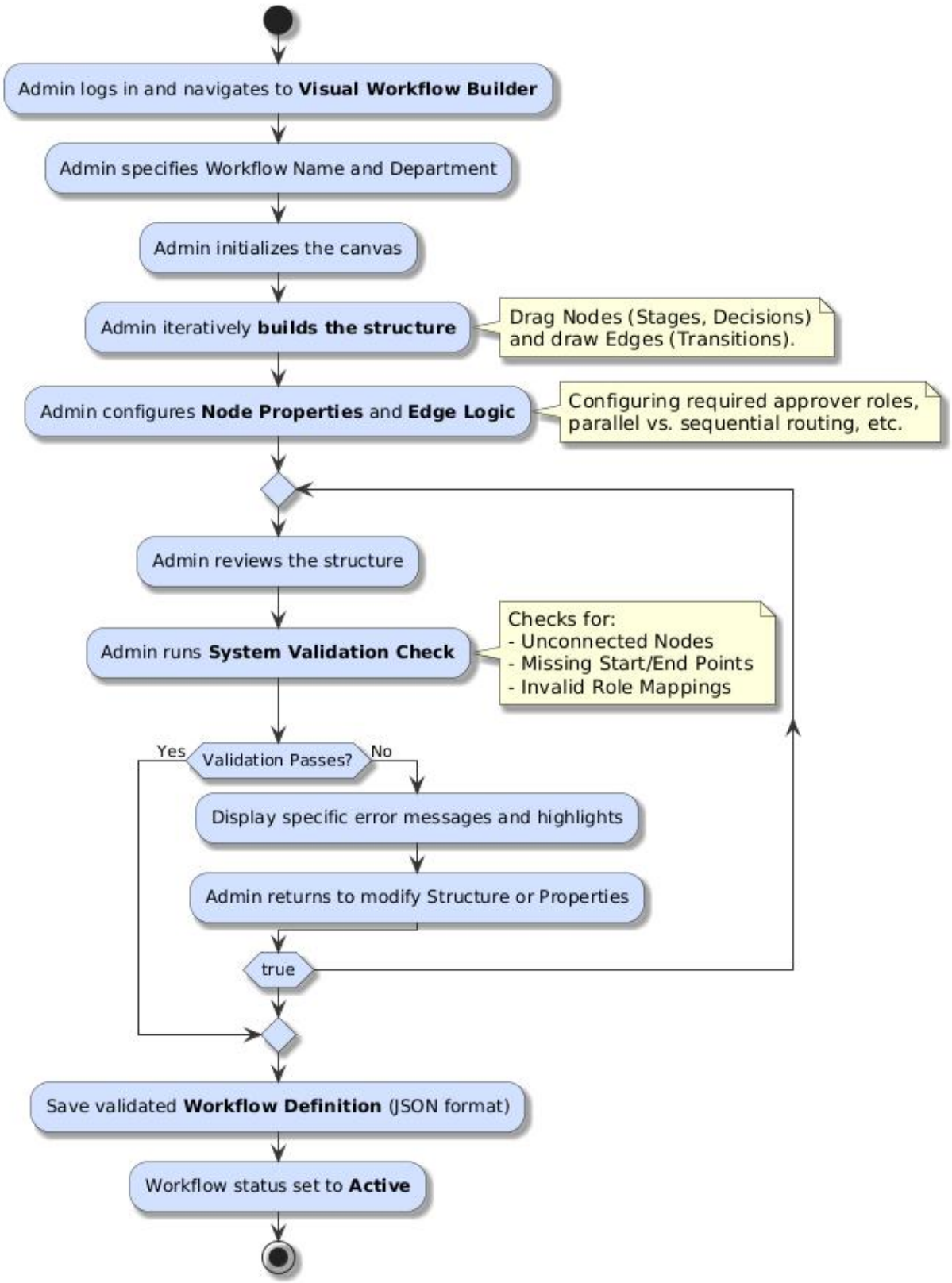

Figure 3.12: Workflow Configuration Activity Diagram

Activity Diagram for Document Status Tracking and History access

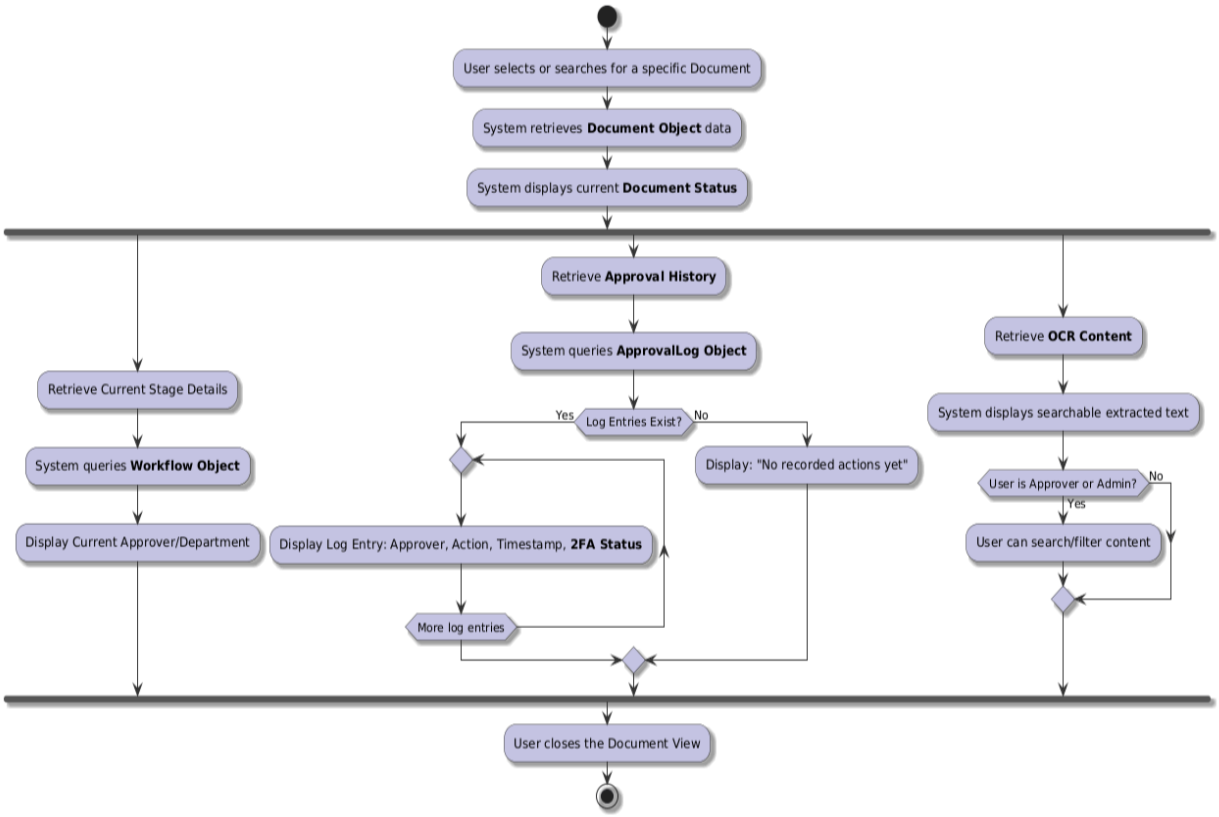

Figure 3.13: Document Status Tracking and History access Activity Diagram

Activity Diagram for Notification Handling

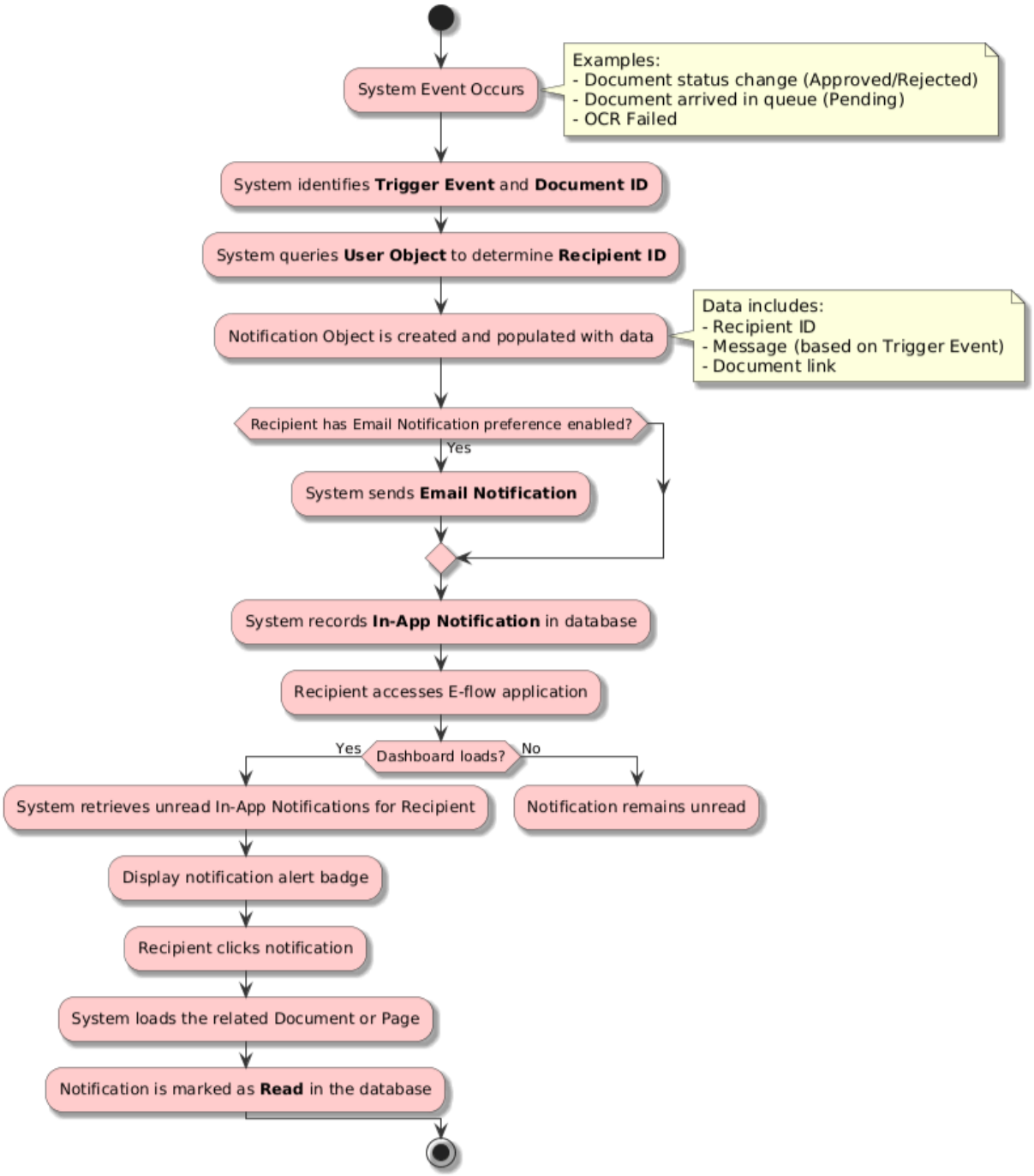

Figure 3.14: Notification Handling Activity diagram

Activity Diagram for Audit log Recording

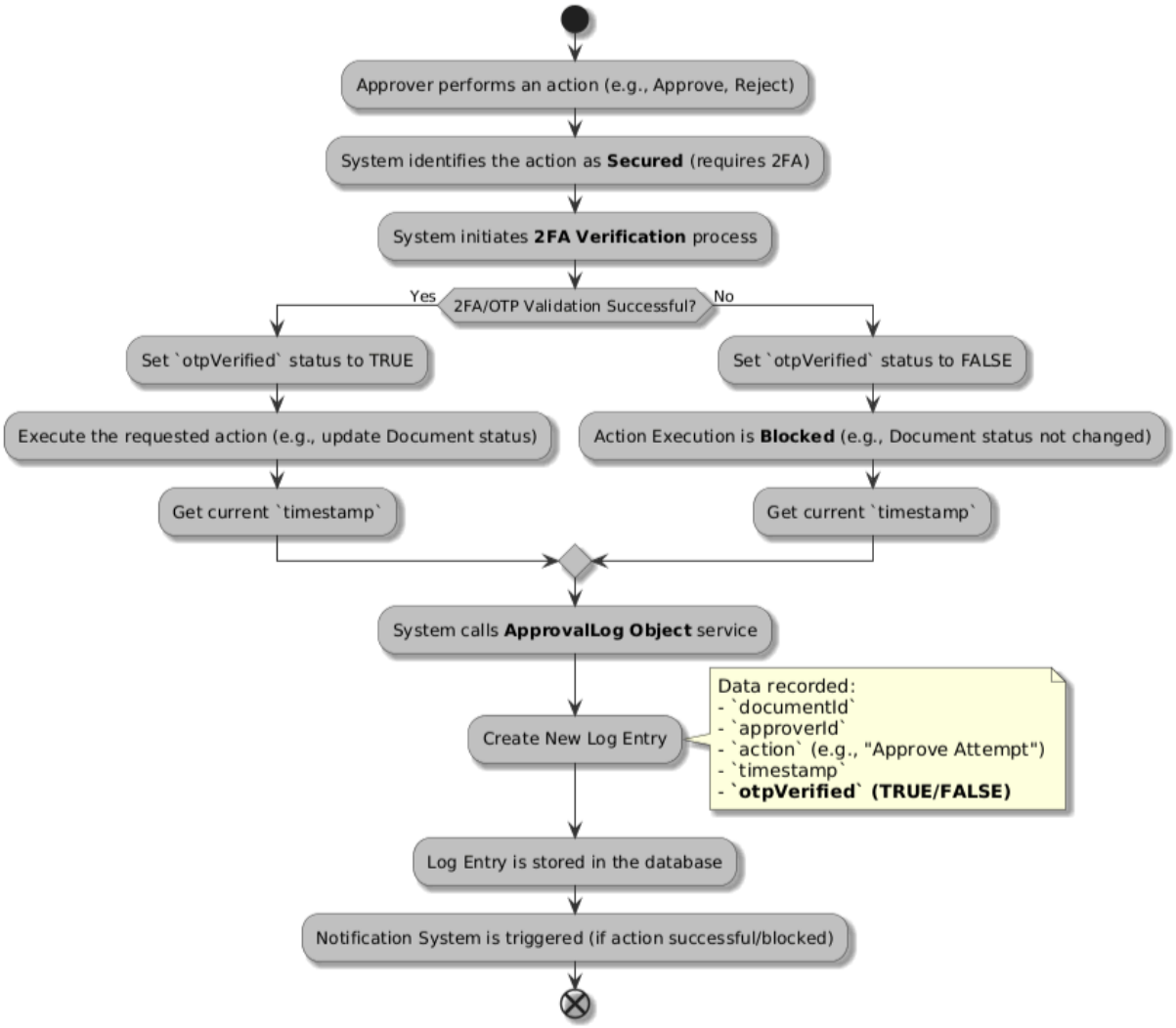

Figure 3.15: Audit log Recording Activity Diagram

3.5.6. State chart diagram

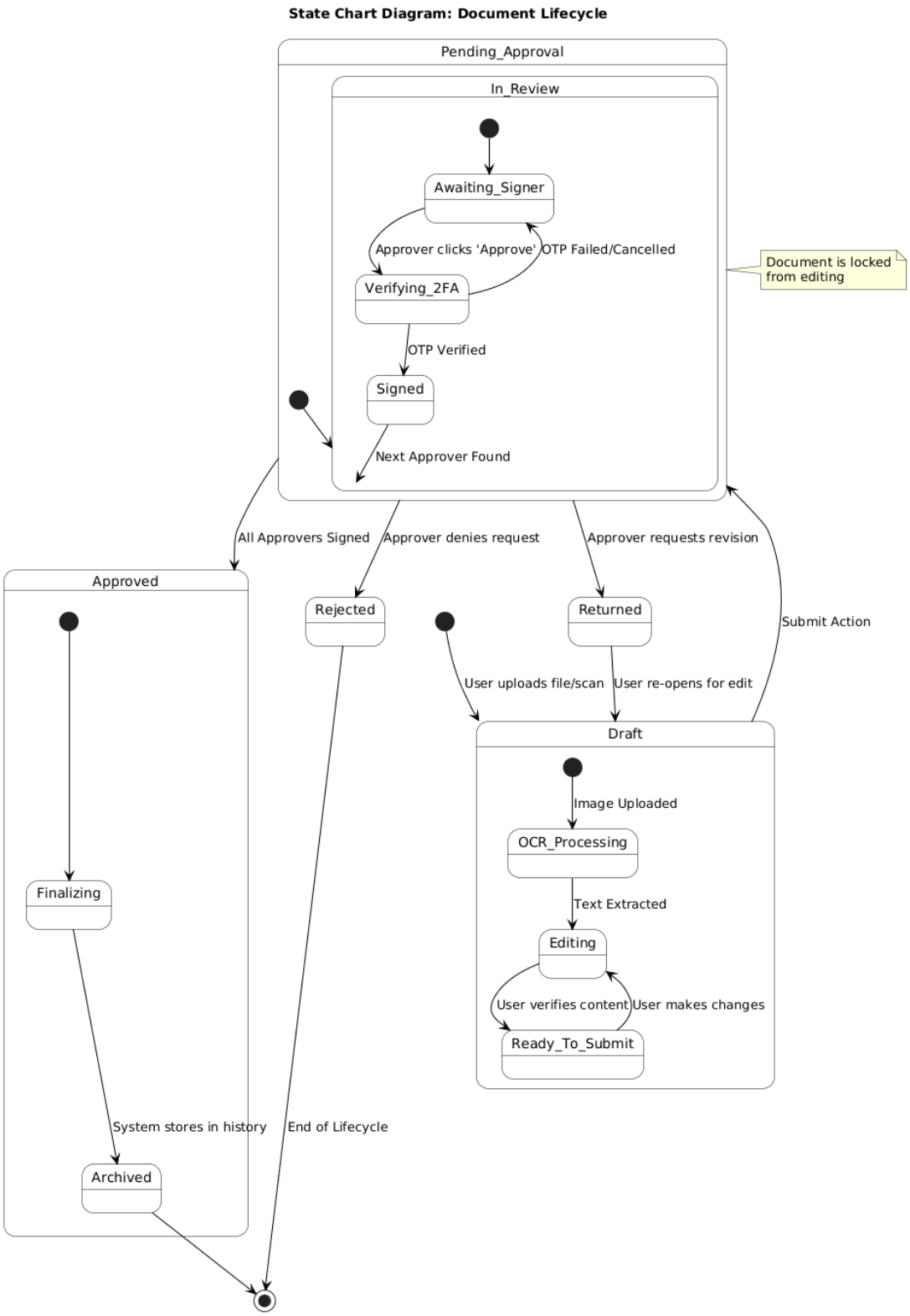

Figure 3.16: Document Lifecycle State chart diagram

Chapter 4: System design

4.1 Overview

This section provides a general description of the system design of the system, E-flow: Intelligent Document Tracking and Workflow Management System for Universities with a Visual Workflow Builder focuses on creating a smooth, organized, and transparent way for universities to handle documents. Instead of relying on scattered manual processes, the system brings everything together in one digital platform where documents can be submitted, processed, tracked, and approved with clarity.

At the center of the solution is a web-based interface built with React.js. This interface provides separate dashboards for students, staff, and administrators. Through it, users can upload documents, follow their progress, approve tasks, and receive notifications whenever an action is required. For administrators, the system offers a Visual Workflow Builder powered by React Flow. This tool lets them design approval steps by simply dragging and arranging nodes, making it easy to create or update processes without writing a single line of code.

Behind the scenes, the backend application built using a server-side framework such as Node.js with Express handles the system's main logic. This includes routing documents based on predefined workflows, managing secure approvals using Two-Factor Authentication (2FA/OTP), keeping track of versions, controlling user access, and generating complete audit trails. Communication between the frontend and backend happens through secure RESTful APIs to ensure fast and reliable data exchange.

All system data including workflows, user information, documents, and approval histories is stored safely in a PostgreSQL database, which ensures that information is organized, secure, and easy to retrieve whenever needed.

The system also relies on several supporting features such as secure approval verification using One-Time Passwords (OTP), notification services for reminders, and role-based permissions to maintain security and accountability. By combining these components and their interactions, the E-flow provides a unified and dependable platform that improves efficiency, reduces delays, and makes document handling at universities far more transparent and convenient for everyone involved.

For scanned and image-based documents, the system integrates an OCR module that extracts textual content and allows user verification before workflow processing.

4.2 Purpose of the System Design

The system design of the E-flow: Intelligent Document Tracking and Workflow Management System for Universities with a Visual Workflow Builder is created to ensure that the final solution meets all functional and non-functional requirements while providing a solid foundation for development and testing. The design supports the system's goals of improving transparency, speeding up document processing, and ensuring smooth communication across departments. Its purpose can be summarized as follows:

➢ Supporting System and Object Models

The design is aligned with the system model and object model of E-flow: Intelligent Document Tracking and Workflow Management System for Universities, clearly defining how different components such as users, documents, workflows, and approval steps interact with each other. By organizing entities and their relationships, the design helps developers and stakeholders easily understand how the system behaves and how processes flow from one stage to another.

➢ Facilitating System Implementation

The design provides a clear blueprint for implementation by outlining the responsibilities of each module, the flow of data, and the interaction between the frontend, backend, and database. This structure allows team members to develop different parts of the system concurrently without confusion. It also ensures that key features such as workflow automation, document tracking, digital signatures, and notifications can be developed in a consistent and organized way.

➢ Supporting Testing and Quality Assurance

By defining clear interfaces, API endpoints, and interactions between components, the system design makes both module-level and full-system testing easier. Testers can validate each part such as workflow routing, user authentication, or audit logging, individually before checking how everything works together. This reduces errors, increases reliability, and ensures that the system performs well before deployment.

➢ Ensuring a Better User Experience

A well-structured design allows the system to be intuitive, responsive, and easy to use. By integrating all features in a seamless way, the design helps ensure that students, staff, and administrators can interact with the system without confusion. Users can submit documents, follow progress, and complete approvals smoothly, ultimately improving administrative efficiency and reducing delays.

4.3 Design Goals

The design of the E-flow: Intelligent Document Tracking and Workflow Management System for Universities is based on key principles that ensure the system is organized, easy to maintain, and able to adapt to future needs. It emphasizes modularity by separating the system into clear components such as the workflow builder, document management module, and backend services. This helps group related functions together, increasing cohesion while keeping coupling low so that updates in one module do not negatively affect others.

The design also applies abstraction and encapsulation to simplify user interactions and protect sensitive operations like authentication, digital signatures, and workflow routing. By hiding unnecessary complexity, the system becomes easier for users to operate and for developers to maintain. Reusability is another goal, allowing shared features such as notifications, access control, and document status tracking to be used across different workflows without duplication.

Overall, these design goals guide the system toward being secure, flexible, and user-friendly, ensuring it can support Universities administrative processes effectively.

4.4. Proposed System Architecture

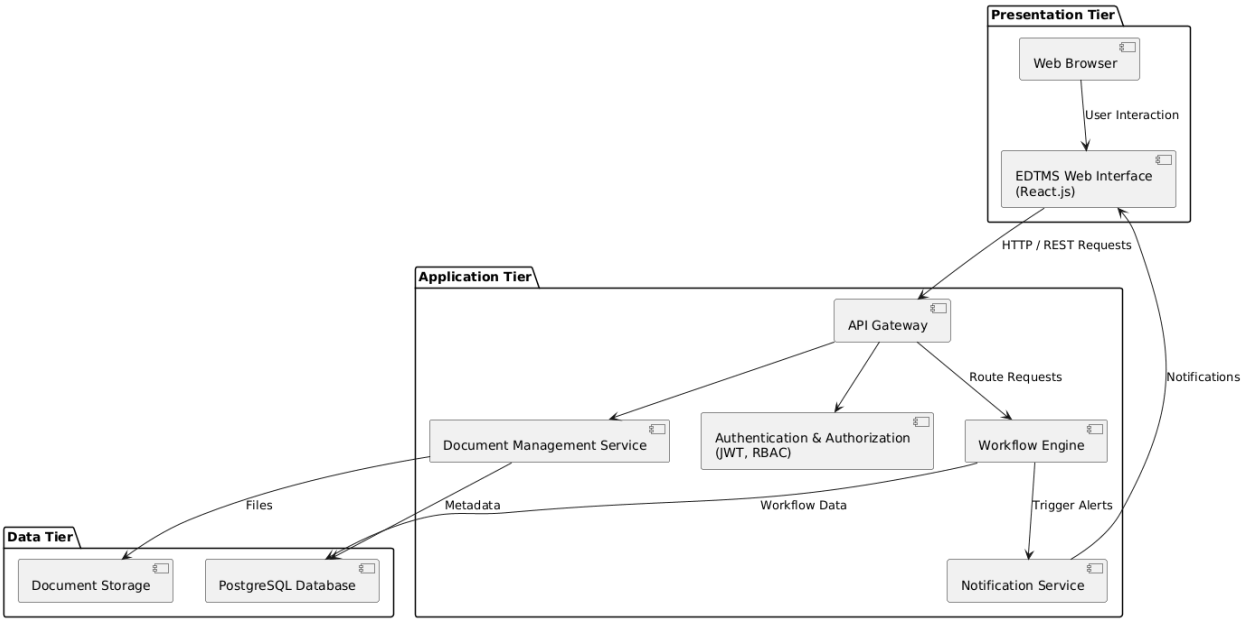

Figure 4.1: 3 Tier System Architecture

4.5. Subsystem Decomposition

The E-flow: Intelligent Document Tracking and Workflow Management System for Universities is decomposed into several subsystems to simplify development, improve maintainability, and support modular implementation. Each subsystem is responsible for a specific set of functions and interacts with others through well-defined interfaces. This decomposition allows the system to be developed, tested, and maintained efficiently.

1. User Interface Subsystem

This subsystem handles all user interactions with the system. It provides role-based dashboards for students, staff, and administrators, allowing users to submit documents, design workflows, approve requests, and track document status. It also displays notifications and system feedback in real time.

Main responsibilities:

- Document submission and viewing
- Workflow design using drag-and-drop tools
- Approval actions and status tracking
- User notifications and alerts

2. Workflow Management Subsystem

The workflow management subsystem controls how documents move through different approval stages. It interprets workflow definitions created by administrators and automatically routes documents based on predefined rules.

Main responsibilities:

- Creating and managing workflow definitions
- Executing approval sequences
- Tracking workflow progress and states
- Handling conditional routing logic

3. OCR Processing Subsystem

This subsystem is responsible for extracting textual content from scanned documents and image-based files submitted to the system. It uses an Optical Character Recognition (OCR) engine to convert non-editable documents into machine-readable text that can be indexed, searched, and processed within workflows.

The OCR Processing Subsystem supports a human-in-the-loop approach, allowing users to review and correct extracted text to improve accuracy before the document proceeds through the workflow.

Main responsibilities:

- Processing scanned PDF and image documents
- Text extraction using OCR technology (e.g., Tesseract.js)
- Allowing user verification and correction of OCR output
- Storing extracted text for search and workflow use

4. Document Management Subsystem

This subsystem is responsible for handling document-related operations. It manages document uploads, storage, version control, and secure approval confirmations using 2FA/OTP while ensuring data integrity and security.

Main responsibilities:

- Document upload and retrieval
- Version control and history management
- Integration with OCR-extracted content
- Secure approval records using 2FA/OTP
- Secure document storage

5. User and Access Control Subsystem

This subsystem ensures that only authorized users can access or modify system resources. It manages authentication, role-based access control, and permission validation across all system components.

Main responsibilities:

- User authentication (login/logout)
- Role and permission management
- Session and token handling
- Access validation for system actions

6. Notification Subsystem

The notification subsystem informs users about important system events. It sends alerts when documents require action, approvals are completed, or workflow statuses change.

Main responsibilities:

- Email notifications
- In-app alerts
- Event-based message triggering

7. Audit and Reporting Subsystem

This subsystem records all system activities to ensure transparency and accountability. It generates logs and reports that can be reviewed by administrators for monitoring and auditing purposes.

Main responsibilities:

- Activity logging and audit trails
- Workflow and document reports
- User action tracking

8. Data Management Subsystem

The data management subsystem stores and manages all persistent system data. It ensures consistency, reliability, and secure access to user data, documents, and workflow information.

Main responsibilities:

- Database operations (CRUD)
- Data integrity and consistency
- Backup and recovery support

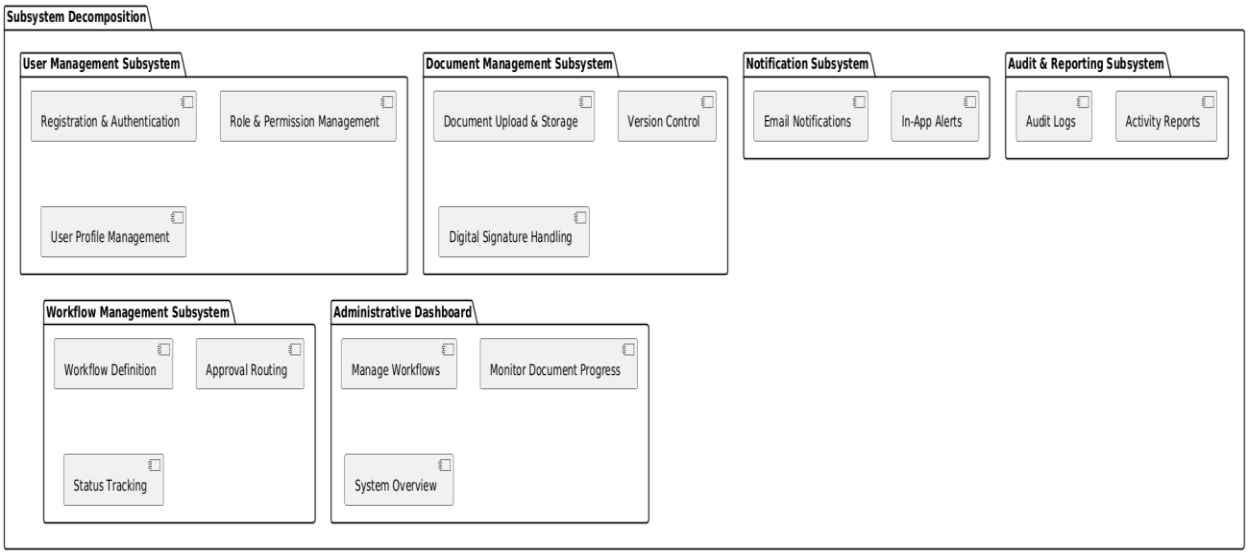

Figure 4.2: Proposed System Architecture

4.6. Subsystem Description

Subsystem Name	**Function**	**Interfaces**	**Interactions**	**Dependencies**
User Management Subsystem	Handles user registration, login, profile management, and role-based access control	Web UI, Authentication API	Interacts with Workflow Management and Notification Subsystems for user actions	Depends on Authentication Service and Database for storing user data

Document Management Subsystem	Manages document upload, storage, version control, and secure approval confirmation using One-Time Password (OTP)	Web UI, Backend API	Works with Workflow Management to provide documents for approvals; interacts with Audit Subsystem for tracking	Depends on Database and File Storage services
OCR Processing Subsystem	Extracts text from scanned documents and images using OCR and supports user verification of extracted content	Document Upload Interface, Backend API	Works with Document Management Subsystem to attach extracted text to documents	OCR Engine (Tesseract.js), Backend Application Services
Workflow Management Subsystem	Defines and executes approval workflows; routes documents through various stages	Backend API, Notification API	Interacts with User, Document, and Notification Subsystems to move documents through workflow steps	Depends on Database for workflow definitions and status tracking
Notification Subsystem	Sends email and in-app notifications for workflow events	Backend API, Web UI	Receives triggers from Workflow and Document Management	Depends on SMTP / email server and Web UI for display

			Subsystems; notifies users	
Audit & Reporting Subsystem	Logs system activity and generates reports for monitoring and auditing	Backend API, Admin Dashboard	Receives data from Workflow, Document, and User Management Subsystems	Depends on Database for storing logs and reporting engine
Administrative Dashboard	Provides administrators with a centralized interface to monitor system operations and workflow status	Web UI	Interacts with all other subsystems to present information and manage workflows	Depends on Workflow, Document, and User Management Subsystems
Data Management Subsystem	Centralized storage and management of persistent data, including users, documents, workflows	Backend API	All subsystems interact with data management to read/write data	Depends on Database server and storage services

Table 13: Subsystem Description

4.7. Persistent Data Management

The E-flow: Intelligent Document Tracking and Workflow Management System for Universities relies on persistent data management to ensure that all system information is securely stored, consistently maintained, and readily accessible throughout the system's operation. Persistent data storage enables the system to retain user records, documents, workflow states, and audit logs even after system shutdowns or failures.

The system uses a PostgreSQL relational database management system (RDBMS) to store structured data. PostgreSQL is selected due to its strong support for relational integrity, transaction management, and JSON-based storage, which is suitable for storing workflow definitions and approval metadata. This includes user accounts, roles and permissions, workflow definitions, document metadata, approval records, and system logs. A relational approach is chosen to maintain data integrity, enforce relationships among entities, and support reliable transaction processing.

In addition to document metadata, the system stores OCR-extracted textual content linked to each document. This enables keyword searching, indexing, and workflow-based processing of scanned documents while preserving the original files.

Actual document files are stored in a secure file storage repository, while only their metadata such as document ID, version number, owner, submission date, and current status is stored in the database. This separation improves performance and simplifies data management while ensuring efficient document retrieval.

The E-flow supports standard CRUD (Create, Read, Update, Delete) operations through the application layer. Data access is handled via well-defined APIs that enforce validation rules, access control, and consistency checks. Role-based access control ensures that users can only view or modify data permitted by their roles.

To support accountability and traceability, the system maintains audit logs that record important actions such as document submissions, approvals, rejections, and user logins. These logs are stored persistently and can be reviewed by administrators for monitoring and reporting purposes.

Regular backup and recovery mechanisms are incorporated to protect against data loss. Periodic backups ensure that the system can be restored in case of hardware failure, accidental deletion, or security incidents.

Overall, the persistent data management design of E-flow ensures data reliability, security, and scalability, supporting efficient document tracking and long-term system operation.

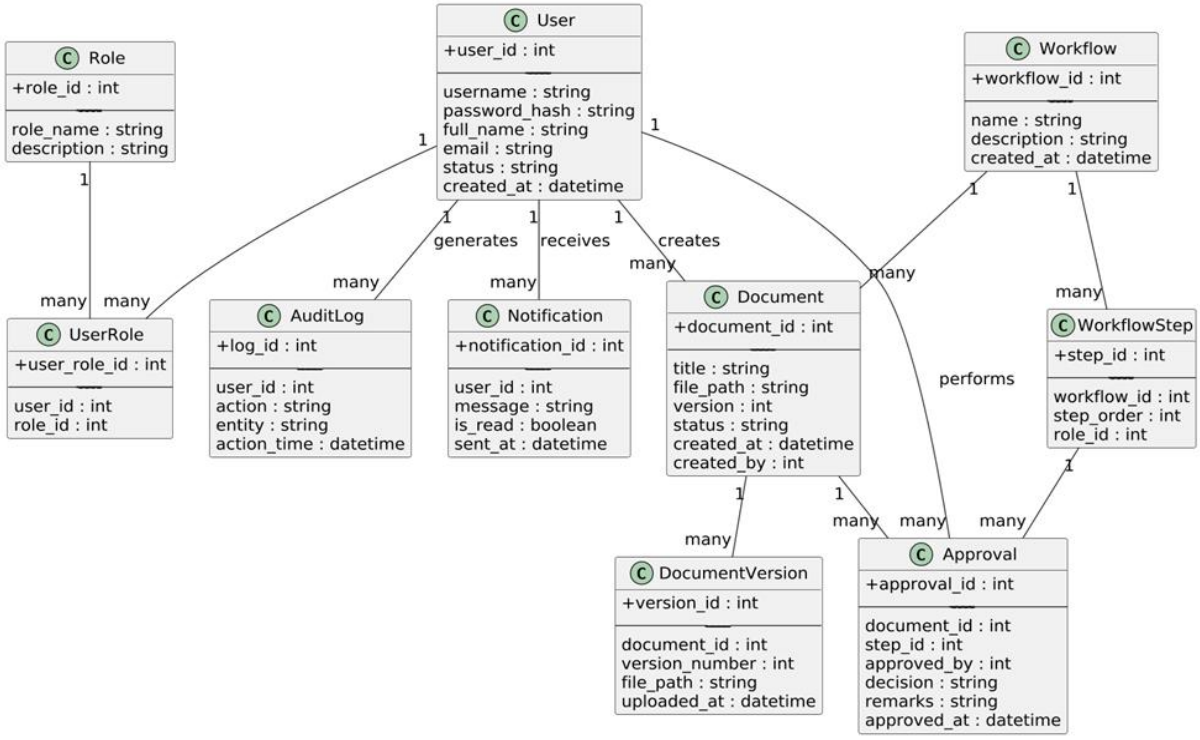

Figure 4.3: Persistent Data Management Diagram

4.8 Component Diagram

The component diagram of the E-flow: Intelligent Document Tracking and Workflow Management System for Universities shows its major software components and the interactions between them. It focuses on how the system is organized into functional modules and how these modules communicate through well-defined interfaces.

The E-flow is composed of a web-based user interface, backend application services, and data management components. The Web Client provides the user-facing functionalities such as document submission, workflow design, approvals, and status tracking. It communicates with the backend through RESTful APIs.

The Backend Application contains the core system logic and is divided into several components, including workflow management, document management, authentication and authorization, notification handling, and audit logging.

Each component performs a specific responsibility and exposes services that can be accessed by other components. This separation improves modularity and allows independent development and testing.

The Data Management Component handles persistent storage using a relational database. All components that require data access interact with the database through the backend services, ensuring controlled and secure data operations.

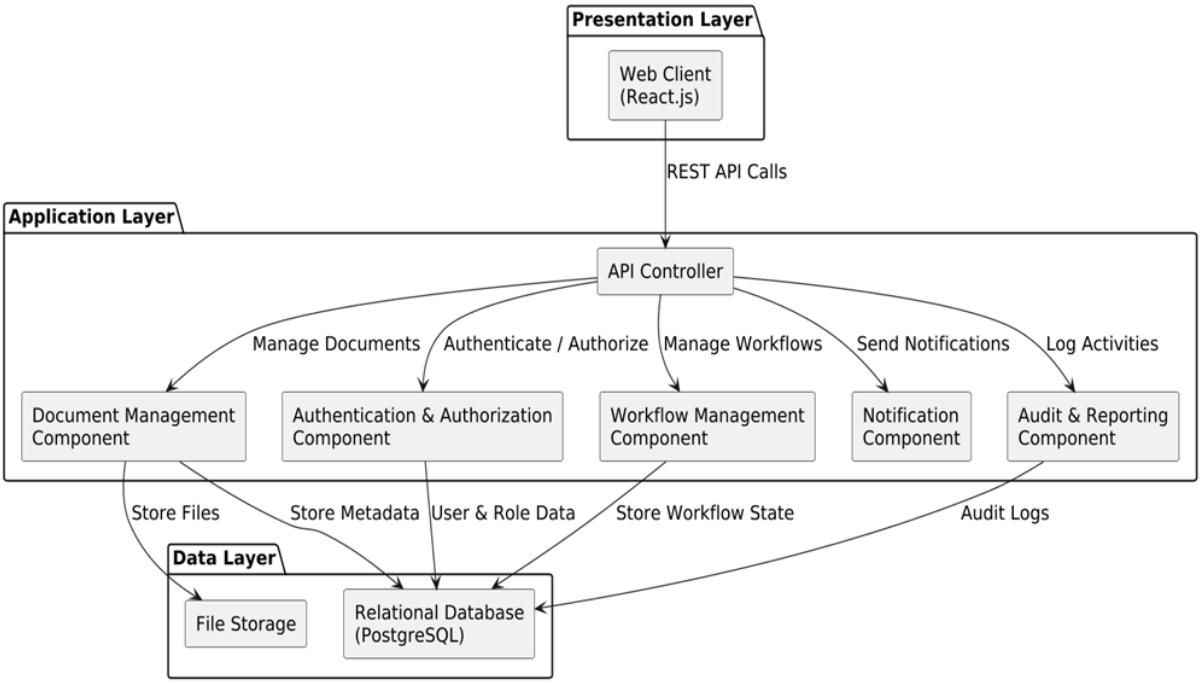

Figure 4.4: Component Diagram

4.9 Database Diagram

The database diagram represents the logical structure of the E-flow: Intelligent Document Tracking and Workflow Management System for Universities database. It illustrates the main database tables, their key attributes, and the relationships between them. The database is designed to support document tracking, workflow automation, approval management, user access control, and audit logging.

The E-flow uses a relational database to ensure data integrity, consistency, and efficient querying. Each table represents a persistent entity in the system, and relationships are defined using primary and foreign keys. This structure allows the system to reliably store and retrieve information related to users, documents, workflows, approvals, and system activities.

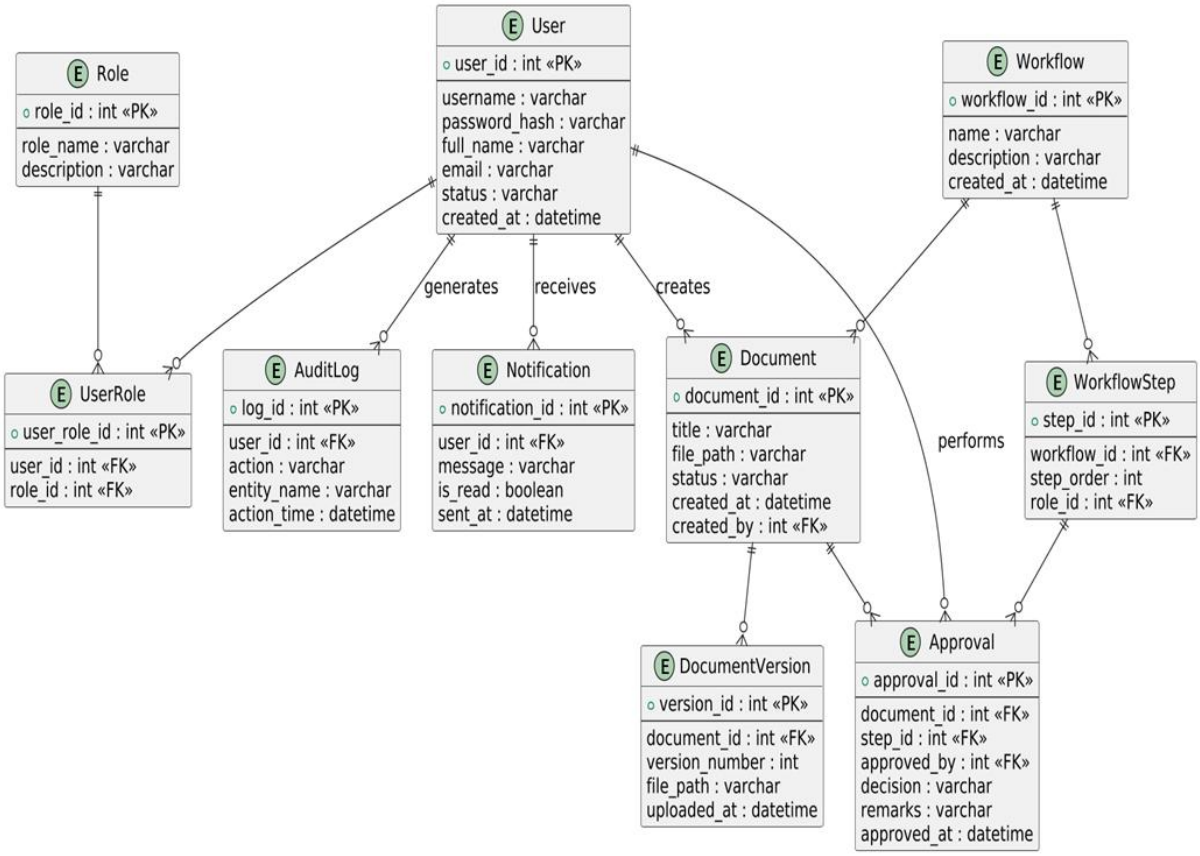

Figure 4.5: Database ER Diagram

4.10 Access Control

The E-flow: Intelligent Document Tracking and Workflow Management System for Universities implements a role-based access control mechanism to ensure that system resources are accessed only by authorized users. Access control is enforced through authentication, authorization, and secure data handling mechanisms. This approach protects sensitive documents, maintains accountability, and ensures proper workflow execution.

Users must be authenticated before accessing the system. Once authenticated, permissions are granted based on assigned roles. Each role defines the actions a user can perform within the system. All access requests are validated by the application layer, and sensitive data is protected using encryption and secure communication protocols.

User Role	User Management	Document Management	Workflow Management	Approval Management	Audit & Feedback	Reporting & Analytics
Student	Update Profile	Submit, View Own Documents	View Workflow Status	N/A	Leave Comments	N/A
Staff	Update Profile	Submit, View Assigned Documents	View Assigned Workflows	Approve / Reject	View Audit Logs	N/A
Department Head	N/A	View Department Documents	Manage Department Workflows	Final Approval	Review Feedback	View Reports
System Administrator	Manage Users	Manage All Documents	Manage Workflows	Override Approvals	View Audit Logs	Generate Reports
System Auditor	N/A	View Documents (Read-only)	View Workflow History	N/A	View Audit Logs	View Reports

Table 14: Users Access Control

Chapter 5: Implementation

5.1 Overview

This chapter describes the implementation phase of the E-flow: Intelligent Document Tracking and Workflow Management System for Universities with a Visual Workflow Builder. The implementation phase focuses on transforming the system design into functional application by integrating frontend interfaces, backend services, database management, and other technologies such as OCR and security mechanisms.

The system is implemented using a web-based architecture consisting of a client-side application, a server- side application and a data storage layer . The frontend is developed by using React.js . One of the key features is implemented at this stage (Visual Workflow Builder).which allows administration to design approval processes through drag and drop components without requiring programming knowledge.

On the server side, we used Node.js with Express, which handles requests processing, workflow execution, document routing , authentication, authorization, and notification triggering. Communication between the client and server is handled by RESTful APIs,
Ensuring secure and efficient data exchange.

The system uses PostgreSQL as the backend database to store user data, workflow definitions, document metadata, approval histories, and audit logs. For scanned and image-based documents, an OCR module was implemented to extract textual content, allowing documents to be indexed and processed within workflows. The extracted text can be reviewed and corrected by users before final submission.

5.2 Coding Standards

In order to ensure code quality and maintainability, established coding standards and best practices were followed throughout the development . These standards will help to reduce errors, improve readability and support collaborative development .

Naming Conventions

- **CamelCase** used for variable and function names in javascript (e.g., getUsersDetails , submitDocument)
- **PascalCase** is applied to react component names (e.g., WorkflowBuilder , DocumentUpload).
- **Snake_case** is used in naming database tables and fields (e.g., user_roles,document_status).

Code structure and modularity

- The application follows modular architecture, where frontend and backend components are separated into logical folders.
- Backend services are divided into controllers, routes, services, and middleware to maintain separation of concerns.
- Reusable components were created for common frontend elements.

Documentation and comments

- Inline comments were added to explain the code base , especially in complex logic.
- Meaningful commit messages were used in version control to track changes effectively .

Frameworks and Libraries

Frontend

- **React.js** – Builds dynamic and responsive user interfaces.
- **React Flow** – Implements the drag-and-drop visual workflow builder.
- **Tailwind CSS / Material UI** – Used for UI styling and responsiveness.

Backend

- **Node.js** – Server-side runtime environment.
- **Express.js** – Framework for building RESTful APIs and handling requests.

Security & Authentication

- **JWT (JSON Web Token)** – User authentication and role-based access control.
- **Crypto (Node.js)** – Secure token and OTP generation.
- **Nodemailer** – Sends OTP emails for Two-Factor Authentication (2FA).

OCR & Document Processing

- **Tesseract.js** – Extracts text from scanned and image-based documents.

Database

- **PostgreSQL** – Stores users, documents, workflows, approvals, and audit logs.

Development & Testing Tools

- **Git & GitHub** – Version control and collaboration.
- **Postman** – API testing and debugging.

Security and best practices

- Input validation and error handling were implemented to prevent invalid data processing.
- Role-based access control (RBAC) was enforced to restrict system actions based on user rules.
- Secure authentication mechanisms and OTP-based approval confirmation were implemented to enhance system security.

5.3 Development Tools

During the development of the system, we used several tools . The selection of these tools was based on reliability, ease of use, and compatibility with the technology stack

Integrated Development Environment (IDE)

- Visual Studio Code (VS Code) was used as the primary IDE due to its lightweight nature, rich extension support, and built-in debugging capabilities.
- Extensions such as ESLint, Prettier, and Git integration improved code quality and productivity.

Frontend Development Tools

- React.js was used for building the user interface and implementing dynamic components such as dashboards and the visual workflow builder.
- React Flow was used to implement the drag-and-drop workflow design feature.
- Browser developer tools were used for UI testing, layout inspection, and performance debugging.

Backend Development Tools

- **Node.js and Express.js** were used for server-side development and API creation.
- **Postman** was used to test and validate RESTful API endpoints during development.
- Middleware tools were used for authentication, authorization, and request validation.

Database and Storage Tools

- PostgreSQL served as the relational database for persistent data storage.
- Database management tools such as pgAdmin were used to manage schemas, queries, and data integrity.
- Secure file storage was used for document handling, with metadata stored in the database.

Version Control

- Git was used for version control to track code changes and manage development progress.
- A structured commit strategy ensured traceability and easy rollback when required.

5.4 Prototype

A working prototype of the system is developed during the implementation phase to demonstrate the feasibility of the proposed system and to validate the selected technologies.

This prototype focuses on the core functionalities of the system, which are user authentication, role-based access control, document submission, workflow configuration, approval processing, and document status tracking.

The prototype is tested using sample document workflows to ensure that documents move correctly between workflow stages and that user roles are enforced properly. Feedback from testing is used to refine navigation, improve workflow clarity, and enhance overall system usability.

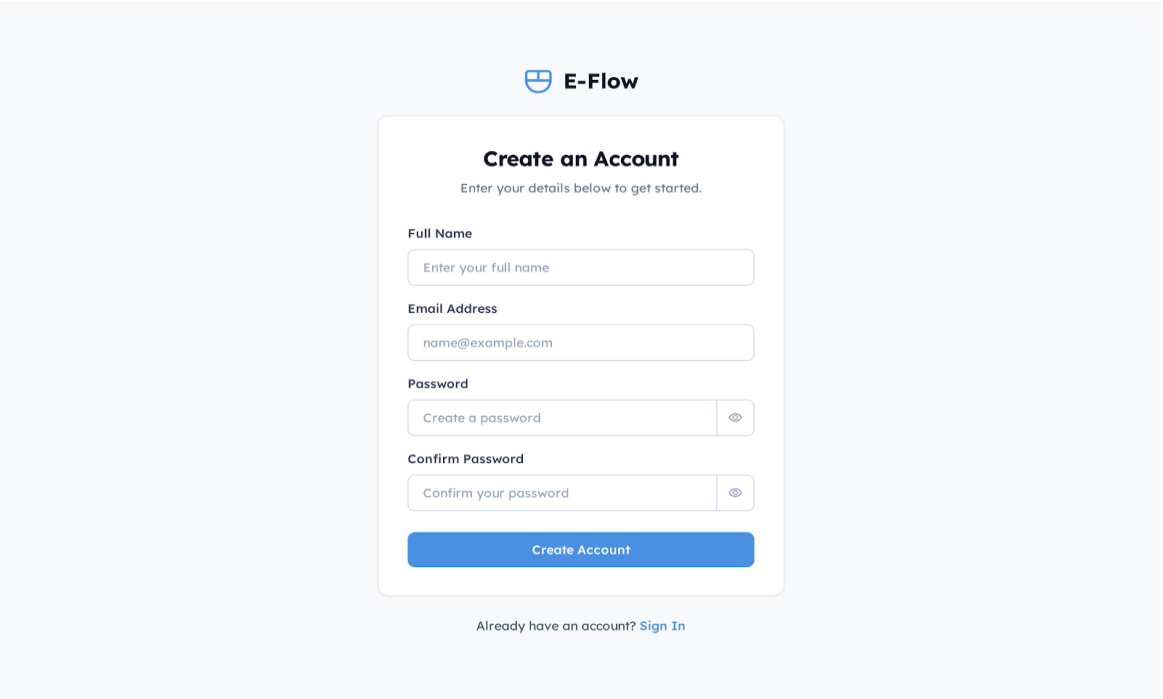

Figure 5.1: User registration

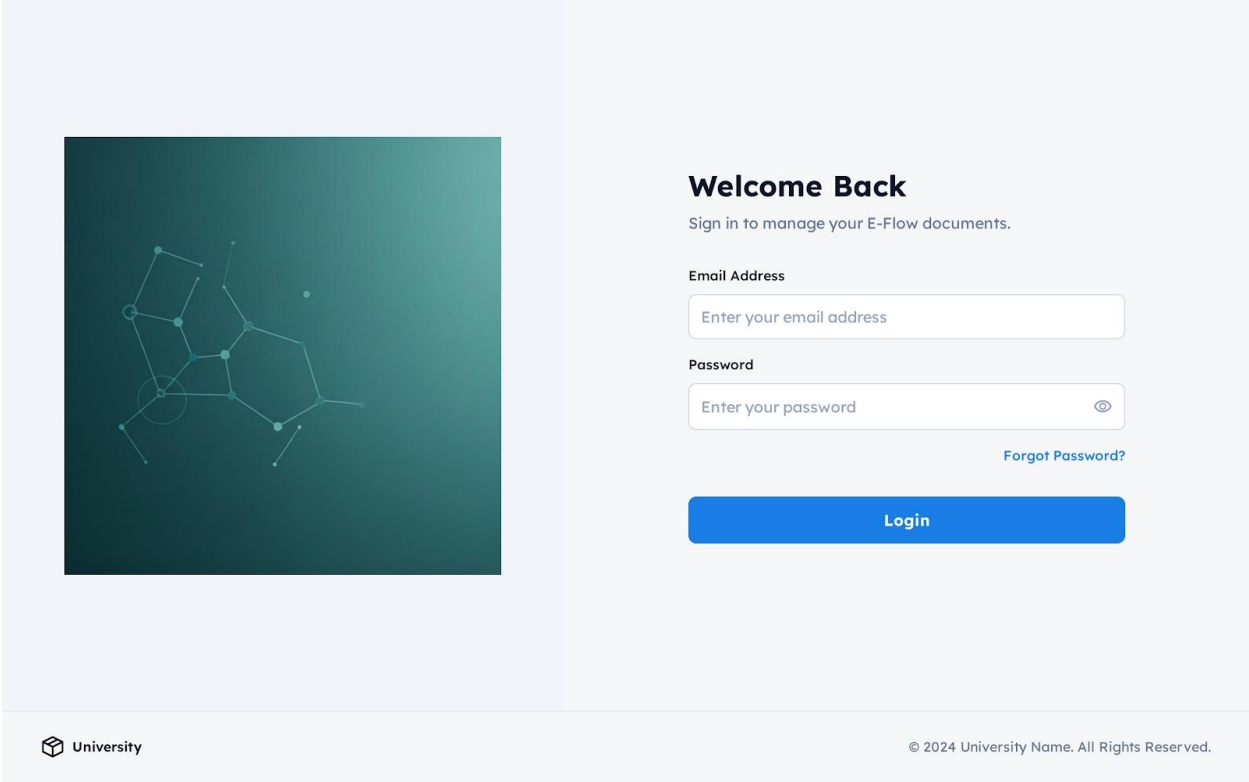

Figure 5.2: User login page

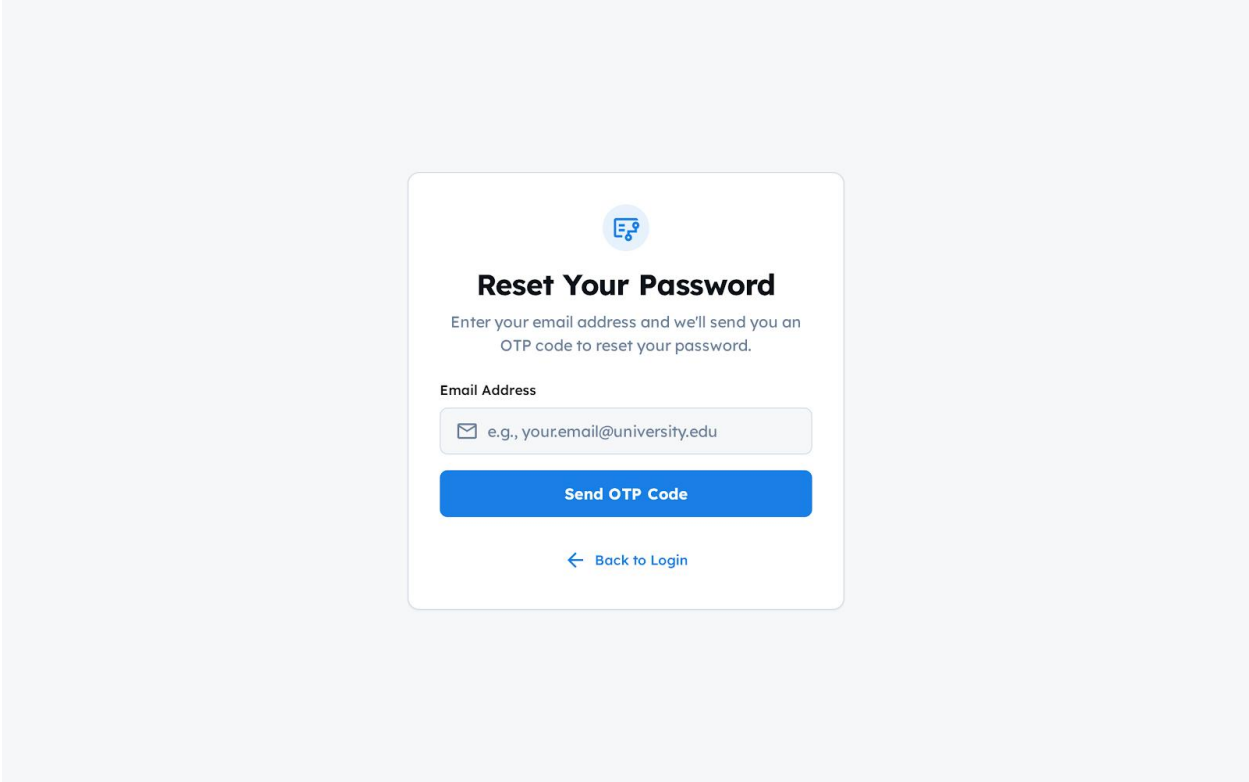

Figure 5.3: password reset page

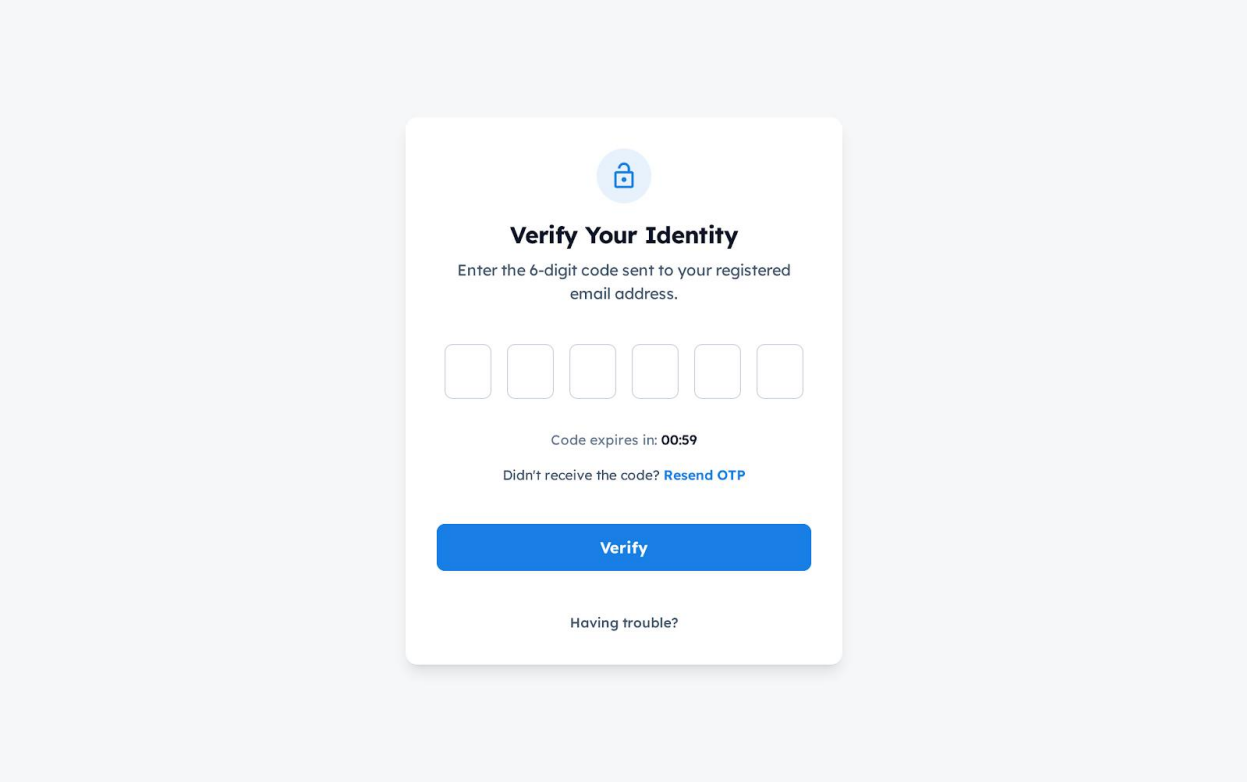

Figure 5.4: password reset identity verification(2fa)

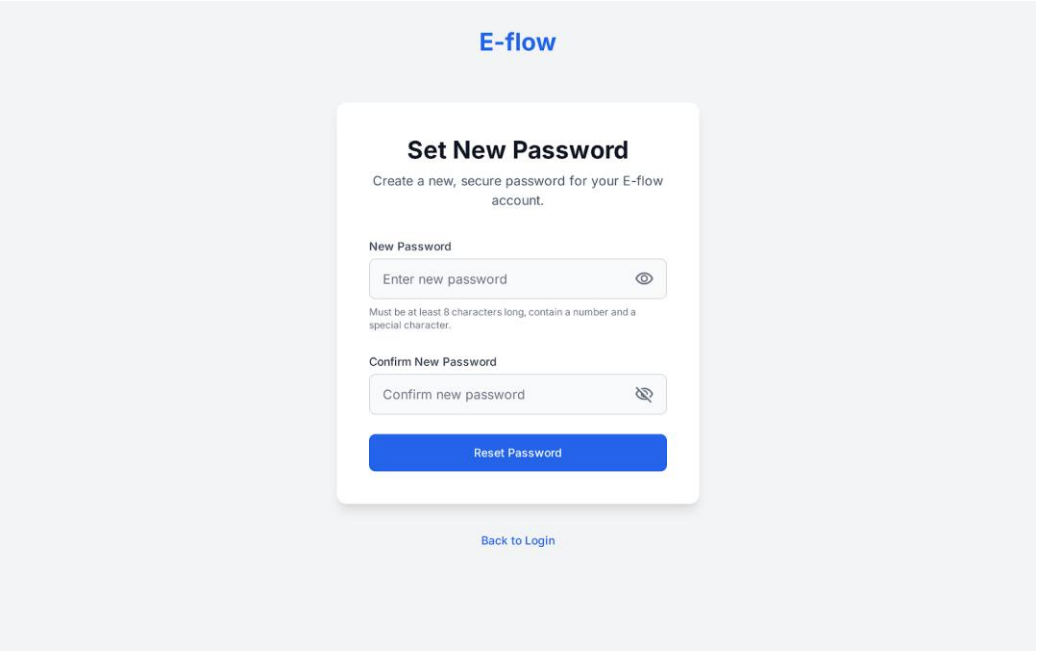

Figure 5.5: set new password page

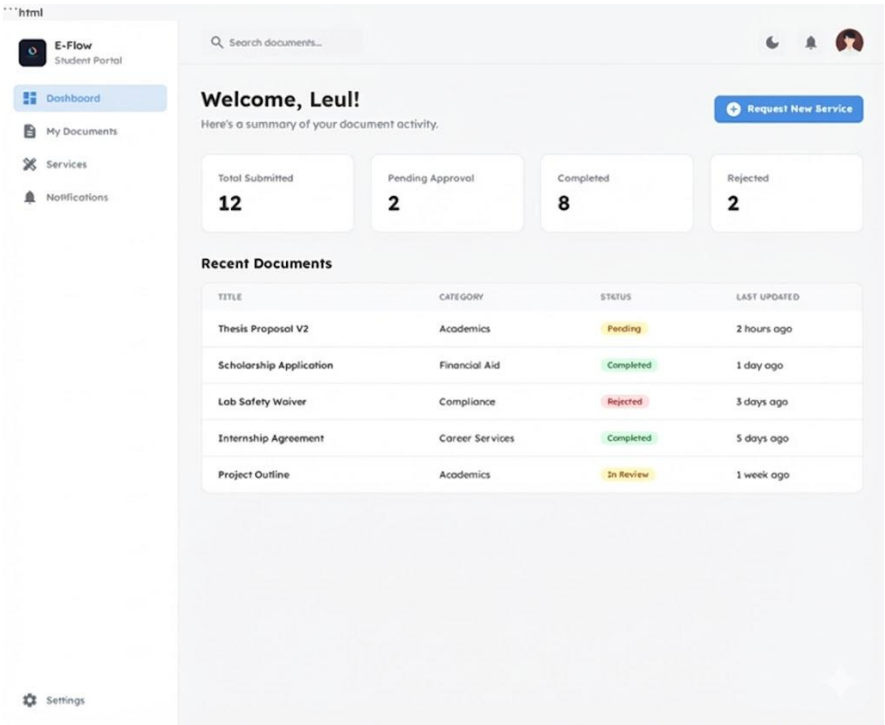

Figure 5.6: student Dashboard

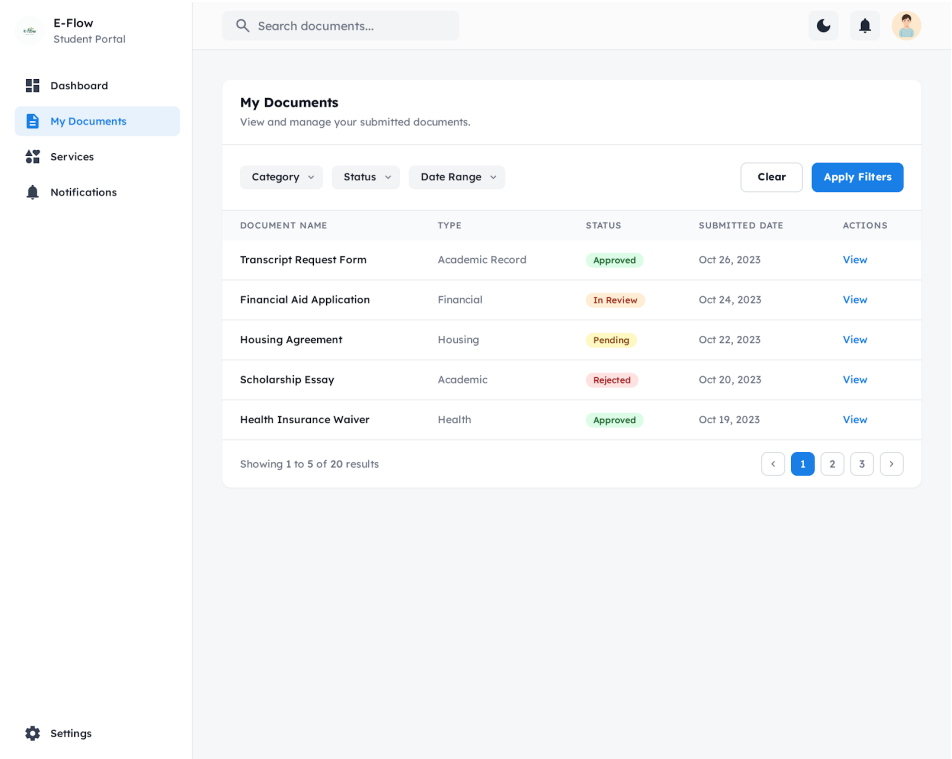

Figure 5.7: student's my documents page

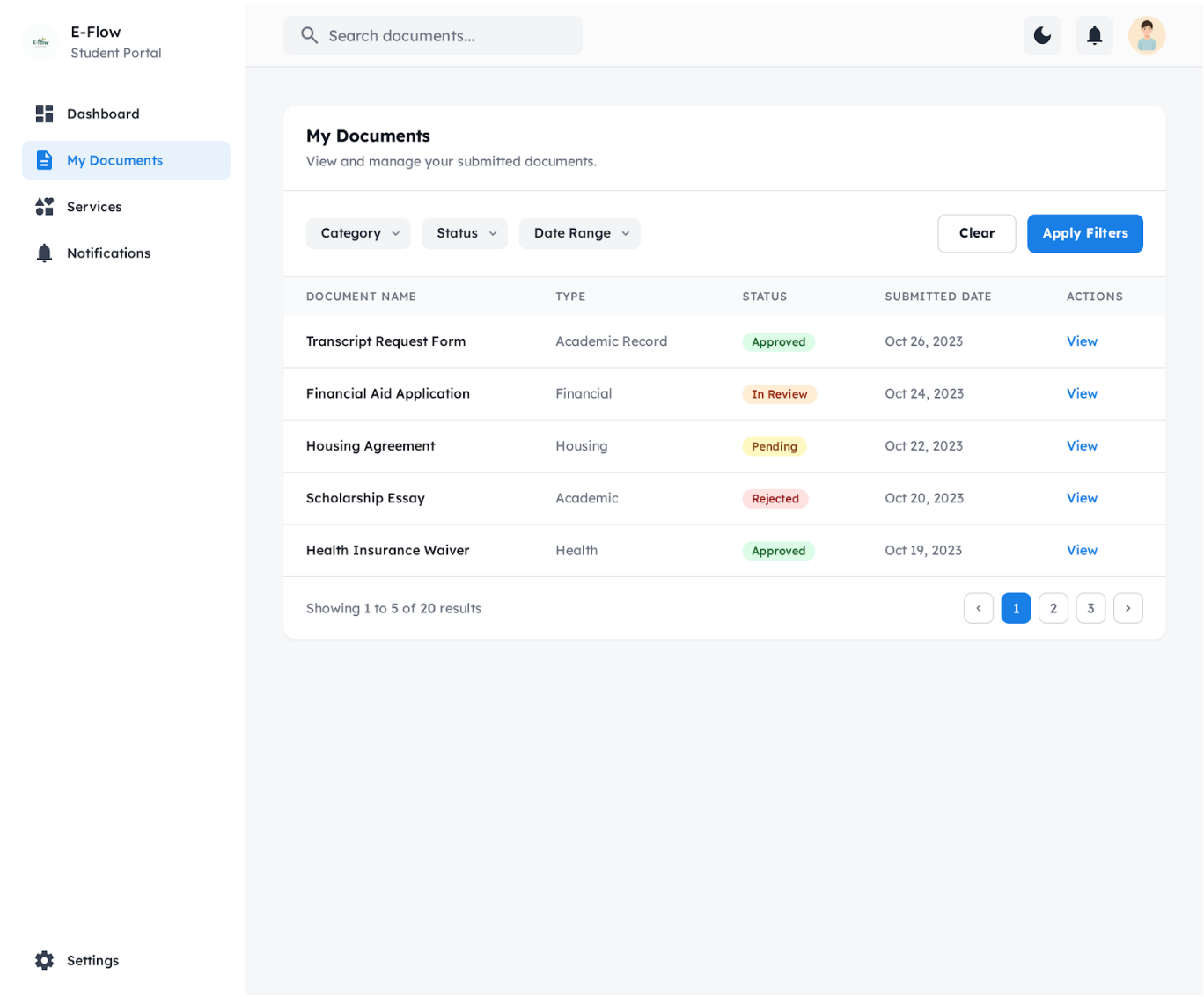

Figure 5.8: Student’s my document page

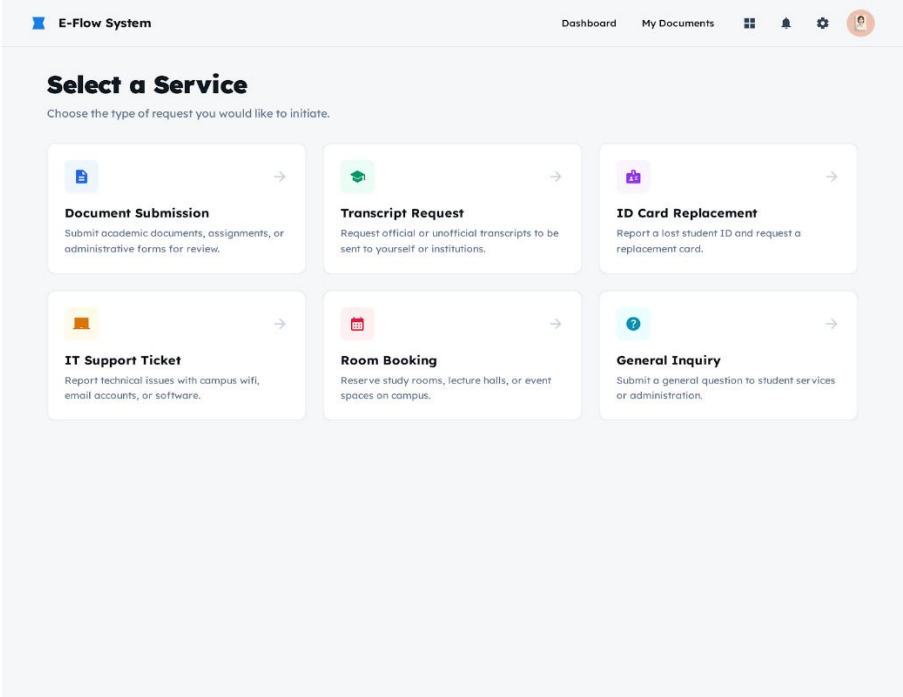

Figure 5.9: Student's Service Page

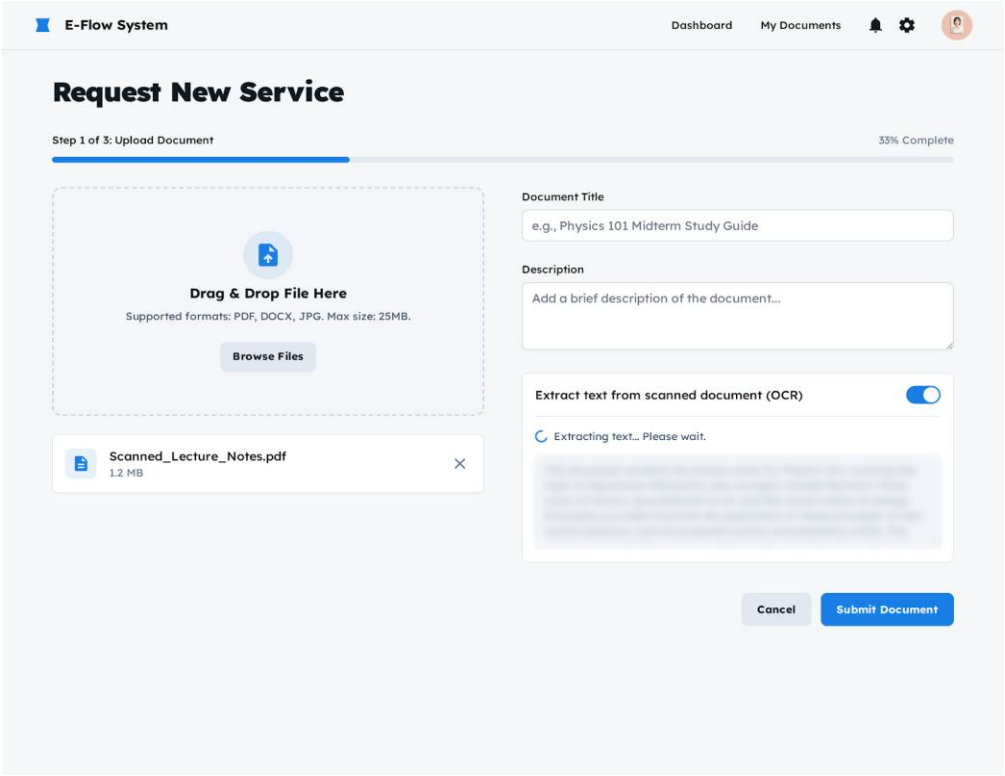

Figure 5.10: students request service page

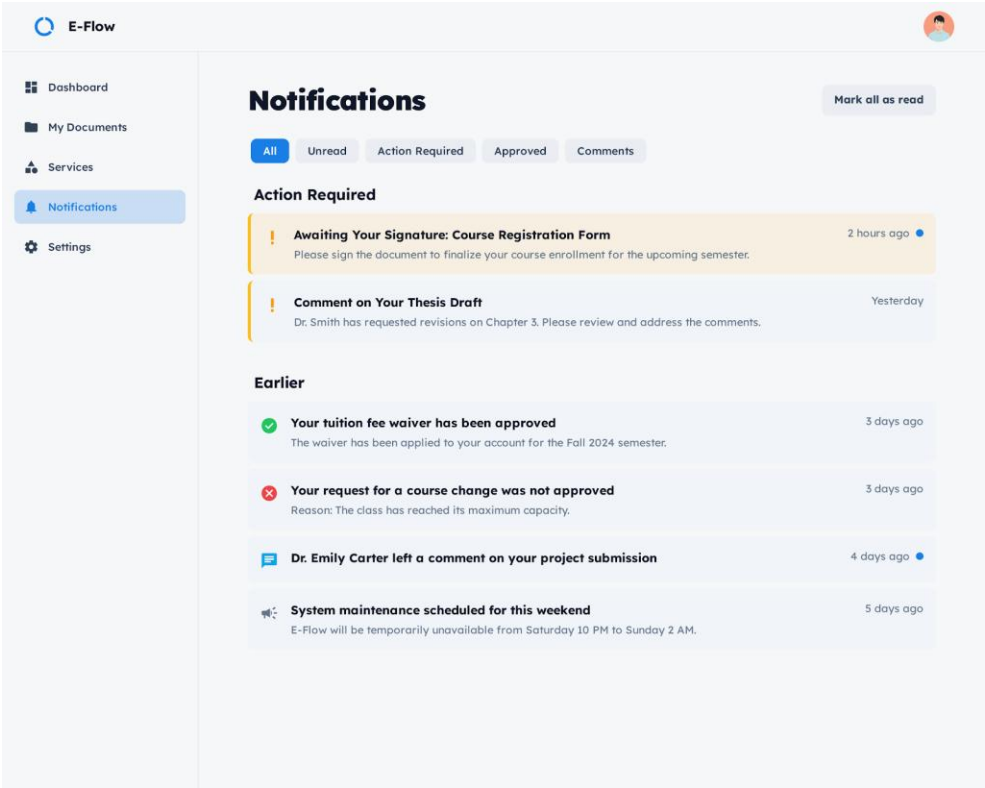

Figure 3.11: student's Notification page

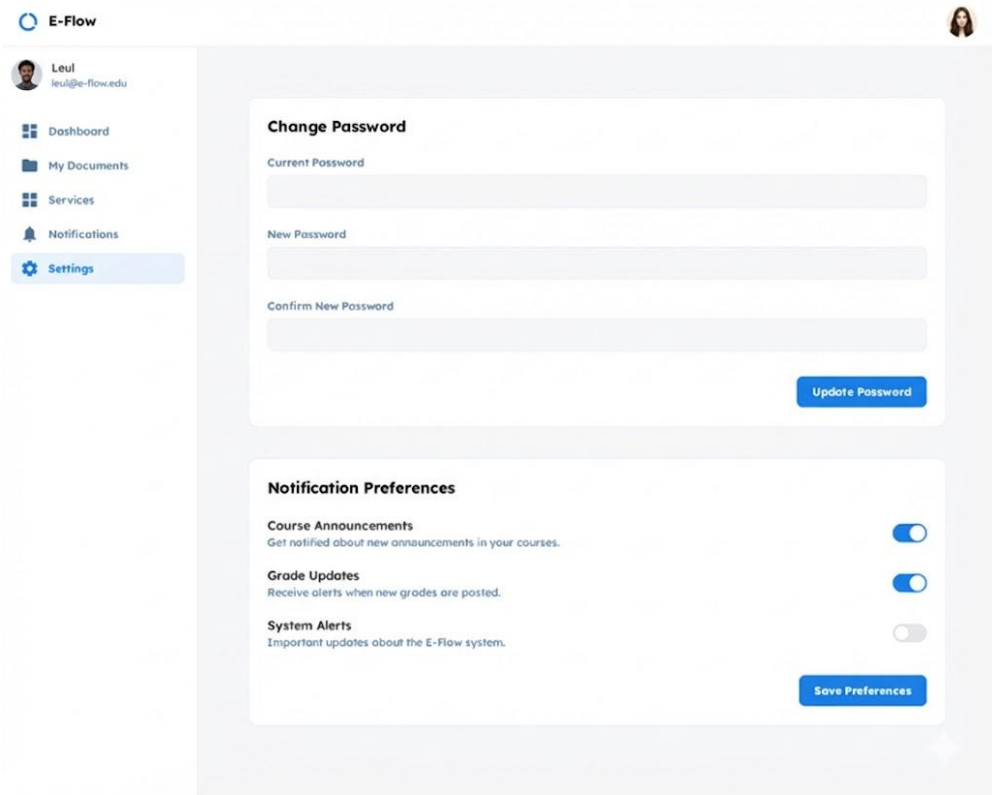

Figure 3.12: Students setting page

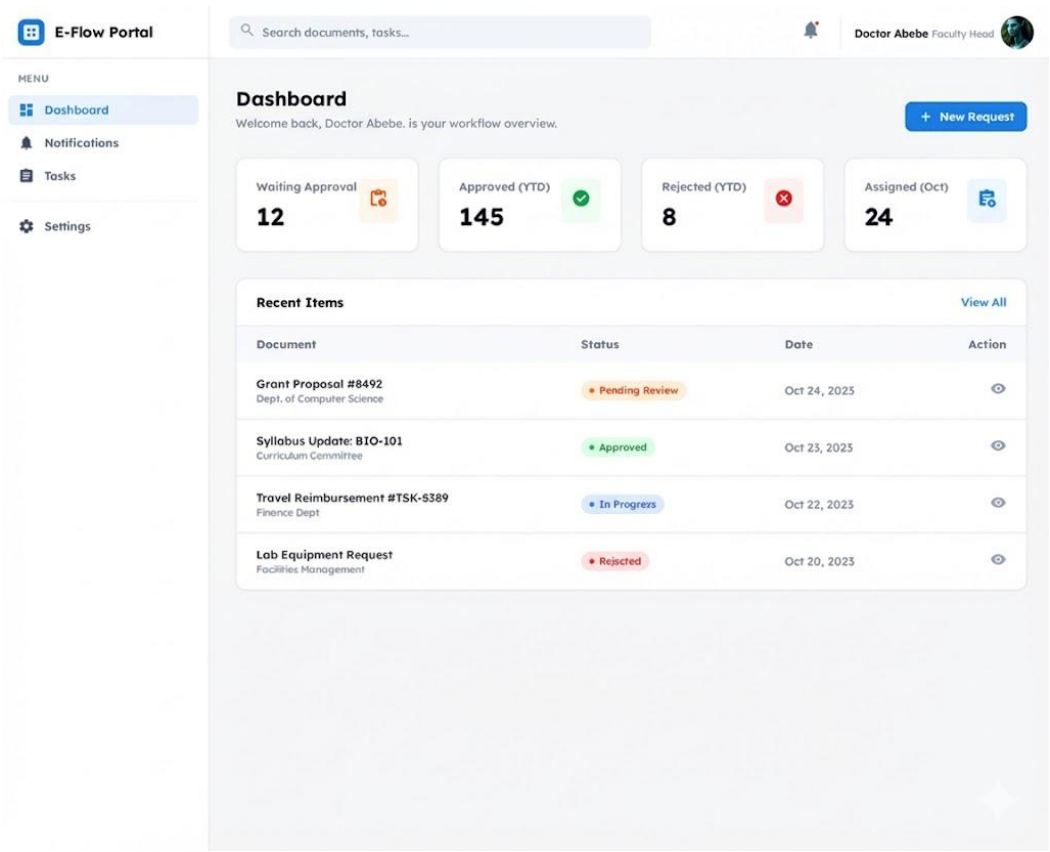

Figure 5.13: Stuff Dashboard

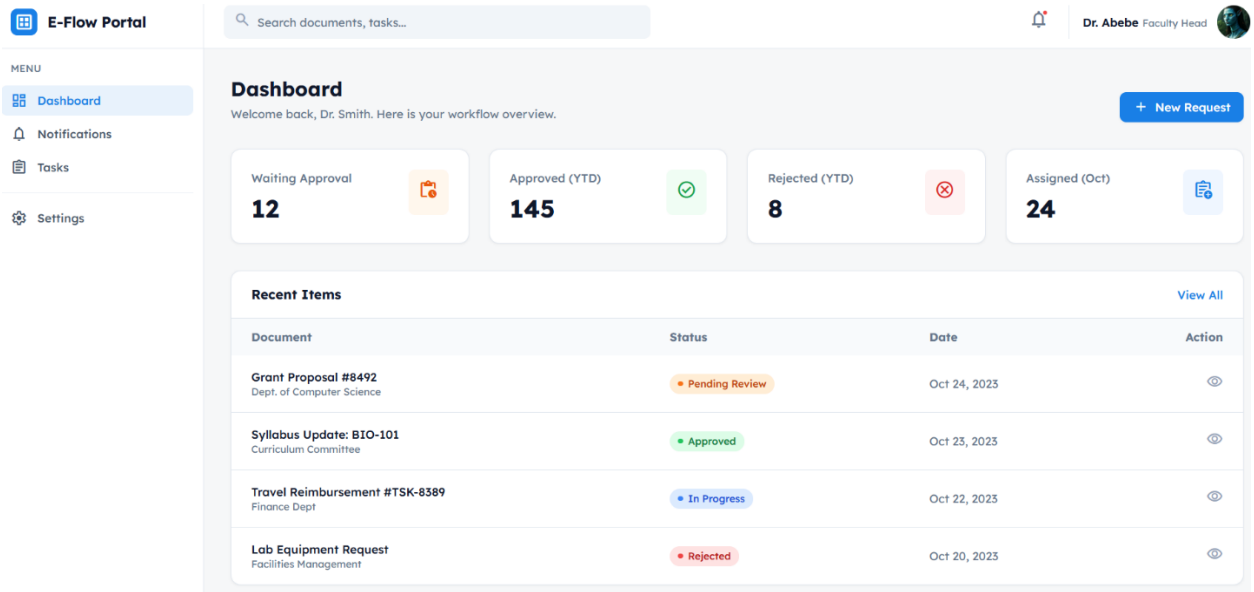

Figure 5.14: stuff's Tasks page

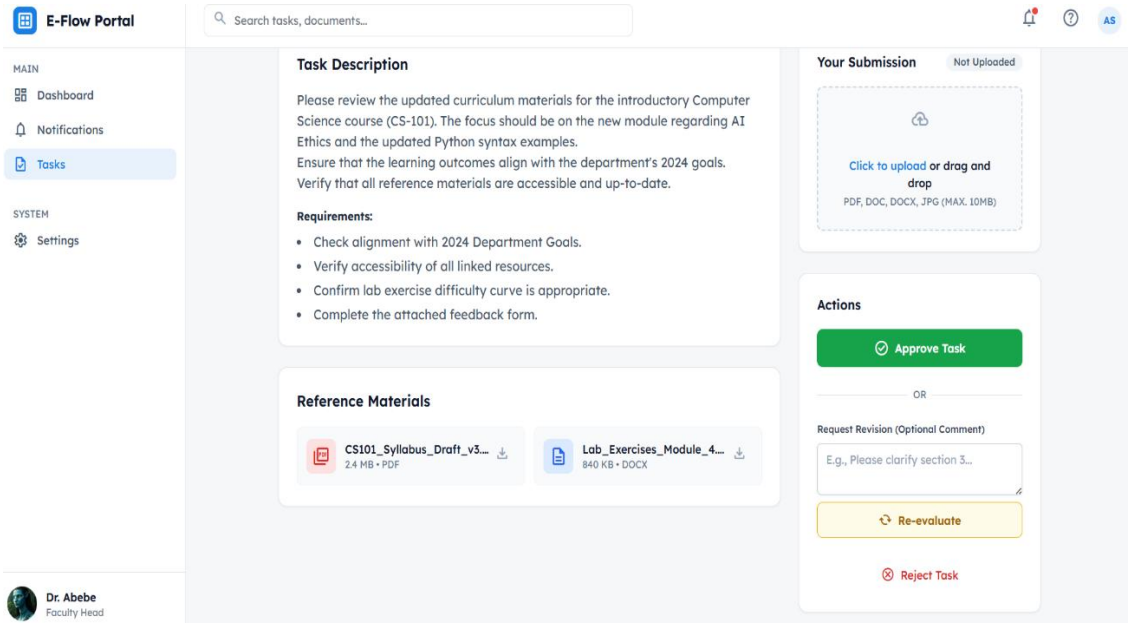

Figure 5.15: Stuff's specific tasks view page

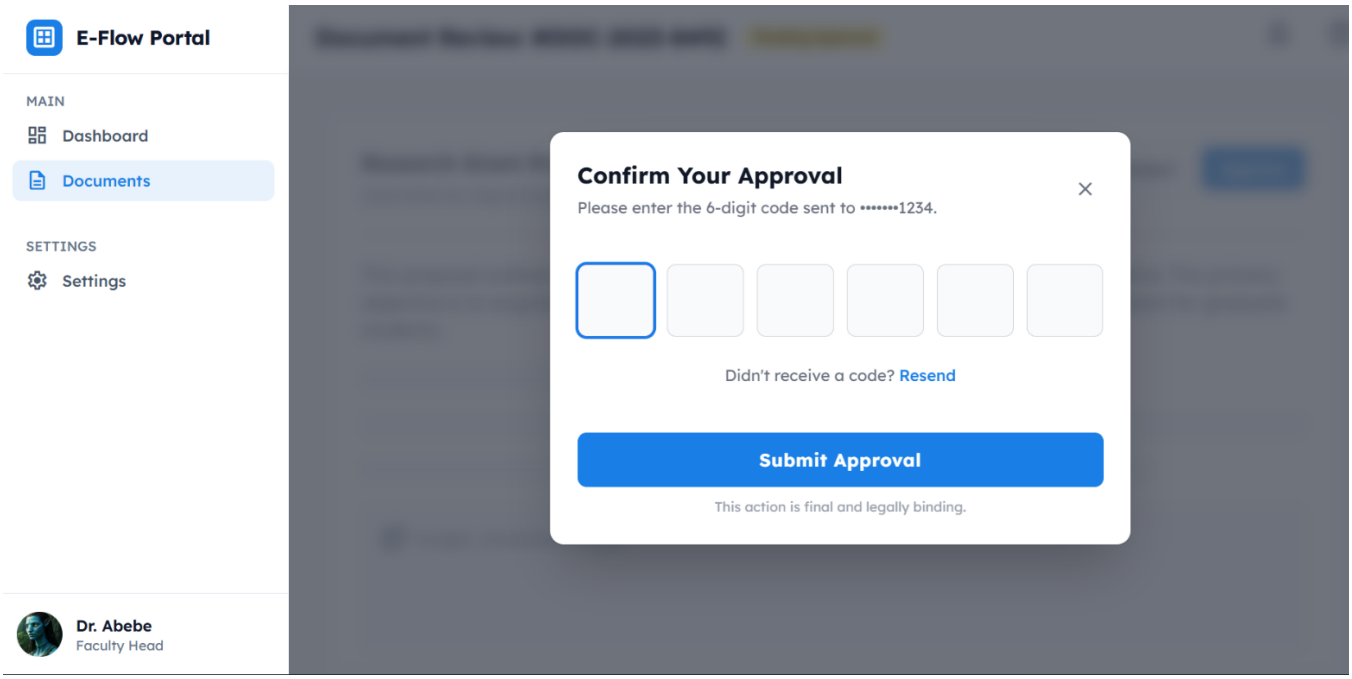

Figure 5.16: Digital signature page

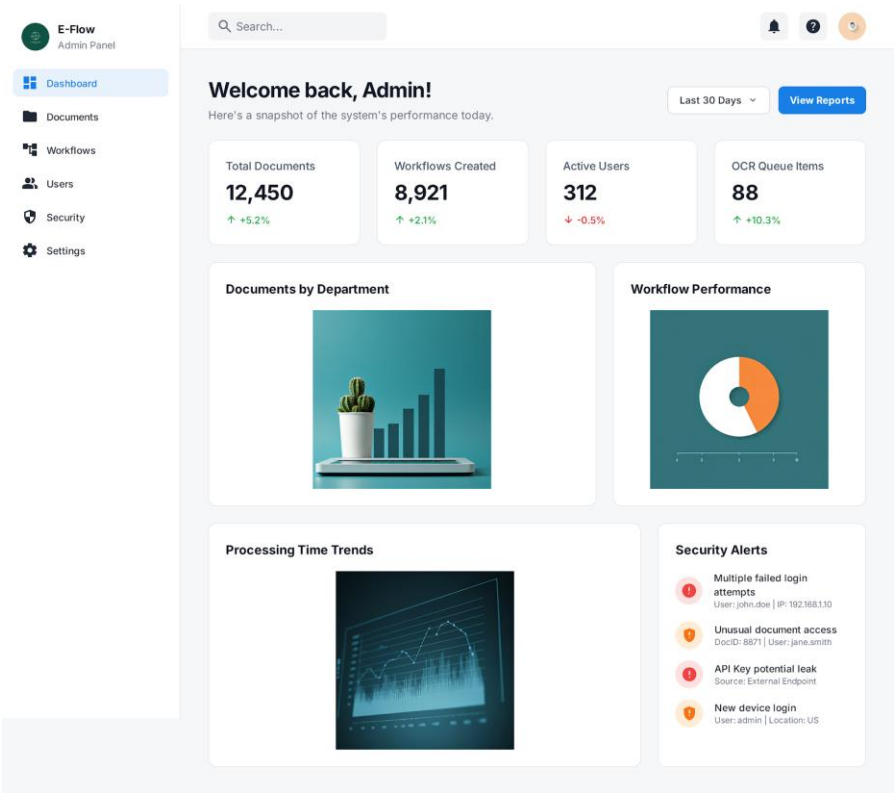

Figure 5.17: Admin Dashboard

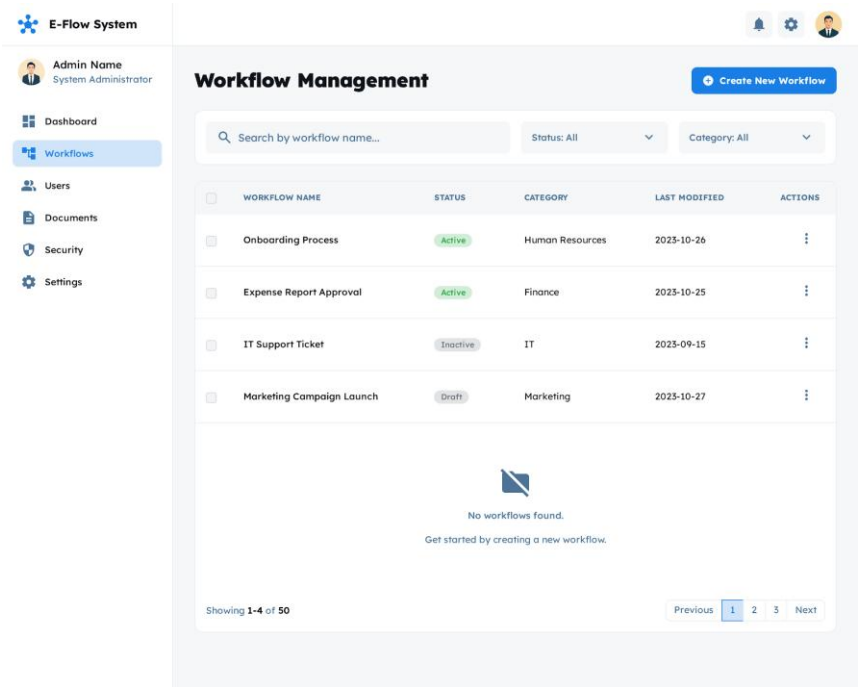

Figure 5.18: Admin Workflow Management page

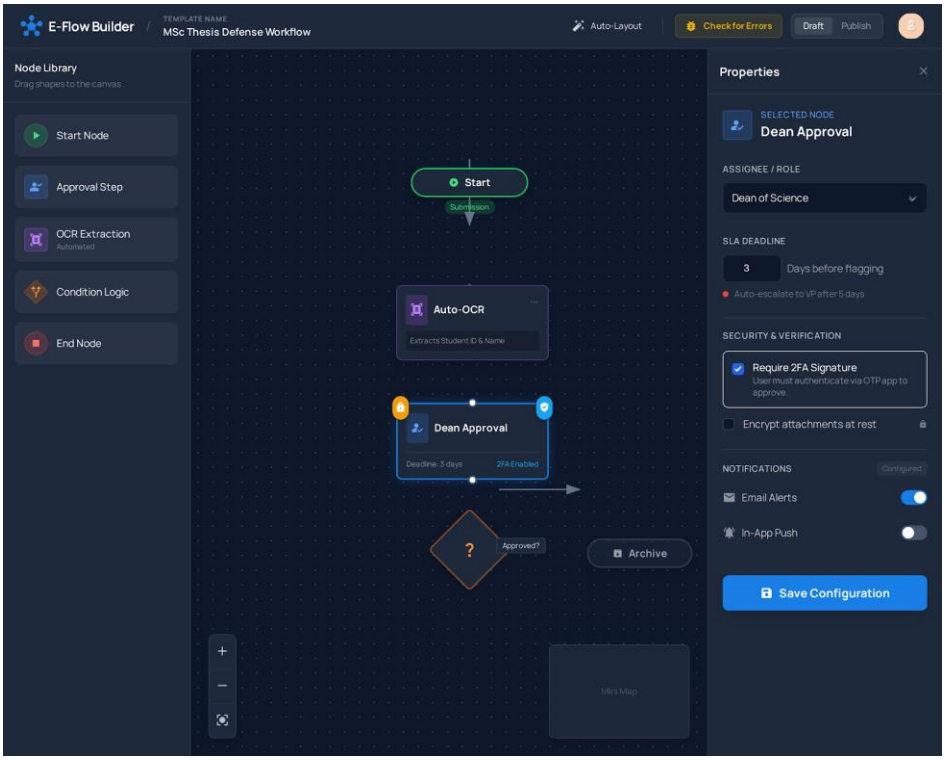

Figure 5.19: Admin workflow Builder

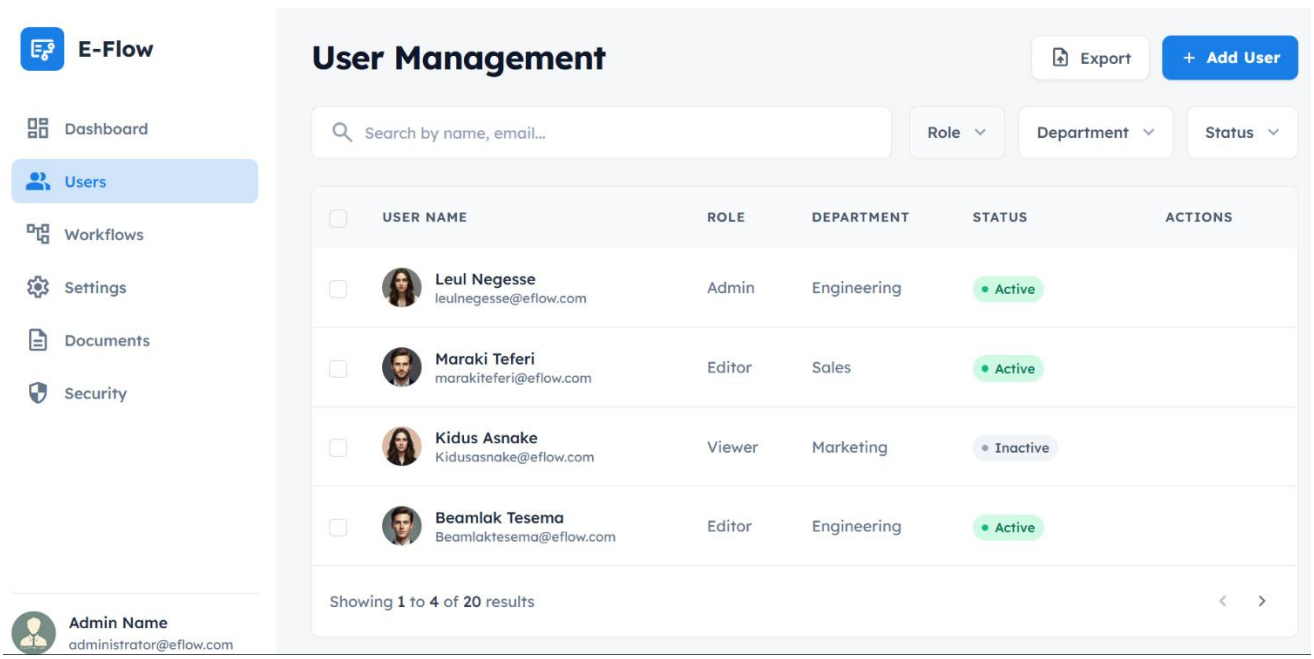

Figure 5.20: Admin User management page

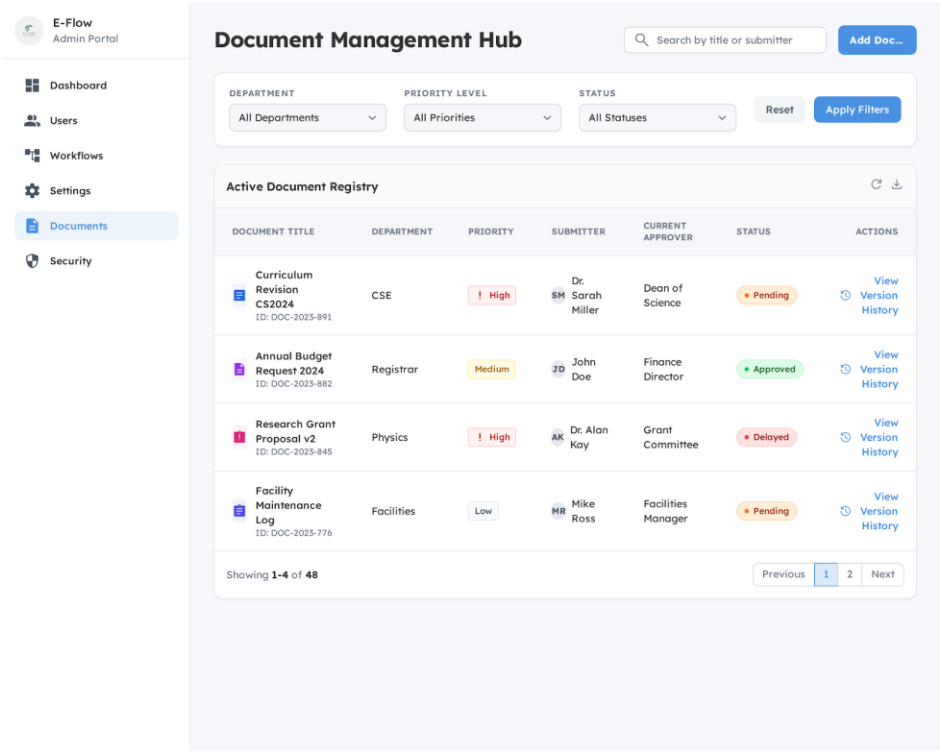

Figure 5.21: Document management hub

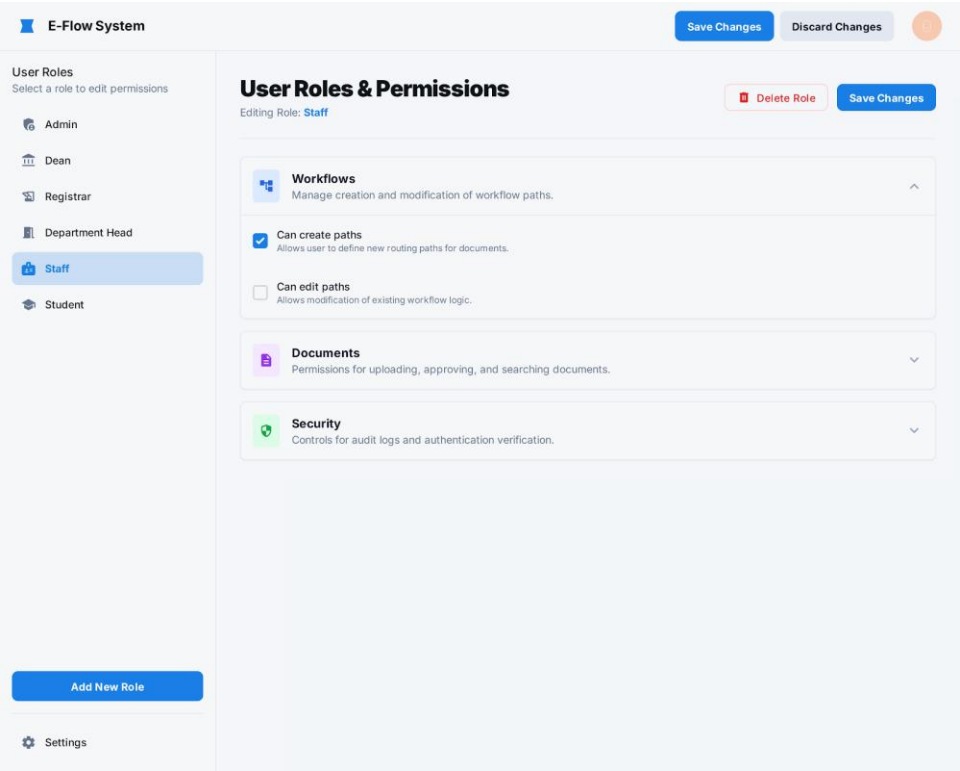

Figure 5.22: Admin Role Management page

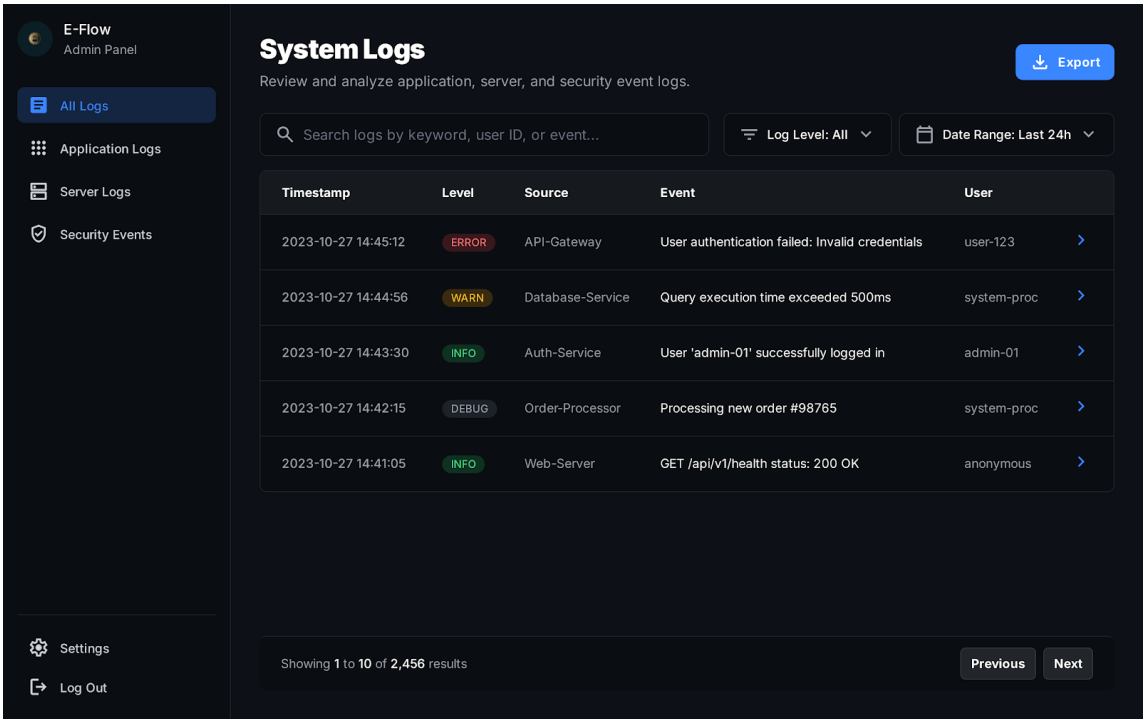

Figure 5.23: System logs page

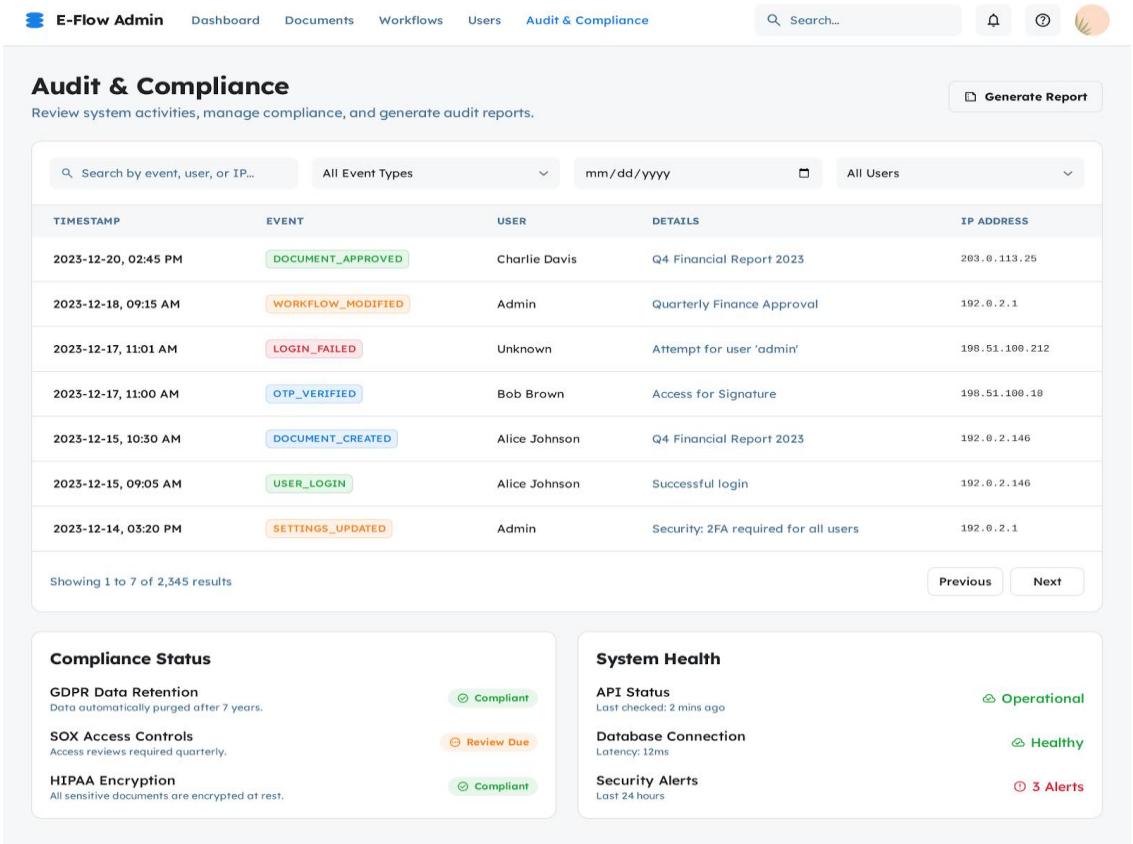

Figure 5.24: Audit and compliance page

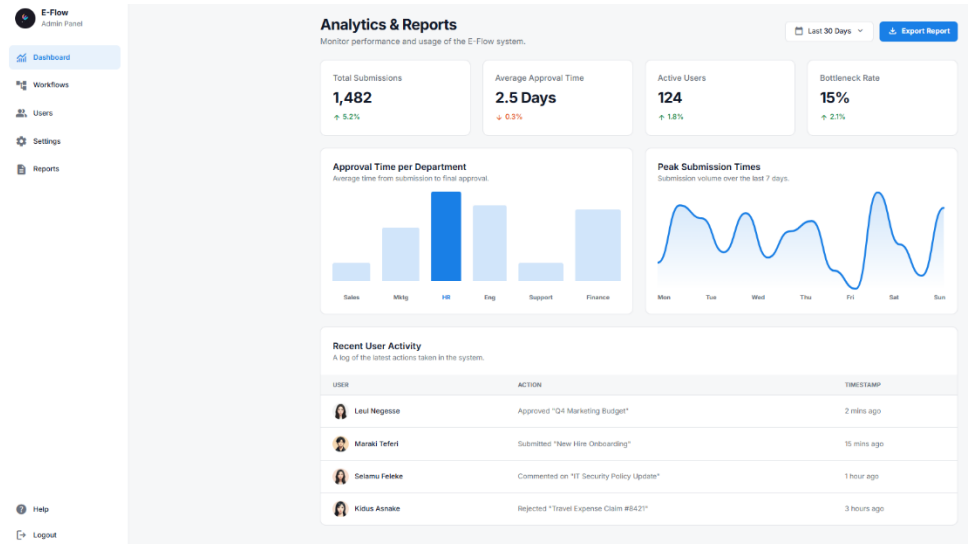

Figure 5.25: Admin Analytics and report page

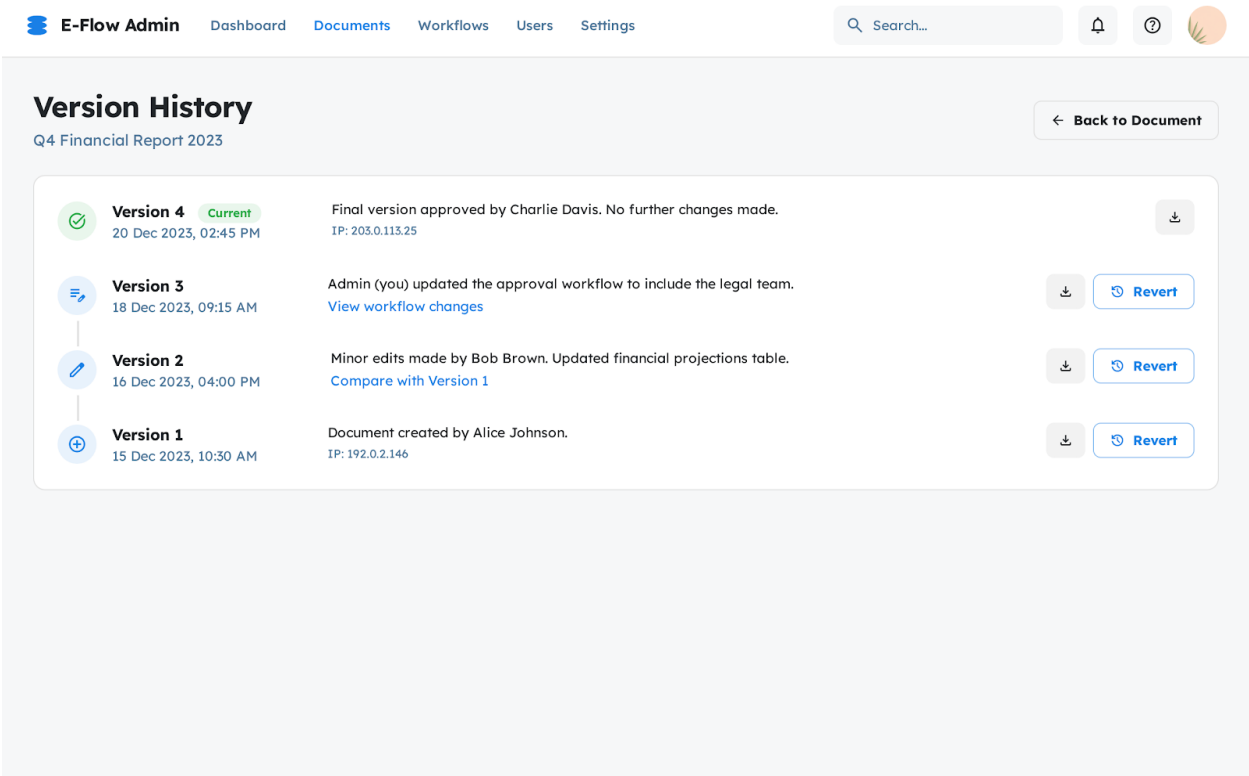

Figure 5.26: Document version history

5.5 Implementation Detail

This section provides a detailed explanation of how the system is implemented using the selected development tools and technologies. The system follows a **three-tier architecture**, which separates the presentation layer, application logic layer, and data storage layer. This architectural approach improves the system's scalability, security, and maintainability.

Key implementation characteristics:

- Separation of concerns between client-side, server-side, and database layers
- Secure client–server communication using RESTful APIs
- Centralized workflow execution and validation on the server-side
- Role-based access control enforced at multiple levels
- Persistent storage and auditability of all system activities

5.5.1 Client-Side Implementation

The client-side of the **E-flow system** is implemented using **React.js**, which provides a component-based framework for building dynamic and interactive web applications. The client-side is responsible for presenting system features to users and enabling interaction with documents and workflows.

User interfaces are designed based on user roles such as administrators, staff, and students. Each role is provided with a customized dashboard displaying relevant documents, pending tasks, and workflow status. Role-based rendering ensures that users only access features permitted by their assigned roles.

The visual workflow builder is implemented using **React Flow**, allowing administrators to create and modify document approval workflows through a drag-and-drop interface. This feature enables flexible workflow design without requiring programming knowledge.

Styling and layout are handled using **Tailwind CSS and Material UI** to ensure a consistent, responsive, and user-friendly interface. Client-side validation is implemented to minimize invalid input before data is sent to the server.

To support document digitization, we integrate **Tesseract.js** on the client-side, allowing users to upload scanned documents and extract text from image-based files. This extracted text is then prepared for further processing and storage by the backend.

Throughout system usage, the client-side continuously interacts with the server by requesting updated document statuses, approval notifications, and workflow progress. These interactions ensure that users receive near real-time feedback on document movement and approval outcomes.

5.5.2 Server-Side Implementation

The server-side is implemented using **Node.js with Express.js** to manage application logic, security, and data processing. The server acts as the central control unit of the system, coordinating interactions between the client-side and the database.

The backend handles user authentication and authorization using **JSON Web Tokens (JWT)**. After successful login, a token is issued to the user and is used to verify identity and enforce role-based access control for all protected operations.

Workflow automation logic is implemented on the server-side. When a document is submitted, the server determines the appropriate workflow path, assigns the document to the next approval

stage, and updates its status accordingly. Approval actions are validated to ensure that only authorized users can perform them.
For secure document approval, the system integrates an email-based verification mechanism using **Nodemailer**. During critical approval actions, a **One-Time Password (OTP)** is generated using cryptographic libraries (**Crypto or pyotp**) and sent to the user's registered email address. The server also manages error handling, input validation, and audit logging. All significant actions, such as document submission, approval, and status changes, are recorded to maintain transparency and accountability.

5.5.3 Database

The database component of our **E-flow system** is implemented using **PostgreSQL**, a reliable and open-source relational database management system. The database is responsible for storing all persistent data required for system operation, including user accounts, roles, document metadata, workflow definitions, approval records, and audit logs.
The database schema is designed to support dynamic workflow management. Tables are created for users, roles, documents, workflows, workflow steps and approval history. Relations between these tables are defined using primary and foreign keys to maintain data consistency and integrity.
PostgreSQL's support for structured data and JSON fields is utilized to store flexible workflow configurations. This allows workflows to be created and modified without requiring changes to the database structure.
All database operations are performed through the server-side application, ensuring controlled and secure access. Indexing is applied to frequently queried fields such as document status and user identifiers to improve system performance. This design ensures efficient data retrieval, accurate document tracking, and reliable audit trail management.